```
{
"data":{
"id":"1",
"type":"Ticket",
"attributes":{
"subject":"Problem with Ticket",
"description":"Hello,\r\n\r\nI have a problem with my ticket.\r\n\r\nCan you please check it?\r\n\r\nThanks\r\n\r\n\r\n
```

Mr. Ralph Gallagher
Director of Operations, Sunset Ridge Recreation

Mr. Todd Martin
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Mr. Todd Martin
Mr. Ralph Gallagher
Mr. Todd Martin
Mr. Ralph Gallagher

```
{
    "_id" : ObjId(0),
    "_rev" : "1-ed1d0d0d0d0d0d0d",
    "_attachments" : {
        "1" : {
            "content_type" : "application/octet-stream",
            "revpos" : 1,
            "digest" : "sha1-ZGRlZGRlZGRlZGRlZGRlZGRlZGQ=",
            "stub" : true,
            "length" : 1024
        },
        "2" : {
            "content_type" : "application/octet-stream",
            "revpos" : 1,
            "digest" : "sha1-ZmZlZmZlZmZlZmZlZmZlZmZlZmY=",
            "stub" : true,
            "length" : 1024
        },
        "3" : {
            "content_type" : "application/octet-stream",
            "revpos" : 1,
            "digest" : "sha1-Y2NjY2NjY2NjY2NjY2NjY2NjY2M=",
            "stub" : true,
            "length" : 1024
        },
        "4" : {
            "content_type" : "application/octet-stream",
            "revpos" : 1,
            "digest" : "sha1-YmJiYmJiYmJiYmJiYmJiYmJiYmI=",
            "stub" : true,
            "length" : 1024
        },
        "5" : {
            "content_type" : "application/octet-stream",
            "revpos" : 1,
            "digest" : "sha1-YWFhYWFhYWFhYWFhYWFhYWFhYWE=",
            "stub" : true,
            "length" : 1024
        },
        "6" : {
            "content_type" : "application/octet-stream",
            "revpos" : 1,
            "digest" : "sha1-YGBgYGBgYGBgYGBgYGBgYGBgYGA=",
            "stub" : true,
            "length" : 1024
        },
        "7" : {
            "content_type" : "application/octet-stream",
            "revpos" : 1,
            "digest" : "sha1-X19fX19fX19fX19fX19fX19fX19f",
            "stub" : true,
            "length" : 1024
        },
        "8" : {
            "content_type" : "application/octet-stream",
            "revpos" : 1,
            "digest" : "sha1-Xl5eXl5eXl5eXl5eXl5eXl5eXl4=",
            "stub" : true,
            "length" : 1024
        },
        "9" : {
            "content_type" : "application/octet-stream",
            "revpos" : 1,
            "digest" : "sha1-XV1dXV1dXV1dXV1dXV1dXV1dXV0=",
            "stub" : true,
            "length" : 1024
        },
        "10" : {
            "content_type" : "application/octet-stream",
            "revpos" : 1,
            "digest" : "sha1-XFxcXFxcXFxcXFxcXFxcXFxcXFw=",
            "stub" : true,
            "length" : 1024
        },
        "11" : {
            "content_type" : "application/octet-stream",
            "revpos" : 1,
            "digest" : "sha1-W1tbW1tbW1tbW1tbW1tbW1tbW1s=",
            "stub" : true,
            "length" : 1024
        },
        "12" : {
            "content_type" : "application/octet-stream",
            "revpos" : 1,
            "digest" : "sha1-WmpaamphamphamphamphamphamphaW8=",
            "stub" : true,
            "length" : 1024
        },
        "13" : {
            "content_type" : "application/octet-stream",
            "revpos" : 1,
            "digest" : "sha1-WWlaWWlaWWlaWWlaWWlaWWlaWWlaWWk=",
            "stub" : true,
            "length" : 1024
        },
        "14" : {
            "content_type" : "application/octet-stream",
            "revpos" : 1,
            "digest" : "sha1-WGhYWGhYWGhYWGhYWGhYWGhYWGhYWGg=",
            "stub" : true,
            "length" : 1024
        },
        "15" : {
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            "revpos" : 1,
            "digest" : "sha1-V2dXV2dXV2dXV2dXV2dXV2dXV2dXV2c=",
            "stub" : true,
            "length" : 1024
        },
        "16" : {
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            "revpos" : 1,
            "digest" : "sha1-VmZWVmZWVmZWVmZWVmZWVmZWVmZWVmY=",
            "stub" : true,
            "length" : 1024
        },
        "17" : {
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            "revpos" : 1,
            "digest" : "sha1-VWVlVWVlVWVlVWVlVWVlVWVlVWVlVWU=",
            "stub" : true,
            "length" : 1024
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            "revpos" : 1,
            "digest" : "sha1-VGRkVGRkVGRkVGRkVGRkVGRkVGRkVGQ=",
            "stub" : true,
            "length" : 1024
        },
        "19" : {
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            "revpos" : 1,
            "digest" : "sha1-U2NjU2NjU2NjU2NjU2NjU2NjU2NjU2M=",
            "stub" : true,
            "length" : 1024
        },
        "20" : {
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            "revpos" : 1,
            "digest" : "sha1-UmJiUmJiUmJiUmJiUmJiUmJiUmJiUmI=",
            "stub" : true,
            "length" : 1024
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        "21" : {
            "content_type" : "application/octet-stream",
            "revpos" : 1,
            "digest" : "sha1-UWFhUWFhUWFhUWFhUWFhUWFhUWFhUWE=",
            "stub" : true,
            "length" : 1024
        },
        "22" : {
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            "revpos" : 1,
            "digest" : "sha1-UGBgUGBgUGBgUGBgUGBgUGBgUGBgUGA=",
            "stub" : true,
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        },
        "23" : {
            "content_type" : "application/octet-stream",
            "revpos" : 1,
            "digest" : "sha1-T19fT19fT19fT19fT19fT19fT19fT19f",
            "stub" : true,
            "length" : 1024
        },
        "24" : {
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            "revpos" : 1,
            "digest" : "sha1-Tl5eTl5eTl5eTl5eTl5eTl5eTl5eTl4=",
            "stub" : true,
            "length" : 1024
        },
        "25" : {
            "content_type" : "application/octet-stream",
            "revpos" : 1,
            "digest" : "sha1-TV1dTV1dTV1dTV1dTV1dTV1dTV1dTV0=",
            "stub" : true,
            "length" : 1024
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            "content_type" : "application/octet-stream",
            "revpos" : 1,
            "digest" : "sha1-TFxcTFxcTFxcTFxcTFxcTFxcTFxcTFw=",
            "stub" : true,
            "length" : 1024
        },
        "27" : {
            "content_type" : "application/octet-stream",
            "revpos" : 1,
            "digest" : "sha1-S1tbS1tbS1tbS1tbS1tbS1tbS1tbS1s=",
            "stub" : true,
            "length" : 1024
        },
        "28" : {
            "content_type" : "application/octet-stream",
            "revpos" : 1,
            "digest" : "sha1-SmpaSmpaSmpaSmpaSmpaSmpaSmpaSW8=",
            "stub" : true,
            "length" : 1024
        },
        "29" : {
            "content_type" : "application/octet-stream",
            "revpos" : 1,
            "digest" : "sha1-SWlaSWlaSWlaSWlaSWlaSWlaSWlaSWk=",
            "stub" : true,
            "length" : 1024
        },
        "30" : {
            "content_type" : "application/octet-stream",
            "revpos" : 1,
            "digest" : "sha1-SGhYSGhYSGhYSGhYSGhYSGhYSGhYSGg=",
            "stub" : true,
            "length" : 1024
        },
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            "revpos" : 1,
            "digest" : "sha1-R2dXR2dXR2dXR2dXR2dXR2dXR2dXR2c=",
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            "length" : 1024
        },
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            "content_type" : "application/octet-stream",
            "revpos" : 1,
            "digest" : "sha1-RmZWRmZWRmZWRmZWRmZWRmZWRmZWRmY=",
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            "length" : 1024
        },
        "33" : {
            "content_type" : "application/octet-stream",
            "revpos" : 1,
            "digest" : "sha1-RWVlRWVlRWVlRWVlRWVlRWVlRWVlRWU=",
            "stub" : true,
            "length" : 1024
        },
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            "content_type" : "application/octet-stream",
            "revpos" : 1,
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            "stub" : true,
            "length" : 1024
        },
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            "revpos" : 1,
            "digest" : "sha1-Q2NjQ2NjQ2NjQ2NjQ2NjQ2NjQ2NjQ2M=",
            "stub" : true,
            "length" : 1024
        },
        "36" : {
            "content_type" : "application/octet-stream",
            "revpos" : 1,
            "digest" : "sha1-QmJiQmJiQmJiQmJiQmJiQmJiQmJiQmI=",
            "stub" : true,
            "length" : 1024
        },
        "37" : {
            "content_type" : "application/octet-stream",
            "revpos" : 1,
            "digest" : "sha1-QWFhQWFhQWFhQWFhQWFhQWFhQWFhQWE=",
            "stub" : true,
            "length" : 1024
        },
        "38" : {
            "content_type" : "application/octet-stream",
            "revpos" : 1,
            "digest" : "sha1-QGBgQGBgQGBgQGBgQGBgQGBgQGBgQGA=",
            "stub" : true,
            "length" : 1024
        },
        "39" : {
            "content_type" : "application/octet-stream",
            "revpos" : 1,
            "digest" : "sha1-P19fP19fP19fP19fP19fP19fP19fP19f",
            "stub" : true,
            "length" : 1024
        },
        "40" : {
            "content_type" : "application/octet-stream",
            "revpos" : 1,
            "digest" : "sha1-Pl5ePl5ePl5ePl5ePl5ePl5ePl5ePl4=",
            "stub" : true,
            "length" : 1024
        },
        "41" : {
            "content_type" : "application/octet-stream",
            "revpos" : 1,
            "digest" : "sha1-PV1dPV1dPV1dPV1dPV1dPV1dPV1dPV0=",
            "stub" : true,
            "length" : 1024
        },
        "42" : {
            "content_type" : "application/octet-stream",
            "revpos" : 1,
            "digest" : "sha1-PFxcPFxcPFxcPFxcPFxcPFxcPFxcPFw=",
            "stub" : true,
            "length" : 1024
        },
        "43" : {
            "content_type" : "application/octet-stream",
            "revpos" : 1,
            "digest" : "sha1-O1tbO1tbO1tbO1tbO1tbO1tbO1tbO1s=",
            "stub" : true,
            "length" : 1024
        },
        "44" : {
            "content_type" : "application/octet-stream",
            "revpos" : 1,
            "digest" : "sha1-OmpaOmpaOmpaOmpaOmpaOmpaOmpaOW8=",
            "stub" : true,
            "length" : 1024
        },
        "45" : {
            "content_type" : "application/octet-stream",
            "revpos" : 1,
            "digest" : "sha1-OWlaOWlaOWlaOWlaOWlaOWlaOWlaOWk=",
            "stub" : true,
            "length" : 1024
        },
        "46" : {
            "content_type" : "application/octet-stream",
            "revpos" : 1,
            "digest" : "sha1-OGhYOGhYOGhYOGhYOGhYOGhYOGhYOGg=",
            "stub" : true,
            "length" : 1024
        },
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            "revpos" : 1,
            "digest" : "sha1-N2dXN2dXN2dXN2dXN2dXN2dXN2dXN2c=",
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        },
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            "stub" : true,
            "length" : 1024
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            "revpos" : 1,
            "digest" : "sha1-NWVlNWVlNWVlNWVlNWVlNWVlNWVlNWU=",
            "stub" : true,
            "length" : 1024
        },
        "50" : {
            "content_type" : "application/octet-stream",
            "revpos" : 1,
            "digest" : "sha1-NGRkNGRkNGRkNGRkNGRkNGRkNGRkNGQ=",
            "stub" : true,
            "length" : 1024
        },
        "51" : {
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            "revpos" : 1,
            "digest" : "sha1-M2NjM2NjM2NjM2NjM2NjM2NjM2NjM2M=",
            "stub" : true,
            "length" : 1024
        },
        "52" : {
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            "revpos" : 1,
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            "stub" : true,
            "length" : 1024
        },
        "53" : {
            "content_type" : "application/octet-stream",
            "revpos" : 1,
            "digest" : "sha1-MWFhMWFhMWFhMWFhMWFhMWFhMWFhMWE=",
            "stub" : true,
            "length" : 1024
        },
        "54" : {
            "content_type" : "application/octet-stream",
            "revpos" : 1,
            "digest" : "sha1-MGBgMGBgMGBgMGBgMGBgMGBgMGBgMGA=",
            "stub" : true,
            "length" : 1024
        },
        "55" : {
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            "revpos" : 1,
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            "stub" : true,
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            "stub" : true,
            "length" : 1024
        },
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            "revpos" : 1,
            "digest" : "sha1-K1tbK1tbK1tbK1tbK1tbK1tbK1tbK1s=",
            "stub" : true,
            "length" : 1024
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            "revpos" : 1,
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            "length" : 1024
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            "revpos" : 1,
            "digest" : "sha1-KWlaKWlaKWlaKWlaKWlaKWlaKWlaKWk=",
            "stub" : true,
            "length" : 1024
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            "revpos" : 1,
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            "stub" : true,
            "length" : 1024
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        "63" : {
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            "revpos" : 1,
            "digest" : "sha1-J2dXJ2dXJ2dXJ2dXJ2dXJ2dXJ2dXJ2c=",
            "stub" : true,
            "length" : 1024
        },
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            "revpos" : 1,
            "digest" : "sha1-JmZWJmZWJmZWJmZWJmZWJmZWJmZWJmY=",
            "stub" : true,
            "length" : 1024
        }
    }
}
```

```
{"id":"0000-00-00T00:00:00.000+00:00",
"_id":"0000-00-00",
"_rev":"1-00000000000000000000000000000000",
"_attachments":{},
"_conflicts":["0000-00-00T00:00:00.000+00:00"],
"_deleted_conflicts":["0000-00-00T00:00:00.000+00:00"],
"_local_seq":"0000-00-00T00:00:00.000+00:00",
"_revisions":{"start":1,"ids":["00000000000000000000000000000000"]},
"_revs_info":[{"rev":"1-00000000000000000000000000000000","status":"available"}],
"_rev_tree":[["1-00000000000000000000000000000000",{"status":"available","children":{}}]],
"_id_conflicts":["0000-00-00T00:00:00.000+00:00"],
"_rev_conflicts":["1-00000000000000000000000000000000"],
"_id_deleted_conflicts":["0000-00-00T00:00:00.000+00:00"],
"_rev_deleted_conflicts":["1-00000000000000000000000000000000"],
"_local_seq_conflicts":["0000-00-00T00:00:00.000+00:00"],
"_local_seq_deleted_conflicts":["0000-00-00T00:00:00.000+00:00"],
"_attachment_stubs":{"0000-00-00":{"content_type":"application/octet-stream","digest":"md5-00000000000000000000000000000000","length":0,"revpos":1,"stub":true}},
"_attachment_data":{"0000-00-00":"AAAA
```

```
var ldt=[[[
```

Chapter 1 Sample Chapter

1.1 Sample Section

1.1.1 Sample Subsection

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nunc ac ante ut ante ultrices ultrices. Nunc ac ante ut ante ultrices ultrices. Nunc ac ante ut ante ultrices ultrices

```
<!-- ~~~
```

```
<div class="page">

<div>
<span id="1" />
<span id="2" />
<span id="3" />
<span id="4" />
<span id="5" />
<span id="6" />
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<span id="8" />
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<span id="10" />
<span id="11" />
<span id="12" />
<span id="13" />
```

```
<?xml version="1.0"?>

<!DOCTYPE
  PLOT_SPECIFICATION SYSTEM "http://www.fr.com/fr/fr.dtd">

<PLOT_SPECIFICATION
  COMMENT "Generated by Gnuplot 4.4 patchlevel 0"
  VERSION "1.0"
  DATE "Wed Jan 13 00:00:00 2010"
  DEVICE "unknown"
  TERMINAL "unknown"
  OUTPUT "/tmp/tmpkbcb8Ia_.png"
>

<?xml version="1.0"?>

<!DOCTYPE
  PLOT_TEXT SYSTEM "http://www.fr.com/fr/fr.dtd">

<PLOT_TEXT
  ID "1"
  TEXT "Gnuplot"
  FONT "Helvetica"
  POINT_SIZE "12.000000"
  ROTATION "0.000000"
  ALIGNMENT "CENTER"
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>

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  FONT "Helvetica"
  POINT_SIZE "12.000000"
  ROTATION "0.000000"
  ALIGNMENT "CENTER"
  COLOR "0.000000,0.000000,0.000000"
>
```

Gnuplot is a command-line driven interactive function plotting program. Originally intended as a means to stimulate students' interest in mathematics, it has been used in scientific fields for over a decade. Gnuplot is capable of plotting 2D functions and data in various styles, and 3D functions and data in the form of surfaces and contour maps. Gnuplot can output to a variety of formats, including PostScript, PNG, and SVG.

Gnuplot is a portable command-line driven interactive data and function plotting utility for UNIX, IBM OS/2, MS Windows, DOS, and many other platforms. It was originally intended for students to explore mathematical functions and data in a friendly way, but has been used for many scientific and engineering applications since then. Gnuplot supports many types of plots in either 2D or 3D. It can plot functions and data read from ASCII files. Gnuplot can output to a variety of formats, including PNG, PostScript, and SVG.

```
<?xml version="1.0"?>
```

```
<div class="page">

<div>
<div>
<div>
```

Ad Clash Report

Ad Clash Report

Ad Clash Report

{
"id": "1707.07187",
"title": "Adversarial Examples for Semantic Segmentation and Object Detection",
"abstract": "Numerous works have recently shown that state-of-the-art classifiers are vulnerable to adversarial perturbations. We extend the study of adversarial examples to more complex vision tasks, namely semantic segmentation and object detection. We show that adversarial examples can also be easily generated for these tasks, and that they transfer across datasets and models. We also show that adversarial training can be used to defend against such examples, and that it leads to improved robustness.",
"authors": "Cihang Xie, Jianyu Wang, Zhishuai Zhang, Yuyin Zhou, Lingxi Xie, Alan Yuille",
"subjects": "Computer Vision and Pattern Recognition (cs.CV); Machine Learning (cs.LG); Machine Learning (stat.ML)",
"comments": "Technical report. Revised version of arXiv:1703.08603 and arXiv:1704.08784",
"doi": "10.48550/arXiv.1707.07187",
"licenses": "Creative Commons Attribution Non-Commercial Share Alike 4.0"
}

Adversarial Examples for Semantic Segmentation and Object Detection

Cihang Xie[1], Jianyu Wang[2], Zhishuai Zhang[1], Yuyin Zhou[1], Lingxi Xie[1], Alan Yuille[1]
[1]Johns Hopkins University, [2]Huazhong University of Science and Technology
{cihang.xie, zhishuai.zhang, yuyin.zhou, lingxi.xie, alan.yuille@jhu.edu, jianyuwang@hust.edu.cn}

Numerous works have recently shown that state-of-the-art classifiers are vulnerable to adversarial perturbations. We extend the study of adversarial examples to more complex vision tasks, namely semantic segmentation and object detection. We show that adversarial examples can also be easily generated for these tasks, and that they transfer across datasets and models. We also show that adversarial training can be used to defend against such examples, and that it leads to improved robustness.

Intro

Cross-Sectional Study of Dietary Patterns and Anemia Among Adolescent Girls in Rural Tribal Hamlets of Gadchiroli, Maharashtra

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```
{"id":"0000-00-00T00:00:00.000Z",
"_id":"5ffb0ffb0ffb0ffb0ffb0ffb",
"index":0,
"hash":"0000000000000000000000000000000000000000000000000000000000000000",
"nonce":"0000000000000000000000000000000000000000000000000000000000000000",
"mrkl_root":"0000000000000000000000000000000000000000000000000000000000000000",
"time":0,"bits":0,"prev_block":"0000000000000000000000000000000000000000000000000000000000000000",
"prev_block_hash":"0000000000000000000000000000000000000000000000000000000000000000",
"transaction_count":0,"size":0,"version":0,"merkle_root":"0000000000000000000000000000000000000000000000000000000000000000",
"block_reward":0,"block_fees":0,"difficulty":0,"total_output":0,"next_block":"",
"next_block_hash":"",
"transactions":[]}

{"id":"0000-00-00T00:00:00.000Z",
"_id":"5ffb0ffb0ffb0ffb0ffb0ffb",
"index":0,
```

```
var _cookie_expiration=7
var _cookie_path="/"
var _cookie_domain="/"
var _cookie_secure=0
var _cookie_enabled=0 var_cookie_force_force=0var
var
var
var
var_cookie_test_expiration=7
var
var_cookie_test_path="/"var_cookie_test_domain="/"
var
var_cookie_test_secure=0var_cookie_test_enabled=0
var
var_cookie_test_force_force=0var
var
var
var
var_cookie_test_expiration=7var_cookie_test_path="/"
var
var_cookie_test_domain="/"var_cookie_test_secure=0
var
var_cookie_test_enabled=0var_cookie_test_force_force=0
```

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Page 1

Design of Electrical Generator operated on Low speed Wind Mill for Village Electrification

Prof. M. Ashok Kumar, Prof. Ch. Ramesh, Prof. Ch. Ramesh, Prof. Ch. Ramesh

```
<?xml version="1.0" ?>
<rss version="2.0" xmlns:dc="http://purl.org/dc/elements/1.1/" xmlns:content="http://purl.org/rss/1.0/modules/content/" xmlns:atom="http://www.w3.org/2005/Atom" xmlns:media="http://search.yahoo.com/mrss/" xmlns:wfw="http://wellformedweb.org/CommentAPI/" xmlns:itunes="http://www.itunes.com/dtd/2.0/" xmlns:sy="http://purl.org/rss/1.0/modules/syndication/" xmlns:slash="http://purl.org/rss/1.0/modules/slash/" xmlns:dsq="http://disqus.com/disqus-internal/" xmlns:opensearch="http://a9.com/-/spec/opensearchrss/1.0/" xmlns:geo="http://www.w3.org/2003/01/geo/wgs84_pos#" xmlns:georss="http://www.georss.org/georss" xmlns:gml="http://www.opengis.net/gml" xmlns:osi="http://rssnamespace.org/feedburner/extensions/1.0" xmlns:feedburner="http://rssnamespace.org/feedburner/extensions/1.0" xmlns:fb="http://www.facebook.com/2008/fbml" xmlns:media="http://search.yahoo.com/mrss/" xmlns:atom10="http://www.w3.org/2005/Atom
```

7.2.2 Equilibrium

Let's return to our model of monopoly power.

```
In [1]: import numpy as np
        import matplotlib.pyplot as plt
        %matplotlib inline
        from sympy import symbols, Eq, solve, lambdify, latex, init_printing, Rational, linsolve, diff, simplify
```

```
Out[1]:
```

Suppose the price-cost mark-up is 0.25 (25%), and the elasticity is 1.5. We know that a firm's profit-maximizing mark-up is $\frac{1}{\epsilon}$, so we can solve for the elasticity. This tells us that we can write the quantity demanded as a function of the price as:

$$q = p^{-\epsilon} = p^{-1.5} \Rightarrow p = q^{-\frac{2}{3}}$$

```
In [2]: q, p = symbols('q p')
        demand = Eq(q, p**(-Rational(3,2)))
        demand
```

```
Out[2]:
```

$$q = \frac{1}{p^{\frac{3}{2}}}$$

```
In [3]: inverse = solve(demand, p)
        inverse
```

```
Out[3]:
```

$$\left[\frac{1}{q^{\frac{2}{3}}} \right]$$

```
In [4]: inverse = inverse[0]
        inverse
```

```
Out[4]:
```

$$\frac{1}{q^{\frac{2}{3}}}$$

```
In [5]: inverse.subs(q, 1)
```

```
Out[5]:
```

$$1$$

```
In [6]: inverse.subs(q, 2)
```

```
Out[6]:
```

$$\frac{1}{2^{\frac{2}{3}}}$$

```
In [7]: inverse.subs(q, 3)
```

```
Out[7]:
```

$$\frac{1}{3^{\frac{2}{3}}}$$

```
In [8]: inverse.subs(q, 4)
```

```
Out[8]:
```

$$\frac{1}{4^{\frac{2}{3}}}$$

```
In [9]: inverse.subs(q, 5)
```

```
Out[9]:
```

$$\frac{1}{5^{\frac{2}{3}}}$$

```
In [10]: %matplotlib inline
         from sympy.plotting import plot, plot_implicit
         plot(inverse, (q, 0, 10))
```

```
Out[10]:
```



```
<sympy.plotting.plot.Plot object at 0x7f8a1a0a0a90>
```

Suppose the price-cost mark-up is 0.25 (25%), and the elasticity is 1.5. We know that a firm's profit-maximizing mark-up is $\frac{1}{\epsilon}$, so we can solve for the elasticity. This tells us that we can write the quantity demanded as a function of the price as:

$$q = p^{-\epsilon} = p^{-1.5} \Rightarrow p = q^{-\frac{2}{3}}$$

```
In [11]: q, p = symbols('q p')
         demand = Eq(q, p**(-Rational(3,2)))
         demand
```

```
Out[11]:
```

$$q = \frac{1}{p^{\frac{3}{2}}}$$

```
In [12]: inverse = solve(demand, p)
         inverse
```

```
Out[12]:
```

$$\left[\frac{1}{q^{\frac{2}{3}}} \right]$$

```
In [13]: inverse = inverse[0]
         inverse
```

```
Out[13]:
```

$$\frac{1}{q^{\frac{2}{3}}}$$

```
In [14]: inverse.subs(q, 1)
```

```
Out[14]:
```

$$1$$

```
In [15]: inverse.subs(q, 2)
```

```
Out[15]:
```

$$\frac{1}{2^{\frac{2}{3}}}$$

```
In [16]: inverse.subs(q, 3)
```

```
Out[16]:
```

$$\frac{1}{3^{\frac{2}{3}}}$$

```
In [17]: inverse.subs(q, 4)
```

```
Out[17]:
```

$$\frac{1}{4^{\frac{2}{3}}}$$

```
In [18]: inverse.subs(q, 5)
```

```
Out[18]:
```

$$\frac{1}{5^{\frac{2}{3}}}$$

```
In [19]: %matplotlib inline
         from sympy.plotting import plot, plot_implicit
         plot(inverse, (q, 0, 10))
```

```
Out[19]:
```



```
<sympy.plotting.plot.Plot object at 0x7f8a1a0a0a90>
```

Suppose the price-cost mark-up is 0.25 (25%), and the elasticity is 1.5. We know that a firm's profit-maximizing mark-up is $\frac{1}{\epsilon}$, so we can solve for the elasticity. This tells us that we can write the quantity demanded as a function of the price as:

$$q = p^{-\epsilon} = p^{-1.5} \Rightarrow p = q^{-\frac{2}{3}}$$

```
In [20]: q, p = symbols('q p')
         demand = Eq(q, p**(-Rational(3,2)))
         demand
```

```
Out[20]:
```

$$q = \frac{1}{p^{\frac{3}{2}}}$$

```
In [21]: inverse = solve(demand, p)
         inverse
```

```
Out[21]:
```

$$\left[\frac{1}{q^{\frac{2}{3}}} \right]$$

```
In [22]: inverse = inverse[0]
         inverse
```

```
Out[22]:
```

$$\frac{1}{q^{\frac{2}{3}}}$$

```
In [23]: inverse.subs(q, 1)
```

```
Out[23]:
```

$$1$$

```
In [24]: inverse.subs(q, 2)
```

```
Out[24]:
```

$$\frac{1}{2^{\frac{2}{3}}}$$

```
In [25]: inverse.subs(q, 3)
```

```
Out[25]:
```

$$\frac{1}{3^{\frac{2}{3}}}$$

```
In [26]: inverse.subs(q, 4)
```

```
Out[26]:
```

$$\frac{1}{4^{\frac{2}{3}}}$$

```
In [27]: inverse.subs(q, 5)
```

```
Out[27]:
```

$$\frac{1}{5^{\frac{2}{3}}}$$

```
In [28]: %matplotlib inline
         from sympy.plotting import plot, plot_implicit
         plot(inverse, (q, 0, 10))
```

```
Out[28]:
```



```
<sympy.plotting.plot.Plot object at 0x7f8a1a0a0a90>
```

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```
<?xml version="1.0" encoding="UTF-8"?>
<rss version="2.0" xmlns:dc="http://purl.org/dc/elements/1.1/" xmlns:sy="http://purl.org/rss/1.0/modules/syndication/" xmlns:slash="http://purl.org/rss/1.0/modules/slash/" xmlns:rdfs="http://www.w3.org/2000/01/rdf-schema#" xmlns:wfw="http://wellformedweb.org/CommentAPI/" xmlns:dcterms="http://purl.org/dc/terms/" xmlns:media="http://search.yahoo.com/mrss/" xmlns:atom="http://www.w3.org/2005/Atom" xmlns:opensearch="http://a9.com/-/spec/opensearchrss/1.0/" xmlns:geo="http://www.w3.org/2003/01/geo/wgs84_pos#" xmlns:georss="http://www.georss.org/georss" xmlns:gml="http://www.opengis.net/gml" xmlns:osm="http://www.openstreetmap.org/dtd/0.6" xmlns:fb="http://www.facebook.com/2008/fbml" xmlns:content="http://purl.org/rss/1.0/modules/content/" xmlns:wfw="http://wellformedweb.org/CommentAPI/" xmlns:dc="http://purl.org/dc/elements/1.1/" xmlns:atom="http://www.w3.org/2005/Atom" xmlns:sy="http://purl.org/rss/1.0/modules/syndication/" xmlns:slash="http://purl.org/rss/1.0/modules/slash/" xmlns:media="http://search.yahoo.com/mrss/
```

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```
{
  "name": "Sample",
  "version": "0.0.0",
  "description": "",
  "keywords": [],
  "author": "",
  "license": "ISC",
  "dependencies": {
    "@ffmpeg-installer/ffmpeg": "1.1.0",
    "@ffmpeg-installer/linux-arm": "1.1.0",
    "@ffmpeg-installer/linux-arm64": "1.1.0",
    "@ffmpeg-installer/linux-ia32": "1.1.0",
    "@ffmpeg-installer/linux-x64": "1.1.0",
    "@ffmpeg-installer/mac": "1.1.0",
    "@ffmpeg-installer/win32": "1.1.0",
```

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Character set encoding: [UTF-8]
Produced by: [Arthur DiBianca and David Widger]
Source: [Gutenberg]

```
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet href="chrome://global/skin/" type="text/xsl"?>
<!--
  This file contains the default icons for Firefox.
  -->
<svg
  xmlns="http://www.w3.org/2000/svg"
  xmlns:xlink="http://www.w3.org/1999/xlink"
  xmlns:rdf="http://www.w3.org/22-rdf-syntax-ns#"
  xmlns:cc="http://creativecommons.org/ns#"
  xmlns:dc="http://purl.org/dc/elements/1.1/"
  xmlns:sodipodi="http://sodipodi.sourceforge.net/DTD/sodipodi-0.dtd"
  xmlns:inkscape="http://www.inkscape.org/namespaces/inkscape"
  id="svg"
  viewBox="0 0 16 16"
  height="16px"
  width="16px">
```

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d615 | Page 5 Conclusion The Ultrasonic Assisted Extraction (UAE) of Ginger Oleoresin from Ginger Rhizome Powder using Ethanol as solvent and optimized UAE. This study concluded following points.
1. Ginger Oleoresin is a valuable natural product.
2.

Table of Contents
Sub-Interface - Layer3 Subinterface
Configure a Layer3 Subinterface on the GUI
Configure a Layer3 Subinterface on the CLI
Verify Layer3 Subinterface Configuration on the GUI
Verify Layer3 Subinterface Configuration on the CLI
Related Topics

Sub-Interface - Layer3 Subinterface

Palo Alto Networks firewalls support Layer 3 routing on subinterfaces to accommodate multiple virtual routers and security zones on a single physical interface. The ability to define subinterfaces is useful when you want to route traffic for multiple virtual routers or security zones through a single physical interface. Each subinterface can be assigned to a different virtual router and security zone. Each subinterface can be configured with its own IP address, static routes, RIP, OSPF, BGP, or multicast routing protocol. Subinterfaces are also useful if you have a firewall that is connected to a switch that is configured for multiple VLANs. You can define a subinterface for each VLAN and assign traffic from each VLAN to a different virtual router and security zone.

Configure a Layer3 Subinterface on the GUI

Step 1. Define the parent interface. Select Network > Interfaces and click the interface name. Select Interface Type Layer3 and configure the IPv4, IPv6 or both addresses, Management Profile, MTU, Link Speed, Link Duplex, and Link State as needed. Click OK.

Step 2. Define the subinterface. Select Network > Interfaces and click Add Subinterface. Enter the Interface Name in the format eth<slot>.<port_number>. Enter the Tag (VLAN ID) and Virtual Router. Select the Security Zone and configure the IPv4, IPv6 or both addresses, Management Profile, MTU, Link Speed, Link Duplex, and Link State as needed. Click OK.

Step 3. Commit the changes. Click Commit.

Verify Layer3 Subinterface Configuration on the GUI

Step 1. Verify the parent interface. Select Network > Interfaces and click the interface name. Select Interface Type Layer3 and configure the IPv4, IPv6 or both addresses, Management Profile, MTU, Link Speed, Link Duplex, and Link State as needed. Click OK.

Step 2. Verify the subinterface. Select Network > Interfaces and click Add Subinterface. Enter the Interface Name in the format eth<slot>.<port_number>. Enter the Tag (VLAN ID) and Virtual Router. Select the Security Zone and configure the IPv4, IPv6 or both addresses, Management Profile, MTU, Link Speed, Link Duplex, and Link State as needed. Click OK.

Step 3. Commit the changes. Click Commit.

Configure a Layer3 Subinterface on the CLI

```
set interfaces ethernet <slot>/<port_number>
set interfaces ethernet <slot>/<port_number> mtu <mtu_size>
set interfaces ethernet <slot>/<port_number> link-speed <speed_value>
set interfaces ethernet <slot>/<port_number> link-duplex <duplex_value>
set interfaces ethernet <slot>/<port_number> link-state <state_value>
set interfaces ethernet <slot>/<port_number> layer3 ip <ipv4_address>/<mask_length>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 <ipv6_address>/<mask_length>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 enable-duplicate-address-detection yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement enable yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement hop-limit <hop_limit_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement lifetime <lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement reachable-time <reachable_time_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement retransmit-time <retransmit_time_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement prefix <ipv6_prefix>/<mask_length> on-link yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement prefix <ipv6_prefix>/<mask_length> autonomous yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement prefix <ipv6_prefix>/<mask_length> preferred-lifetime <preferred_lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement prefix <ipv6_prefix>/<mask_length> valid-lifetime <valid_lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement dns-server <dns_server_address_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement dns-suffix <dns_suffix_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement enable yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery neighbor <ipv6_address_value> <mac_address_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery duplicate-address-detection clone-limit <clone_limit_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery duplicate-address-detection freeze-interface-id yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type prefix-only
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address identifier <identifier_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address identifier <identifier_value> eui-64 yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address identifier <identifier_value> randomize yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address lifetime <lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address lifetime <lifetime_value> no-expiry yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address lifetime <lifetime_value> on-link yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address lifetime <lifetime_value> preferred yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address lifetime <lifetime_value> autonomous yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address lifetime <lifetime_value> preferred-lifetime <preferred_lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address lifetime <lifetime_value> valid-lifetime <valid_lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool identifier <identifier_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool identifier <identifier_value> eui-64 yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool identifier <identifier_value> randomize yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool lifetime <lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool lifetime <lifetime_value> no-expiry yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool lifetime <lifetime_value> on-link yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool lifetime <lifetime_value> preferred yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool lifetime <lifetime_value> autonomous yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool lifetime <lifetime_value> preferred-lifetime <preferred_lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool lifetime <lifetime_value> valid-lifetime <valid_lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude identifier <identifier_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude identifier <identifier_value> eui-64 yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude identifier <identifier_value> randomize yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> no-expiry yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> on-link yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> preferred yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> autonomous yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> preferred-lifetime <preferred_lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> valid-lifetime <valid_lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude identifier <identifier_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude identifier <identifier_value> eui-64 yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude identifier <identifier_value> randomize yes
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set interfaces ethernet <slot>/<port_number> layer3 
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$\int \frac{1}{x^2 - 2x + 2}\,dx$ - Integral of $\frac{1}{x^2 - 2x + 2}$ with respect to x | Wolfram|Alpha Integral Calculator, Antiderivative

$\int \frac{1}{x^2 - 2x + 2}\,dx$ - Integral of $\frac{1}{x^2 - 2x + 2}$ with respect to x

Input interpretation:

$\int \frac{1}{x^2 - 2x + 2}\,dx$

Result:

$\tan^{-1}(x - 1)$

Indefinite integral:

$\int \frac{1}{x^2 - 2x + 2}\,dx = \tan^{-1}(x - 1) + \text{constant}$

Alternate form:

$\int \frac{1}{x^2 - 2x + 2}\,dx = \frac{1}{2} i \log(-i x + i + 1) - \frac{1}{2} i \log(i x - i + 1) + \text{constant}$

Alternate form assuming x is real:

$\int \frac{1}{x^2 - 2x + 2}\,dx = \tan^{-1}(x - 1) + \text{constant}$

Alternate form assuming x is complex:

$\int \frac{1}{x^2 - 2x + 2}\,dx = \frac{1}{2} i \log(-i x + i + 1) - \frac{1}{2} i \log(i x - i + 1) + \text{constant}$

Derivative:

$\frac{d}{dx}\left(\int \frac{1}{x^2 - 2x + 2}\,dx\right) = \frac{1}{x^2 - 2x + 2}$

Alternate form:

$\frac{d}{dx}\left(\int \frac{1}{x^2 - 2x + 2}\,dx\right) = \frac{1}{x^2 - 2x + 2}$

Alternate form assuming x is real:

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Alternate form assuming x is complex:

$\frac{d}{dx}\left(\int \frac{1}{x^2 - 2x + 2}\,dx\right) = \frac{1}{x^2 - 2x + 2}$

Definite integral:

$\int_0^1 \frac{1}{x^2 - 2x + 2}\,dx = \frac{\pi}{4}$

Alternate form:

$\int_0^1 \frac{1}{x^2 - 2x + 2}\,dx = \frac{\pi}{4}$

Alternate form assuming x is real:

$\int_0^1 \frac{1}{x^2 - 2x + 2}\,dx = \frac{\pi}{4}$

Alternate form assuming x is complex:

$\int_0^1 \frac{1}{x^2 - 2x + 2}\,dx = \frac{\pi}{4}$

Alternate form:

$\int_0^1 \frac{1}{x^2 - 2x + 2}\,dx = \frac{1}{2} i \log(-i + i + 1) - \frac{1}{2} i \log(i - i + 1)$

Alternate form assuming x is real:

$\int_0^1 \frac{1}{x^2 - 2x + 2}\,dx = \frac{\pi}{4}$

Alternate form assuming x is complex:

$\int_0^1 \frac{1}{x^2 - 2x + 2}\,dx = \frac{1}{2} i \log(-i + i + 1) - \frac{1}{2} i \log(i - i + 1)$

Indefinite integral:

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$\int \frac{1}{x^2 - 2x + 2}\,dx = \tan^{-1}(x - 1) + \text{constant}$

Alternate form assuming x is complex:

$\int \frac{1}{x^2 - 2x + 2}\,dx = \frac{1}{2} i \log(-i x + i + 1) - \frac{1}{2} i \log(i x - i + 1) + \text{constant}$

Derivative:

$\frac{d}{dx}\left(\int \frac{1}{x^2 - 2x + 2}\,dx\right) = \frac{1}{x^2 - 2x + 2}$

Alternate form:

$\frac{d}{dx}\left(\int \frac{1}{x^2 - 2x + 2}\,dx\right) = \frac{1}{x^2 - 2x + 2}$

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$\frac{d}{dx}\left(\int \frac{1}{x^2 - 2x + 2}\,dx\right) = \frac{1}{x^2 - 2x + 2}$

Alternate form assuming x is complex:

$\frac{d}{dx}\left(\int \frac{1}{x^2 - 2x + 2}\,dx\right) = \frac{1}{x^2 - 2x + 2}$

Definite integral:

$\int_0^1 \frac{1}{x^2 - 2x + 2}\,dx = \frac{\pi}{4}$

Alternate form:

$\int_0^1 \frac{1}{x^2 - 2x + 2}\,dx = \frac{\pi}{4}$

Alternate form assuming x is real:

$\int_0^1 \frac{1}{x^2 - 2x + 2}\,dx = \frac{\pi}{4}$

Alternate form assuming x is complex:

$\int_0^1 \frac{1}{x^2 - 2x + 2}\,dx = \frac{\pi}{4}$

Alternate form:

$\int_0^1 \frac{1}{x^2 - 2x + 2}\,dx = \frac{1}{2} i \log(-i + i + 1) - \frac{1}{2} i \log(i - i + 1)$

Alternate form assuming x is real:

$\int_0^1 \frac{1}{x^2 - 2x + 2}\,dx = \frac{\pi}{4}$

Alternate form assuming x is complex:

$\int_0^1 \frac{1}{x^2 - 2x + 2}\,dx = \frac{1}{2} i \log(-i + i + 1) - \frac{1}{2} i \log(i - i + 1)$

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```
<?xml version="1.0" encoding="UTF-8" ?>
<rss version="2.0" xmlns:dc="http://purl.org/dc/elements/1.1/" xmlns:sy="http://purl.org/rss/1.0/modules/syndication/" xmlns:slash="http://purl.org/rss/1.0/modules/slash/" xmlns:wfw="http://wellformedweb.org/CommentAPI/" xmlns:dc="http://purl.org/dc/elements/1.1/" xmlns:atom="http://www.w3.org/2005/Atom" xmlns:sy="http://purl.org/rss/1.0/modules/syndication/" xmlns:slash="http://purl.org/rss/1.0/modules/slash/" xmlns:media="http://search.yahoo.com/mrss/" xmlns:content="http://purl.org/rss/1.0/modules/content
```

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Section 1.10.32 of "de Finibus Bonorum et Malorum", written by Cicero in 45 BC

"Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos qui ratione voluptatem sequi nesciunt. Neque porro quisquam est, qui dolorem ipsum quia dolor sit amet, consectetur, adipisci velit, sed quia non numquam eius modi tempora incidunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim ad minima veniam, quis nostrum exercitationem ullam corporis suscipit laboratoriosam, nisi ut aliquid ex ea commodi consequatur? Quis autem vel eum iure reprehenderit qui in ea voluptate velit esse quam nihil molestiae consequatur, vel illum qui dolorem eum fugiat, quo voluptas nulla pariatur?"

1914 translation by H. Rackham

"But I must explain to you how all this mistaken idea of denouncing pleasure and praising pain was born and I will give you a complete account of the system, and expound the actual teachings of the great explorer of the truth, the master-builder of human happiness. No one rejects, dislikes, or avoids pleasure itself, because it is pleasure, but because those who do not know how to pursue pleasure rationally encounter consequences that are extremely painful. Nor again is there anyone who loves or pursues or desires to obtain pain of itself, because it is pain, but because occasionally circumstances occur in which toil and pain can procure him some great pleasure. To take a trivial example, which of us ever undertakes laborious physical exercise, except to obtain some advantage from it? But who has any right to find fault with a man who chooses to enjoy a pleasure that has no annoying consequences, or one who avoids a pain that produces no resultant pleasure?"

1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 |

```
<mx:Text id="mainMenuTitle"
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  fx:Bind="{$ctx.isGameInProgress ? 'Resume Game' : 'New Game'}" fx:Event="$ctx.startNewGame" />
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<mx:Panel id="confirmDialog" />
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<mx:Panel id="mapPanel" />
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<mx:Panel id="shopPanel" />
<mx:Panel id="craftingPanel" />
<mx:Panel id="settingsPanel" />
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<mx:Panel id="skillTreePanel" />
<mx:Panel id="partyPanel" />
<mx:Panel id="worldMapPanel" />
<mx:Panel id="minimapPanel" />
<mx:Panel id="chatPanel" />
<mx:Panel id="notificationPanel" />
<mx:Panel id="tooltipPanel" />
<mx:Panel id="contextMenuPanel" />
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<mx:Panel id="debugPanel" />
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<mx:Panel id="screenFadeOverlay" />
<mx:Panel id="loadingTipsPanel" />
<mx:Panel id="pauseTimeIndicator" />
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<mx:Panel id="levelUpStatsPanel" />
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一种电缆 专利说明书

[19]中华人民共和国专利局
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[74][illegible]
[54]发明名称 一种电缆

[57]摘要 [illegible]

[illegible]

이사회 의장 주주 제안서

제1조 (제안의 목적)

제1조 (제안의 목적) 본 제안서는 주주 김지훈이 이사회 의장 선임을 제안하기 위해 작성되었다. 의장 선임은 이사회 운영 전반에 대한 주주 권리 행사의 가장 직접적인 수단이다. 의장은 의제 설정, 회의 진행, 의결 정리, 이사회 의사록 작성 등 이사회 운영 전반에 대한 권한을 가진다. 의장 선임 제안은 이사회 구성 변경 없이 이사회 운영 방식을 바꾸는 가장 빠른 경로다. 이사 선임 제안은 시간이 걸리지만, 의장 선임은 다음 이사회 한 번이면 된다.

제2조 (의장 선임 제안의 근거)

제2조 (의장 선임 제안의 근거) 의장 선임 제안은 세 가지 근거에 기한다. 첫째, 현 의장 김민수는 이사회 의사록 작성 의무를 지키지 않았다. 둘째, 의장 선임 제안은 이사회 구성 변경 없이 이사회 운영 방식을 바꾸는 가장 빠른 경로다. 셋째, 의장은 의제 설정, 회의 진행, 의결 정리, 이사회 의사록 작성 등 이사회 운영 전반에 대한 권한을 가진다.

제3조 (의사록 미작성 사실)

제3조 (의사록 미작성 사실) 2024년 3월 이사회 의사록 미작성 사실은 이사회 의사록 작성 의무를 지키지 않았다는 것을 보여준다. 의사록 작성 의무는 이사회 의장의 의무다. 의사록 미작성 사실은 의장 김민수가 이사회 의사록 작성 의무를 지키지 않았다는 것을 보여준다.

제4조 (의장 김민수의 의사록 작성 의무 불이행)

제4조 (의장 김민수의 의사록 작성 의무 불이행) 의장 김민수는 이사회 의사록 작성 의무를 지키지 않았다. 의사록 작성 의무는 이사회 의장의 의무다. 의장 김민수는 이사회 의사록 작성 의무를 지키지 않았다.

제5조 (의장 선임 제안의 효과)

제5조 (의장 선임 제안의 효과) 의장 선임 제안은 이사회 구성 변경 없이 이사회 운영 방식을 바꾸는 가장 빠른 경로다. 이사 선임 제안은 시간이 걸리지만, 의장 선임은 다음 이사회 한 번이면 된다. 의장 선임 제안은 이사회 구성 변경 없이 이사회 운영 방식을 바꾸는 가장 빠른 경로다.

[illegible]

[illegible]

[illegible]

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<?xml version="1.0"
<!DOCTYPE
<!-- Generated by IcoMoon.io -->
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<title>ios-cloud-download</title>
<path d="M
256.000,
256.000,
256.000,
```

1.1 Introduction

1.1.1 Why Study Differential Equations?

Differential equations are equations involving derivatives of unknown functions. For example, the equation

$$\frac{dy}{dx} = 2x + 3$$

is a differential equation. Here, y is a function of x, and $\frac{dy}{dx}$ represents the rate at which y changes with respect to x.

Differential equations are fundamental in mathematics because they describe how quantities change over time or in response to other variables. For example, Newton's Second Law, $F = ma$, can be rewritten as $F = m\frac{d^2x}{dt^2}$, where $x(t)$ is the position of an object at time t. This equation allows us to predict the motion of the object by solving for $x(t)$.

Differential equations are used in physics, engineering, economics, biology, and many other fields to model dynamic systems. For example, in electrical circuits, the current $I(t)$ in a circuit with resistance R, inductance L, and voltage $V(t)$ is described by the equation $L\frac{dI}{dt} + RI = V(t)$. Solving this equation allows us to predict how the current changes over time.

1.1.2 Ordinary Differential Equations (ODEs)

An ordinary differential equation (ODE) is a differential equation in which the unknown function depends on a single variable. For example, the equation $\frac{dy}{dx} = 2x + 3$ is an ODE because y depends only on x.

ODEs are used to model systems where the rate of change of a quantity depends on the quantity itself and other factors. For example, the population of a city may be modeled by the ODE $\frac{dP}{dt} = kP$, where $P(t)$ is the population at time t and k is a constant representing the growth rate.

ODEs are used in physics, engineering, economics, biology, and many other fields to model dynamic systems. For example, in electrical circuits, the current $I(t)$ in a circuit with resistance R, inductance L, and voltage $V(t)$ is described by the ODE $L\frac{dI}{dt} + RI = V(t)$. Solving this ODE allows us to predict how the current changes over time.

1.1.3 Partial Differential Equations (PDEs)

A partial differential equation (PDE) is a differential equation in which the unknown function depends on multiple variables. For example, the equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ is a PDE because u depends on both x and t.

PDEs are used to model systems where the rate of change of a quantity depends on the quantity itself and other factors. For example, the heat equation $\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2}$ describes how the temperature $u(x,t)$ changes over time and space, where k is a constant representing the thermal conductivity.

PDEs are used in physics, engineering, economics, biology, and many other fields to model dynamic systems. For example, in electrical circuits, the current $I(t)$ in a circuit with resistance R, inductance L, and voltage $V(t)$ is described by the PDE $L\frac{\partial I}{\partial t} + RI = V(t)$. Solving this PDE allows us to predict how the current changes over time and space.

Ch. 2 Lect 2 Part 1 - Chapter Summary

Ch. 2 Lect 2 Part 1 - Chapter Summary

Ch. 2 Lect 2 Part 1 - Chapter Summary

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Accepted: 3 November 2020

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    "type": "articles",
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}
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Chapter 1 Introduction to Computer Science: Using Ruby 1.9.2 and Ruby 2.0 - 1.1 Introduction 1.1.1 Computer Hardware 1.1.2 Computer Software 1.1.3 Computer Programs 1.1.4 Computer Science 1.1.5 Computer Science Versus Computer Programming 1.1.6 Computer Science Versus Software Engineering 1.1.7 Computer Science Versus Information Technology 1.1.8 Computer Science Versus Other Disciplines 1.1.9 Computer Science Versus Mathematics 1.1.10 Computer Science Versus Science 1.1.11 Computer Science Versus Art 1.1.12 Computer Science Versus Philosophy 1.1.13 Computer Science Versus Engineering 1.1.14 Computer Science Versus Business 1.1.15 Computer Science Versus Medicine 1.1.16 Computer Science Versus Law 1.1.17 Computer Science Versus Sociology 1.1.18 Computer Science Versus Ethics 1.1.19 Computer Science Versus Politics 1.1.20 Computer Science Versus Economics 1.1.21 Computer Science Versus Psychology 1.1.22 Computer Science Versus Linguistics 1.1.23 Computer Science Versus Sociology 1.1.24 Computer Science Versus Ethics 1.1.25 Computer Science Versus Politics 1.1.26 Computer Science Versus Economics 1.1.27 Computer Science Versus Psychology 1.1.28 Computer Science Versus Linguistics 1.1.29 Computer Science Versus Sociology 1.1.30 Computer Science Versus Ethics 1.1.31 Computer Science Versus Politics 1.1.32 Computer Science Versus Economics 1.1.33 Computer Science Versus Psychology 1.1.34 Computer Science Versus Linguistics

Ch. 2 Lab Report - Angular Momentum

Name: Ryan Ramos

Partners: Ryan Ramos, Ryan Ramos, Ryan Ramos,

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Equation 1 of 1: Gaussian Integral Squared

Type: Definite Integral

Source: Gaussian Integral Squared

Tags: [Gaussian Integral, Gaussian Integral Squared]

Notes: Gaussian Integral Squared. Gaussian Integral Squared. Gaussian Integral Squared.

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Effect of Ultrasound Treatment on Antibacterial Activity of Aqueous Extract of Aloe Vera Gel

Ali Ahmadi , Mohammad Ali Ebrahimzadeh , Mohammad Ali Ghasemi , Ali Ghasemi , Ali Ghasemi ,

Chapter Chapter 1

Section Introduction

Subsection Introduction to Linear Algebra

Linear Algebra

Linear Algebra

Linear Algebra is the study of vectors and linear transformations between vector spaces.

Example 1.1.1

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Proof.

Let $u = (x_1, y_1, z_1)$ and $v = (x_2, y_2, z_2)$ be arbitrary elements of V and let $c \in \mathbb{R}$ be arbitrary.

Then $u + v = (x_1 + x_2, y_1 + y_2, z_1 + z_2)$ and

$$T(u + v) = (x_1 + x_2 + y_1 + y_2, 2(z_1 + z_2) - (x_1 + x_2)) = (x_1 + y_1, 2z_1 - x_1) + (x_2 + y_2, 2z_2 - x_2) = T(u) + T(v).$$

Also, $cu = (cx_1, cy_1, cz_1)$ and

$$T(cu) = (cx_1 + cy_1, 2cz_1 - cx_1) = c(x_1 + y_1, 2z_1 - x_1) = cT(u).$$

Therefore T satisfies both conditions of linearity and is a linear transformation. $\square$

Example 1.1.2

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x + 1).$$

Then T is not a linear transformation because $T(0, 0, 0) = (0, 1) \neq (0, 0)$.

Example 1.1.3

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.4

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.5

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.6

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.7

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.8

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.9

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.10

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.11

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.12

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.13

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.14

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.15

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.16

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.17

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.18

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

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Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.125

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.126

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.127

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.128

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.129

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.130

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.131

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.132

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.133

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.134

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.135

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.136

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.137

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.138

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.139

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.140

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.141

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.142

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.143

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.144

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.145

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.146

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.147

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.148

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.149

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.150

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.151

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.152

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.153

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.154

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.155

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.156

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.157

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let

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Abstract: In this paper, we study the Cauchy problem for the nonlinear Kirchhoff-Carrier wave equation in a bounded domain. We prove that the solution exists globally in time and decays to zero as time tends to infinity.

Keywords: Kirchhoff-Carrier wave equation, Cauchy problem, Global existence, Decay.

MSC: 35L70, 35L20, 35L05, 35L15, 35L71, 35L72, 35L76, 35L90, 35B40, 35B41, 35B45, 35B65, 35G20, 35G25, 35G30, 35G31, 35G50, 35G55, 35G60, 35G61

Topic:

Question:

Find a unit tangent vector to the curve $r(t) = \langle \sin(t), \cos(t), \sqrt{2}t \rangle$ at the point $(0, 1, \pi\sqrt{2}/4)$.

Solution:

Step 1: Find the derivative $r'(t)$ of the position vector $r(t)$.

$$r'(t) = \langle \cos(t), -\sin(t), \sqrt{2} \rangle$$

Step 2: Evaluate $r'(t)$ at the point $t = \pi/4$ (since $\sin(\pi/4) = \cos(\pi/4) = 1/\sqrt{2}$).

$$r'(\pi/4) = \langle 1/\sqrt{2}, -1/\sqrt{2}, \sqrt{2} \rangle$$

Step 3: Find the magnitude of $r'(\pi/4)$.

$$|r'(\pi/4)| = \sqrt{(1/\sqrt{2})^2 + (-1/\sqrt{2})^2 + (\sqrt{2})^2} = \sqrt{1/2 + 1/2 + 2} = \sqrt{3}$$

Step 4: Divide $r'(\pi/4)$ by its magnitude to find the unit tangent vector.

$$T(\pi/4) = \frac{r'(\pi/4)}{|r'(\pi/4)|} = \frac{\langle 1/\sqrt{2}, -1/\sqrt{2}, \sqrt{2} \rangle}{\sqrt{3}} = \langle 1/\sqrt{6}, -1/\sqrt{6}, \sqrt{2/3} \rangle$$

Step 5: Rationalize the denominators.

$$T(\pi/4) = \langle \sqrt{6}/6, -\sqrt{6}/6, \sqrt{6}/3 \rangle$$

Step 6: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 7: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 8: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 9: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 10: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 11: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 12: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 13: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 14: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 15: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 16: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 17: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 18: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 19: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 20: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 21: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 22: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 23: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 24: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 25: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 26: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 27: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 28: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 29: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 30: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 31: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 32: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 33: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 34: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 35: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 36: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 37: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 38: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 39: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 40: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 41: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 42: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 43: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 44: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 45: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 46: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 47: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 48: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 49: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 50: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 51: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 52: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 53: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 54: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 55: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 56: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 57: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 58: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 59: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 60: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 61: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 62: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 63: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 64: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 65: Verify the unit tangent vector.

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```
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```

```
# -*- coding: utf-8 -*-
```

# Generated by Rust (https://www.rust-lang.org)
import numpy as np
import numpy.linalg as LA
import matplotlib.pyplot as plt
from matplotlib import rcParams, rc
from mpl_toolkits.mplot3d import Axes3D
from scipy.integrate import odeint, ode

If we want LaTeX fonts
rcParams['font.family'] = 'serif'
rcParams['font.serif'] = ['Computer Modern']

# Parameters
g = 9.81
L = 1.0
m = 1.0

# Initial conditions: theta and dtheta/dt
y0 = [3.14159265358979, 0.0]

# Time points
t = np.linspace(0, 10, 101)

# Define the ODE system
def f(y, t):
    theta = y[0]
    dtheta = y[1]
    ddtheta = -g/L*np.sin(theta)
    return [dtheta, ddtheta]

# Solve the ODE
y = odeint(f, y0, t)

# Extract theta and dtheta/dt
theta = y[:, 0]
dtheta = y[:, 1]

# Create figure
fig = plt.figure(1, figsize=(8, 6.4))

# Plot theta vs time
ax1 = fig.add_subplot(211)
ax1.plot(t, theta, 'b')
ax1.set_ylabel('θ')

# Plot dtheta/dt vs time
ax2 = fig.add_subplot(212)
ax2.plot(t, dtheta, 'r')
ax2.set_ylabel('$\dot{\theta}$')
ax2.set_xlabel('t')

# Save figure
plt.savefig('simple_pendulum.png', dpi=100)
```

```
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```

NO-FLY-REPORTING/17

Non-American Jewish Foreign-born

Ted Cruz

Ted Cruz's Comments on Donald Trump Article

Ted Cruz's Comments on Donald Trump Article Ted Cruz's Comments on Donald Trump Article Ted Cruz's Comments on Donald Trump Article

Ted Cruz's Comments on Donald Trump Article

Ted Cruz's Comments on Donald Trump Article

Ad Corrigendum Supplementary

Page 1 of 1 Generated on 2025-07-14

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Section 1

Section 2

Section 3

Chapter Eighteen

Sections

Introduction
Sections and Subsections
Section Numbers
Section Titles
Section Cross-References
Section Cross-References in Other Documents
Section Cross-References in Other Documents Using Section Numbers
Section Cross-References in Other Documents Using Section Titles
Section Cross-References in Other Documents Using Section Numbers and Section Titles
Section Cross-References in Other Documents Using Section Numbers and Section Titles Without Section Numbers
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NO. 7147 MONTMORENCY

Sun Journal, 1907-07-19 (Vol. 1 Iss. 1)

Image 1

Text

Sun Journal, 1907-07-19 (Vol. 1 Iss. 1)

Image 2

Text

Sun Journal, 1907-07-19 (Vol. 1 Iss. 1) **THE MONTMORENCY SUN.** Vol. 1. **MONTMORENCY, MONT., SATURDAY JULY 19 1907.** No. 1. **PRICE 5 CENTS.** Number 1. Volume 1. Montmorency, Mont., Saturday, July 19, 1907. Price 5 cents. **Sun Publishing Co.** Publishers. **Sun Publishing Co.** Publishers. **Sun Publishing Co.** Publishers

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MR-TCAP-MAPMODULE

Operations-Generic-ASN1Protocol

Operation

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Type: [Class:Universal,Tag:Sequence,Value:Sequence]

Type: [Class:Universal,Tag:Integer,Value:2]

$$\sin(\frac{\pi}{2}) = \frac{\sqrt{2}}{2}$$

Type: [Class:Universal,Tag:Integer,Value:0]

Type: [Class:Universal,Tag:Sequence,Value:Sequence]

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Type: [Class:Universal,Tag:Sequence,Value:Sequence]

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Type: [Class:Universal,Tag:Sequence,Value:Sequence]

Type: [Class:Universal,Tag:Integer,Value:0]

$$\sin(\frac{\pi}{2}) = \frac{\sqrt{2}}{2}$$

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"verified":true,"
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"favourites_count":1000,"
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Topic:

Topic: Fluid Flow Equations

Topic: Fluid Flow Equations for Incompressible Flow

Topic: Fluid Flow Equations for Incompressible Flow: Bernoulli's Equation, Laminar Flow and Poiseuille's Equation, Viscous Forces, Pipe Networks

Topic: Fluid Flow Equations for Incompressible Flow: Bernoulli's Equation

Topic: Fluid Flow Equations for Incompressible Flow: Bernoulli's Equation

Topic:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0 \qquad \frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{v} = -\frac{\nabla p}{\rho} + \frac{\mu}{\rho}\nabla^2 \mathbf{v} + \mathbf{g} \qquad \nabla \cdot \mathbf{v} = 0$$

$$\nabla \cdot (\rho \mathbf{v}) = \rho \nabla \cdot \mathbf{v} + \mathbf{v} \cdot \nabla \rho$$

$$\frac{\partial \rho}{\partial t} + \rho \nabla \cdot \mathbf{v} + \mathbf{v} \cdot \nabla \rho = 0$$

$$\frac{\partial \rho}{\partial t} + \rho \nabla \cdot \mathbf{v} = 0$$

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v})

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Abstract
This study investigates the effect of pulsed laser ablation in liquid (PLAL) parameters on the synthesis of gold nanoparticles (AuNPs) using a Nd:YAG laser. The influence of laser fluence (2–6 J/cm²), pulse repetition rate (10–50 Hz), and ablation time (5–30 min) on the size, shape, and optical properties of AuNPs was evaluated. Results showed that increasing laser fluence and repetition rate reduced average particle size from 18 nm to 9 nm, while prolonged ablation time led to aggregation and broadening of the size distribution. UV-Vis spectroscopy confirmed the formation of AuNPs with surface plasmon resonance peaks between 520 and 540 nm. The optimal conditions for producing small, spherical AuNPs were identified as 4 J/cm², 30 Hz, and 15 min. These findings provide a systematic approach for tailoring AuNP properties by controlling laser parameters, which is crucial for their application in biomedicine and catalysis.

Keywords
laser ablation, gold nanoparticles, pulsed laser ablation in liquid, Nd:YAG laser, surface plasmon resonance, UV-Vis spectroscopy, nanomaterials, optimization, size control, optical properties

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Author's contribution: All authors contributed equally to this work.
Author Ajibola Sunday Ogunyebi, Author Ajibola Sunday Ogunyebi, Author Ajibola Sunday Ogunyebi,

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Hello!

Hello!

Received: 12th Jan. 2024; Accepted: 18th Jan. 2024; Published: 25th Jan. 2024

Kumar
Kumar Ashish Kumar Ashish Kumar Ashish

DOI: 10.1007/s00401-
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Revised: 13 November 2017
Accepted: 14 November 2017

Received: 2 April 2020
Accepted: 14 May 2020
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Ruo-Jia Cui[1,2,3], Hongxia Duan[1,2], Bin Zhao[1,2], Mengqiao Zhou[1,2], Rui Shi[1,2], Xiao-Yu Zhao[1,2], Yi-Yang Li[1,2], Xin-Quan Wang[1,2], Lei Li[1,2] & Zhi-Jian Qin[1,2]

Lycopene is a red carotenoid of great importance because it is responsible for the pigment of tomatoes and other red fruits and vegetables and because it has been associated with many health benefits. Although the lycopene biosynthetic pathway has been established, the regulation of lycopene accumulation in plants is poorly understood. Here we report that the transcription factor RIPENING INHIBITOR (RIN) binds to promoters of genes encoding lycopene biosynthesis enzymes in tomato fruits, activates their expression, and promotes lycopene accumulation. Ethylene and light inhibit the binding of RIN to the promoters of these genes, and ethylene also inhibits the expression of RIN. Our results suggest that RIN is a master regulator of lycopene biosynthesis in tomato fruits, and that ethylene and light inhibit lycopene accumulation in tomato fruits by inhibiting the expression of RIN and the binding of RIN to the promoters of genes encoding lycopene biosynthesis enzymes.

https://doi.org/10.1038/s41586-020-2412-9

Lycopene is a red carotenoid that is responsible for the pigment of tomatoes and other red fruits and vegetables. Lycopene is a precursor of provitamin A and has been associated with many health benefits, such as a reduced risk of cardiovascular disease and certain cancers[1–3]. Although the lycopene biosynthetic pathway has been established, the regulation of lycopene accumulation in plants is poorly understood. Tomato fruits are a major source of lycopene, and lycopene accumulation in tomato fruits is a complex process that is affected by many factors, such as ripening, ethylene, light, temperature, and genotype[4–7]. Ripening-inhibitor (rin) is a recessive mutant that has been used extensively in studies of tomato fruit ripening. The rin mutant fruits do not ripen and do not accumulate lycopene, even in the presence of ethylene[8]. The RIN gene encodes a MADS-box transcription factor that is a master regulator of fruit ripening in tomato[9–12]. Although the RIN protein has been shown to bind to the promoters of several genes that are involved in ethylene biosynthesis and signal transduction, the genes that are directly regulated by RIN and the mechanism by which RIN regulates lycopene accumulation are unknown. Here we show that RIN binds to promoters of genes encoding lycopene biosynthesis enzymes in tomato fruits, activates their expression, and promotes lycopene accumulation. Ethylene and light inhibit the binding of RIN to the promoters of these genes, and ethylene also inhibits the expression of RIN. Our results suggest that RIN is a master regulator of lycopene biosynthesis in tomato fruits, and that ethylene and light inhibit lycopene accumulation in tomato fruits by inhibiting the expression of RIN and the binding of RIN to the promoters of genes encoding lycopene biosynthesis enzymes.

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MATLAB/SIMULINK

Lecture Notes Prepared by Dr. Ramesh Kumar Ayyala, Assistant Professor, Department of Mathematics, Indian Institute of Technology Jodhpur. Please report any errors or typos and discuss this at Academia.edu/Mathematics Reddit. Use Ctrl+Shift+F to find a term or phrase.

Tags: #MATLAB #SIMULINK #Lecture-Notes #Prof-Dr-Ramesh-Kumar-Ayyala #Indian-Institute-of-Technology-Jodhpur #Mathematics #Reddit #Academia.edu

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Hello everyone, in this video we will discuss about **MATLAB/SIMULINK** and its uses in Control Theory and Control Systems.

Tags: #Control-Theory

Tags: #Control-Theory, #Control-Systems, #MATLAB/SIMULINK #Image

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$$\lim_{x \to 0} \frac{\sin(x)}{x} = 1$$

Tags: #MATLAB/SIMULINK

Tags: #Control-Theory, #Control-Systems, #MATLAB/SIMULINK, #Lecture-Notes, #Prof-Dr-Ramesh-Kumar-Ayyala, #Lecture-Notes-Prof-Dr-Ramesh-Kumar-Ayyala

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Tags: #MATLAB/SIMULINK

Tags: #Control-Theory, #Control-Systems, #MATLAB/SIMULINK, #Lecture-Notes, #Prof-Dr-Ramesh-Kumar-Ayyala, #Lecture-Notes-Prof-Dr-Ramesh-Kumar-Ayyala

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Tags: #Control-Theory, #Control-Systems, #MATLAB/SIMULINK #Image

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$$\lim_{x \to 0} \frac{\sin(x)}{x} = 1 \quad \lim_{x \to 0} \frac{\sin(x)}{x} = 1 \quad \ldots

DOI: 10.1007/s00401-
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Research Article

Jack Lewis Rhodes and Jon C. Cummins

Abstract

Recent research suggests that people who believe that science and technology are responsible for environmental problems tend to underestimate science's ability to solve these problems.

Keywords

Received: 2 April 2020; Revised: 14 July 2020; Accepted: 27 August 2020

Introduction

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Methods

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Results

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Discussion

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DOI: 10.1007/s00401-
Received: 5 July 2017
Revised: 19 December 2017
Accepted: 20 December 2017
Online First: 4 January 2018

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Quantum Mechanical Simulation of Electron Density Distribution in Hydrogen Molecular Ion (H₂⁺) Using Variational Methods

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Abstract

This study presents a variational approach to simulate the electron density distribution in the hydrogen molecular ion (H_2^+) using trial wavefunctions and numerical optimization techniques. The ground state energy and electron density are calculated for various internuclear distances and compared with exact results. The results show that the variational method provides accurate electron density distributions with minimal computational cost. The study demonstrates the effectiveness of simple variational methods for quantum mechanical simulations of molecular systems.

Keywords: Quantum Mechanics, Variational Method, Hydrogen Molecular Ion, Electron Density, Ground State Energy, Computational Chemistry, Molecular Simulation, Trial Wavefunction, Optimization, Hartree-Fock Method

1. Introduction

The hydrogen molecular ion (H_2^+) is the simplest molecular system and serves as a fundamental benchmark for quantum mechanical calculations. Despite its simplicity, it exhibits all the essential features of chemical bonding, including electron sharing between two nuclei and the formation of a stable molecular orbital. The study of H_2^+ provides insights into the nature of chemical bonds and serves as a test case for more complex molecular systems.

The electron density distribution in H_2^+ is crucial for understanding the nature of the chemical bond. The electron density determines the strength of the bond, the bond length, and the vibrational frequencies. Accurate calculation of electron density is essential for predicting molecular properties and understanding chemical reactivity.

Variational methods are powerful tools for calculating molecular properties. These methods provide upper bounds to the ground state energy and allow for the calculation of electron density distributions. The variational method is particularly useful for systems where exact solutions are not available.

This study presents a variational approach to simulate the electron density distribution in H_2^+. The study compares the results with exact calculations and demonstrates the effectiveness of simple variational methods for quantum mechanical simulations.

2. Theoretical Background

2.1 Schrödinger Equation

The time-independent Schrödinger equation for H_2^+ is:

$$H\psi = E\psi$$

where H is the Hamiltonian operator, ψ is the wavefunction, and E is the energy.

2.2 Hamiltonian Operator

The Hamiltonian operator for H_2^+ is:

$$H = -\frac{\nabla_1^2}{2} - \frac{\nabla_2^2}{2} - \frac{1}{r_{a1}} - \frac{1}{r_{b1}} + \frac{1}{R}$$

where ∇_1^2 and ∇_2^2 are the Laplacian operators for electrons 1 and 2, r_{a1} and r_{b1} are the distances from electron 1 to nuclei a and b, and R is the internuclear distance.

2.3 Variational Principle

The variational principle states that for any trial wavefunction ψ_{trial}, the energy E_{trial} satisfies:

$$E_{trial} = \frac{\langle \psi_{trial} | H | \psi_{trial} \rangle}{\langle \psi_{trial} | \psi_{trial} \rangle} \geq E_0 \quad (1)$$

where E_0 is the ground state energy.

2.4 Trial Wavefunction

The trial wavefunction used in this study is:

$$\psi_{trial} = \frac{1}{\sqrt{2(1+S)}}(\phi_a + \phi_b) \quad (2)$$

where ϕ_a and ϕ_b are hydrogen 1s orbitals centered on nuclei a and b, and S is the overlap integral between these orbitals.

3. Methodology

3.1 Computational Approach

The computational approach involves the following steps:

1. Calculation of overlap integrals
2. Calculation of kinetic energy integrals
3. Calculation of potential energy integrals
4. Optimization of the variational parameter
5. Calculation of electron density distribution

3.2 Overlap Integral

The overlap integral between two hydrogen 1s orbitals is:

$$S = \int \phi_a \phi_b \, d\tau = \left(1 + \frac{R}{a_0} + \frac{R^2}{3a_0^2}\right) e^{-R/a_0} \quad (3)$$

where a_0 is the Bohr radius.

3.3 Kinetic Energy Integral

The kinetic energy integral is:

$$T = \int \phi_a \left(-\frac{\nabla^2}{2}\right) \phi_b \, d\tau = \frac{1}{2}\left(1 + \frac{R}{a_0}\right) e^{-R/a_0} \quad (4)$$

3.4 Potential Energy Integral

The potential energy integral is:

$$V = \int \phi_a \left(-\frac{1}{r_{a1}} - \frac{1}{r_{b1}}\right) \phi_b \, d\tau = -\left(1 + \frac{1}{R}\right) e^{-2R/a_0} \quad (5)$$

3.5 Optimization

The variational parameter is optimized using the golden section search method. The optimization minimizes the energy E_{trial} with respect to the variational parameter.

3.6 Electron Density Calculation

The electron density is calculated from the trial wavefunction as:

$$\rho(r) = |\psi_{trial}|^2$$

4. Results and Discussion

4.1 Ground State Energy

The ground state energy calculated using the variational method for various internuclear distances is compared with exact results. The results show that the variational method provides accurate energy values with minimal computational cost.

4.2 Electron Density Distribution

The electron density distribution calculated using the variational method shows the expected features of a chemical bond, including electron sharing between the two nuclei and the formation of a molecular orbital. The electron density is maximum between the nuclei and decreases with distance from the nuclei.

4.3 Comparison with Exact Results

The results are compared with exact results from literature. The variational method provides accurate results for the ground state energy and electron density distribution. The accuracy of the method depends on the quality of the trial wavefunction and the optimization of the variational parameter.

5. Conclusion

This study demonstrates the effectiveness of simple variational methods for quantum mechanical simulations of molecular systems. The variational method provides accurate results for the ground state energy and electron density distribution with minimal computational cost. The study shows that simple trial wavefunctions can provide accurate results when properly optimized.

The results show that the variational method provides accurate results for the ground state energy and electron density distribution. The accuracy of the method depends on the quality of the trial wavefunction and the optimization of the variational parameter.

The study demonstrates the effectiveness of simple variational methods for quantum mechanical simulations of molecular systems. The variational method provides accurate results for the ground state energy and electron density distribution with minimal computational cost.

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MR-TCAST-MQ2000B0B0-V1

MR-TCAST-MQ2000B0B0-V1 Product Specification

Table of Contents

1 Product Overview 1.1 Product Introduction 1.2 Product Features 1.3 Product Specifications 1.4 Product Appearance 1.5 Product Dimensions 1.6 Product Interfaces 1.7 Product Indicators 2 Hardware Specifications 2.1 Hardware Architecture 2.2 Power Supply 2.3 Power Consumption 2.4 Cooling Design 2.5 Environmental Requirements 2.6 Mechanical Specifications 2.7 Electromagnetic Compatibility 2.8 Reliability 3 Software Specifications 3.1 Operating System 3.2 Management Interfaces 3.3 Protocol Support 3.4 Security Features 3.5 Firmware Upgrade 4 Ordering Information 4.1 Product Model Description 4.2 Accessories 4.3 Spare Parts 5 Appendix 5.1 Glossary 5.2 Revision History

1 Product Overview

1.1 Product Introduction

MR-TCAST-MQ2000B0B0-V1 is a high-performance industrial-grade multi-port Ethernet switch designed for demanding industrial automation, smart manufacturing, and critical infrastructure applications. The device provides 2 Gigabit Ethernet ports and 8 Gigabit Ethernet ports, supporting Layer 2 switching functions and offering comprehensive security and management capabilities. It adopts a fanless design with wide temperature operation capability, ensuring stable operation in harsh industrial environments.

1.2 Product Features

- Hardware Platform: Dual-core ARM Cortex-A53 processor, 512MB DDR3 memory, 128MB NAND Flash
- Ethernet Ports: 2 Gigabit Ethernet ports (1000Base-T/SFP combo), 8 Gigabit Ethernet ports (1000Base-T)
- Switching Capacity: 20Gbps
- Forwarding Rate: 14.88Mpps
- MAC Address Table: 16K entries
- VLAN: Supports 4094 VLANs, supports port-based VLAN, 802.1Q VLAN, QinQ
- Spanning Tree: Supports STP, RSTP, MSTP
- Link Aggregation: Supports static LAG and LACP, up to 8 groups, 8 members per group
- Multicast: Supports IGMP Snooping v1/v2/v3, MLD Snooping v1/v2, multicast VLAN
- QoS: Supports 802.1p, DSCP, 8 queues per port, SP, WRR, WFQ scheduling
- Security: Supports port security, MAC address binding, DHCP Snooping, ARP Inspection, IP Source Guard, 802.1X authentication, RADIUS, TACACS+
- Management: Supports CLI, Web, SNMP v1/v2c/v3, Telnet, SSH, NTP, syslog
- Protocols: Supports IPv4/IPv6 dual stack, ARP, ICMP, DHCP, DNS, TFTP, FTP, HTTP, HTTPS
- Reliability: Supports ring network protection, port mirroring, cable diagnostics, watchdog
- Power Supply: Dual power input, supports 12V/24V DC, supports power failure alarm
- Operating Temperature: -40°C ~ +75°C
- Storage Temperature: -40°C ~ +85°C
- Operating Humidity: 5% ~ 95% RH (non-condensing)
- Protection Rating: IP40
- Dimensions: 440mm × 44mm × 180mm (W × H × D)
- Weight: 1.5kg
- Installation: 19-inch rack-mount, desktop, DIN rail

1.3 Product Specifications

Item	Specifications
Model	MR-TCAST-MQ2000B0B0-V1
Ethernet Ports	2 Gigabit Ethernet ports (1000Base-T/SFP combo), 8 Gigabit Ethernet ports (1000Base-T)
Switching Capacity	20Gbps
Forwarding Rate	14.88Mpps
MAC Address Table	16K entries
VLAN	Supports 4094 VLANs, supports port-based VLAN, 802.1Q VLAN, QinQ
Spanning Tree	Supports STP, RSTP, MSTP
Link Aggregation	Supports static LAG and LACP, up to 8 groups, 8 members per group
Multicast	Supports IGMP Snooping v1/v2/v3, MLD Snooping v1/v2, multicast VLAN
QoS	Supports 802.1p, DSCP, 8 queues per port, SP, WRR, WFQ scheduling
Security	Supports port security, MAC address binding, DHCP Snooping, ARP Inspection, IP Source Guard, 802.1X authentication, RADIUS, TACACS+
Management	Supports CLI, Web, SNMP v1/v2c/v3, Telnet, SSH, NTP, syslog
Protocols	Supports IPv4/IPv6 dual stack, ARP, ICMP, DHCP, DNS, TFTP, FTP, HTTP, HTTPS
Reliability	Supports ring network protection, port mirroring, cable diagnostics, watchdog
Power Supply	Dual power input, supports 12V/24V DC, supports power failure alarm
Operating Temperature	-40°C ~ +75°C
Storage Temperature	-40°C ~ +85°C
Operating Humidity	5% ~ 95% RH (non-condensing)
Protection Rating	IP40
Dimensions	440mm × 44mm × 180mm (W × H × D)
Weight	1.5kg
Installation	19-inch rack-mount, desktop, DIN rail

1.4 Product Appearance

1.5 Product Dimensions

1.6 Product Interfaces

Front Panel

- 2 Gigabit Ethernet ports (1000Base-T/SFP combo)
- 8 Gigabit Ethernet ports (1000Base-T)
- Console port
- Power indicator
- System indicator
- Port indicators

Rear Panel

- Power terminal block
- Grounding screw

1.7 Product Indicators

Indicator	Status	Description
Power	Solid green	Power normal
	Off	Power abnormal
System	Solid green	System normal
	Blinking green	System starting
	Solid red	System fault
	Off	No power
Port	Solid green	Link established
	Blinking green	Data transmission
	Off	Link not established

2 Hardware Specifications

2.1 Hardware Architecture

- Processor: Dual-core ARM Cortex-A53 processor
- Memory: 512MB DDR3
- Flash: 128MB NAND Flash
- Switch Chip: Integrated Layer 2 switch chip
- Ethernet Ports: 2 Gigabit Ethernet ports (1000Base-T/SFP combo), 8 Gigabit Ethernet ports (1000Base-T)
- Console Port: 1 console port
- Power Input: Dual power input, supports 12V/24V DC, supports power failure alarm
- Indicators: Power indicator, system indicator, port indicators
- Buttons: Reset button

2.2 Power Supply

- Power Input: Dual power input, supports 12V/24V DC, supports power failure alarm
- Power Consumption: Typical 15W, maximum 25W
- Power Terminal Block: Supports 2-pin terminal block connection
- Grounding Screw: Provides grounding screw for safety and lightning protection

2.3 Power Consumption

- Typical: 15W
- Maximum: 25W
- Idle: 8W

2.4 Cooling Design

- Fanless design, no fan, no noise, no dust, no maintenance
- Natural convection heat dissipation, suitable for long-term stable operation
- Operating temperature: -40°C ~ +75°C, suitable for harsh industrial environments
- Storage temperature: -40°C ~ +85°C
- Operating humidity: 5% ~ 95% RH (non-condensing)
- Protection rating: IP40, suitable for indoor installation

2.5 Environmental Requirements

- Operating temperature: -40°C ~ +75°C
- Storage temperature: -40°C ~ +85°C
- Operating humidity: 5% ~ 95% RH (non-condensing)
- Protection rating: IP40
- Installation environment: Indoor, avoid direct sunlight, ensure good ventilation

2.6 Mechanical Specifications

- Dimensions: 440mm × 44mm × 180mm (W × H × D)
- Weight: 1.5kg
- Installation: 19-inch rack-mount, desktop, DIN rail
- Rack-mount ears: Standard 19-inch rack-mount ears, supports front and rear installation
- Chassis: Metal chassis, sturdy and durable

2.7 Electromagnetic Compatibility

- Conducted emissions: Meets industrial-grade EMC requirements
- Radiated emissions: Meets industrial-grade EMC requirements
- Electrostatic discharge: Supports ESD protection
- Surge protection: Supports surge protection
- Electromagnetic interference: Meets industrial-grade EMC requirements

2.8 Reliability

- Mean Time Between Failures (MTBF): Over 100,000 hours
- Operating temperature: -40°C ~ +75°C, suitable for harsh industrial environments
- Storage temperature: -40°C ~ +85°C
- Operating humidity: 5% ~ 95% RH (non-condensing)
- Protection rating: IP40, suitable for indoor installation
- Power failure alarm: Supports power failure alarm, automatically sends alarm when power is lost
- Watchdog: Supports hardware watchdog, automatically restores system when system hangs
- Port mirroring: Supports port mirroring for fault diagnosis
- Cable diagnostics: Supports cable diagnostics, detects cable faults
- Ring network protection: Supports ring network protection, ensures network continuity

3 Software Specifications

3.1 Operating System

- Operating system: Proprietary operating system
- Kernel: Linux kernel
- Management interfaces: CLI, Web, SNMP, Telnet, SSH
- Protocol stack: Supports IPv4/IPv6 dual stack
- Firmware upgrade: Supports online firmware upgrade, supports dual firmware backup

3.2 Management Interfaces

- CLI: Supports command-line interface, supports Telnet and SSH remote management
- Web: Supports web management, supports HTTP and HTTPS
- SNMP: Supports SNMP v1/v2c/v3, supports MIB-II, supports private MIB
- Telnet: Supports Telnet remote management
- SSH: Supports SSH encrypted management
- NTP: Supports NTP time synchronization
- Syslog: Supports syslog, supports local and remote syslog
- Firmware upgrade: Supports online firmware upgrade, supports dual firmware backup
- Configuration backup: Supports configuration backup and restore
- Batch configuration: Supports batch configuration, supports configuration templates

3.3 Protocol Support

- Layer 2 protocols: Supports ARP, ICMP, DHCP, DNS, TFTP, FTP, HTTP, HTTPS
- Spanning tree: Supports STP, RSTP, MSTP
- Link aggregation: Supports static LAG and LACP, up to 8 groups, 8 members per group
- VLAN: Supports 4094 VLANs, supports port-based VLAN, 802.1Q VLAN, QinQ
- Multicast: Supports IGMP Snooping v1/v2/v3, MLD Snooping v1/v2, multicast VLAN
- QoS: Supports 802.1p, DSCP, 8 queues per port, SP, WRR, WFQ scheduling
- Security: Supports port security, MAC address binding, DHCP Snooping, ARP Inspection, IP Source Guard, 802.1X authentication, RADIUS, TACACS+
- Management: Supports CLI, Web, SNMP v1/v2c/v3, Telnet, SSH, NTP, syslog
- Protocols: Supports IPv4/IPv6 dual stack, ARP, ICMP, DHCP, DNS, TFTP, FTP, HTTP, HTTPS
- Reliability: Supports ring network protection, port mirroring, cable diagnostics, watchdog

3.4 Security Features

- Port security: Supports port security, limits MAC address learning
- MAC address binding: Supports MAC address binding, prevents unauthorized access
- DHCP Snooping: Supports DHCP Snooping, prevents DHCP spoofing attacks
- ARP Inspection: Supports ARP Inspection, prevents ARP spoofing attacks
- IP Source Guard: Supports IP Source Guard, prevents IP address spoofing
- 802.1X authentication: Supports 802.1X authentication, supports RADIUS authentication
- RADIUS: Supports RADIUS authentication, supports accounting
- TACACS+: Supports TACACS+ authentication, supports accounting
- Port mirroring: Supports port mirroring for fault diagnosis
- Cable diagnostics: Supports cable diagnostics, detects cable faults
- Ring network protection: Supports ring network protection, ensures network continuity

3.5 Firmware Upgrade

- Online upgrade: Supports online firmware upgrade, no need to restart the device
- Dual firmware backup: Supports dual firmware backup, automatically switches to backup firmware when upgrade fails
- Configuration backup: Supports configuration backup and restore
- Batch configuration: Supports batch configuration, supports configuration templates

4 Ordering Information

4.1 Product Model Description

MR-TCAST-MQ2000B0B0-V1

- MR: Industrial Ethernet switch
- TCAST: Product series
- MQ2000: 2 Gigabit Ethernet ports + 8 Gigabit Ethernet ports
- B0: Standard configuration
- B0: Standard configuration
- V1: Version 1

4.2 Accessories

- Power adapter: 12V/24V DC power adapter
- Rack-mount ears: Standard 19-inch rack-mount ears
- Grounding screw: Provides grounding screw for safety and lightning protection
- Console cable: Console cable for configuration
- Ethernet cable: Ethernet cable for connection
- Installation manual: Installation manual
- Warranty card: Warranty card

4.3 Spare Parts

- Power adapter: 12V/24V DC power adapter
- Rack-mount ears: Standard 19-inch rack-mount ears
- Grounding screw: Provides grounding screw for safety and lightning protection
- Console cable: Console cable for configuration
- Ethernet cable: Ethernet cable for connection
- Installation manual: Installation manual
- Warranty card: Warranty card

5 Appendix

5.1 Glossary

- Switch: Network switch, forwards data packets between network devices
- Ethernet: Ethernet, wired network protocol
- Gigabit Ethernet: Gigabit Ethernet, 1000Mbps Ethernet
- Port: Port, network interface
- VLAN: Virtual Local Area Network, logical network segmentation
- QinQ: Double VLAN, supports VLAN stacking
- STP: Spanning Tree Protocol, prevents network loops
- RSTP: Rapid Spanning Tree Protocol, fast convergence
- MSTP: Multiple Spanning Tree Protocol, supports multiple spanning trees
- LAG: Link Aggregation Group, bundles multiple physical links into one logical link
- LACP: Link Aggregation Control Protocol, dynamic link aggregation
- IGMP: Internet Group Management Protocol, multicast management protocol
- MLD: Multicast Listener Discovery, IPv6 multicast management protocol
- QoS: Quality of Service, supports traffic prioritization and bandwidth allocation
- ACL: Access Control List, supports packet filtering and access control
- 802.1X: Port-based Network Access Control, supports user authentication
- RADIUS: Remote Authentication Dial-In User Service, supports centralized authentication
- TACACS+: Terminal Access Controller Access-Control System Plus, supports centralized authentication
- SNMP: Simple Network Management Protocol, supports network management
- MIB: Management Information Base, supports SNMP management
- CLI: Command Line Interface, supports command-line management
- Web: Supports web management, supports HTTP and HTTPS
- Telnet: Supports Telnet remote management
- SSH: Supports SSH encrypted management
- NTP: Supports NTP time synchronization
- Syslog: Supports syslog, supports local and remote syslog
- Firmware upgrade: Supports online firmware upgrade, supports dual firmware backup
- Configuration backup: Supports configuration backup and restore
- Batch configuration: Supports batch configuration, supports configuration templates

5.2 Revision History

Version	Date	Description
V1.0	2026-04-01	Initial release

Tags: ancient civilizations, ancient language speakers, Lost Cemetery of Ancient Language Speakers, Joseph Francis Alcazar, Cornelius Ashburn, Dr. Ellis, Lucinda Hawkes, Samuel Hawkes, Tom Hawkes, Professor Alcazar, Professor Ashburn, Professor Ellis, Professor Hawkes, Professor Lucinda Hawkes, Professor Samuel Hawkes, Professor Tom Hawkes, Professor Ashburn's Assistant, Professor Ellis's Assistant, Professor Hawkes's Assistant, Professor Lucinda Hawkes's Assistant, Professor Samuel Hawkes's Assistant, Professor Tom Hawkes's Assistant, Professor Ashburn's Assistant's Assistant, Professor Ellis's Assistant's Assistant, Professor Hawkes's Assistant's Assistant, Professor Lucinda Hawkes's Assistant's Assistant, Professor Samuel Hawkes's Assistant's Assistant, Professor Tom Hawkes's Assistant's Assistant, Professor Ashburn's Assistant's Assistant's Assistant, Professor Ellis's Assistant's Assistant's Assistant, Professor Hawkes's Assistant's Assistant's Assistant, Professor Lucinda 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Ref: Fluid Mechanics

Date: Monday, June 1, 2026 at 10:00:00 AM Coordinated Universal Time 7

Fluid Mechanics

Topic: Laplace transformation for Partial differential equations

Subtopic: Laplace transformation for Partial differential equations

Keywords: Laplace transform, Partial differential equations, Fluid mechanics, Heat transfer, Signal processing, Control systems, Electrical engineering, Mathematical modeling, Boundary value problems, Initial value problems, Differential equations, Transfer functions, Stability analysis, Frequency domain analysis, Time domain analysis, Integral transforms, Complex analysis, Engineering mathematics, Applied mathematics, Solution methods, Analytical methods, Boundary conditions, Initial conditions, Transient analysis, Steady state analysis, Dynamic systems, Linear systems, Nonlinear systems, Partial differential equations, Ordinary differential equations, Boundary value problems, Initial value problems, Laplace transform, Fourier transform, Inverse transform, Convolution theorem, Heaviside step function, Dirac delta function, Green's function

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Fig.2 Sensitivity analysis of Fermentation Capacity

Fig.2

Author: Prof. Dr. Ali Ekber Kocaoğlu, Dr. Ali Ekber Kocaoğlu, Dr. Ali Ekber Kocaoğlu,

MR-TCOT-MQ2MR0NQ7VT

MR-TCOT-MQ2MR0NQ7VT-Mathematics

Mr. Chandra Sir, CHEMISTRY(Organic Chemistry)

Q1.

Which of the following is the correct IUPAC name of the compound shown below?

(1) 2-methyl-3-methoxy-1,4-dimethylcyclohexane

(2) 1-methoxy-2-methyl-3,4-dimethylcyclohexane

(3) 1-methoxy-2-methyl-3,4-dimethylcyclohexane(4) 1-methoxy-2-methyl-3,4-dimethylcyclohexane

Ans. 1-methoxy-2-methyl-3,4-dimethylcyclohexane

Sol.

Ans.

Q2.

Consider the following data for a first order reaction:Rate = 0.693 [A]Time (in min) 0 100 200 300 400 500 600 700 800 900 1000[A] (in M) 0.100 0.050 0.025 0.012 0.006 0.003 0.001 0.000 0.000 0.000 0.000Rate (in M min^{-1}) 0.0693 0.0347 0.0173 0.0083 0.0042 0.0021 0.0007 0.000 0.000 0.000 0.000

Which of the following plots will be correct?

(1)

(2)

(3)

(4)

Ans.

Q3.

Consider the following statements:

Statement-I: Rate of a first order reaction depends upon the rate constant and the concentration of reactant.

Statement-II: Half life period of a first order reaction depends upon the rate constant and the concentration of reactant.

Choose the correct option:

(1) Statement-I is false but Statement-II is true(2) Both Statement-I and Statement-II are false

(3) Both Statement-I and Statement-II are true(4) Statement-I is true but Statement-II is false

Ans.

Q4.

Consider the following data:

Rate = 0.693 [A]Time (in min) 0 100 200 300 400 500 600 700 800 900 1000[A] (in M) 0.100 0.050 0.025 0.012 0.006 0.003 0.001 0.000 0.000 0.000 0.000Rate (in M min^{-1}) 0.0693 0.0347 0.0173 0.0083 0.0042 0.0021 0.0007 0.000 0.000 0.000 0.000

Which of the following plots will be correct?

(1)

(2)

(3)

(4)

Ans.

Q5.

Consider the following statements:

Statement-I: Rate of a first order reaction depends upon the rate constant and the concentration of reactant.

Statement-II: Half life period of a first order reaction depends upon the rate constant and the concentration of reactant.

Choose the correct option:

(1) Statement-I is false but Statement-II is true(2) Both Statement-I and Statement-II are false

(3) Both Statement-I and Statement-II are true(4) Statement-I is true but Statement-II is false

Ans.

Q6.

Consider the following statements:

Statement-I: Rate of a first order reaction depends upon the rate constant and the concentration of reactant.

Statement-II: Half life period of a first order reaction depends upon the rate constant and the concentration of reactant.

Choose the correct option:

(1) Statement-I is false but Statement-II is true(2) Both Statement-I and Statement-II are false

(3) Both Statement-I and Statement-II are true(4) Statement-I is true but Statement-II is false

Ans.

Q7.

Consider the following data:

Rate = 0.693 [A]Time (in min) 0 100 200 300 400 500 600 700 800 900 1000[A] (in M) 0.100 0.050 0.025 0.012 0.006 0.003 0.001 0.000 0.000 0.000 0.000Rate (in M min^{-1}) 0.0693 0.0347 0.0173 0.0083 0.0042 0.0021 0.0007 0.000 0.000 0.000 0.000

Which of the following plots will be correct?

(1)

(2)

(3)

(4)

Ans.

Q8.

Consider the following statements:

Statement-I: Rate of a first order reaction depends upon the rate constant and the concentration of reactant.

Statement-II: Half life period of a first order reaction depends upon the rate constant and the concentration of reactant.

Choose the correct option:

(1) Statement-I is false but Statement-II is true(2) Both Statement-I and Statement-II are false

(3) Both Statement-I and Statement-II are true(4) Statement-I is true but Statement-II is false

Ans.

Q9.

Consider the following data:

Rate = 0.693 [A]Time (in min) 0 100 200 300 400 500 600 700 800 900 1000[A] (in M) 0.100 0.050 0.025 0.012 0.006 0.003 0.001 0.000 0.000 0.000 0.000Rate (in M min^{-1}) 0.0693 0.0347 0.0173 0.0083 0.0042 0.0021 0.0007 0.000 0.000 0.000 0.000

Which of the following plots will be correct?

(1)

(2)

(3)

(4)

Ans.

Q10.

Consider the following statements:

Statement-I: Rate of a first order reaction depends upon the rate constant and the concentration of reactant.

Statement-II: Half life period of a first order reaction depends upon the rate constant and the concentration of reactant.

Choose the correct option:

(1) Statement-I is false but Statement-II is true(2) Both Statement-I and Statement-II are false

(3) Both Statement-I and Statement-II are true(4) Statement-I is true but Statement-II is false

Ans.

Q11.

Consider the following data:

Rate = 0.693 [A]Time (in min) 0 100 200 300 400 500 600 700 800 900 1000[A] (in M) 0.100 0.050 0.025 0.012 0.006 0.003 0.001 0.000 0.000 0.000 0.000Rate (in M min^{-1}) 0.0693 0.0347 0.0173 0.0083 0.0042 0.0021 0.0007 0.000 0.000 0.000 0.000

Which of the following plots will be correct?

(1)

(2)

(3)

(4)

Ans.

Q12.

Consider the following statements:

Statement-I: Rate of a first order reaction depends upon the rate constant and the concentration of reactant.

Statement-II: Half life period of a first order reaction depends upon the rate constant and the concentration of reactant.

Choose the correct option:

(1) Statement-I is false but Statement-II is true(2) Both Statement-I and Statement-II are false

(3) Both Statement-I and Statement-II are true(4) Statement-I is true but Statement-II is false

Ans.

Q13.

Consider the following data:

Rate = 0.693 [A]Time (in min) 0 100 200 300 400 500 600 700 800 900 1000[A] (in M) 0.100 0.050 0.025 0.012 0.006 0.003 0.001 0.000 0.000 0.000 0.000Rate (in M min^{-1}) 0.0693 0.0347 0.0173 0.0083 0.0042 0.0021 0.0007 0.000 0.000 0.000 0.000

Which of the following plots will be correct?

(1)

(2)

(3)

(4)

Ans.

Q14.

Consider the following statements:

Statement-I: Rate of a first order reaction depends upon the rate constant and the concentration of reactant.

Statement-II: Half life period of a first order reaction depends upon the rate constant and the concentration of reactant.

Choose the correct option:

(1) Statement-I is false but Statement-II is true(2) Both Statement-I and Statement-II are false

(3) Both Statement-I and Statement-II are true(4) Statement-I is true but Statement-II is false

Ans.

Q15.

Consider the following data:

Rate = 0.693 [A]Time (in min) 0 100 200 300 400 500 600 700 800 900 1000[A] (in M) 0.100 0.050 0.025 0.012 0.006 0.003 0.001 0.000 0.000 0.000 0.000Rate (in M min^{-1}) 0.0693 0.0347 0.0173 0.0083 0.0042 0.0021 0.0007 0.000 0.000 0.000 0.000

Which of the following plots will be correct?

(1)

(2)

(3)

(4)

Ans.

Q16.

Consider the following statements:

Statement-I: Rate of a first order reaction depends upon the rate constant and the concentration of reactant.

Statement-II: Half life period of a first order reaction depends upon the rate constant and the concentration of reactant.

Choose the correct option:

(1) Statement-I is false but Statement-II is true(2) Both Statement-I and Statement-II are false

(3) Both Statement-I and Statement-II are true(4) Statement-I is true but Statement-II is false

Ans.

Q17.

Consider the following data:

Rate = 0.693 [A]Time (in min) 0 100 200 300 400 500 600 700 800 900 1000[A] (in M) 0.100 0.050 0.025 0.012 0.006 0.003 0.001 0.000 0.000 0.000 0.000Rate (in M min^{-1}) 0.0693 0.0347 0.0173 0.0083 0.0042 0.0021 0.0007 0.000 0.000 0.000 0.000

Which of the following plots will be correct?

(1)

(2)

(3)

(4)

Ans.

Q18.

Consider the following statements:

Statement-I: Rate of a first order reaction depends upon the rate constant and the concentration of reactant.

Statement-II: Half life period of a first order reaction depends upon the rate constant and the concentration of reactant.

Choose the correct option:

(1) Statement-I is false but Statement-II is true(2) Both Statement-I and Statement-II are false

(3) Both Statement-I and Statement-II are true(4) Statement-I is true but Statement-II is false

Ans.

Q19.

Consider the following data:

Rate = 0.693 [A]Time (in min) 0 100 200 300 400 500 600 700 800 900 1000[A] (in M) 0.100 0.050 0.025 0.012 0.006 0.003 0.001 0.000 0.000 0.000 0.000Rate (in M min^{-1}) 0.0693 0.0347 0.0173 0.0083 0.0042 0.0021 0.0007 0.000 0.000 0.000 0.000

Which of the following plots will be correct?

(1)

(2)

(3)

(4)

Ans.

Q20.

Consider the following statements:

Statement-I: Rate of a first order reaction depends upon the rate constant and the concentration of reactant.

Statement-II: Half life period of a first order reaction depends upon the rate constant and the concentration of reactant.

Choose the correct option:

(1) Statement-I is false but Statement-II is true(2) Both Statement-I and Statement-II are false

(3) Both Statement-I and Statement-II are true(4) Statement-I is true but Statement-II is false

Ans.

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[illegible]

[illegible]

[illegible]

Keywords: Fluid Mechanics, Lecture Series, Transcript, Transcription, Text, Lecture Transcript, Transcription, Text, Lecture Transcript, Transcription, Text, Lecture

MR-TIGER-MAGNUM-MAXIMUS

Mr-Tiger-Magnum-Maximus-Supreme-Gamma-Omega-Sigma-Alpha

Mr-Tiger-Magnum-Maximus-Supreme-Gamma-Omega-Sigma-Alpha

Mr-Tiger-Magnum-Maximus-Supreme-Gamma-Omega-Sigma-Alpha

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Mr-Tiger-Magnum-Maximus-Supreme-Gamma-Omega-Sigma-Alpha

Mr-Tiger-Magnum-Maximus-Supreme-Gamma-Omega-Sigma-Alpha

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Mr-Tiger-Magnum-Maximus-Supreme-Gamma-Omega-Sigma-Alpha

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Fabian Ruehle [1] [2] [3] [4] [5] [6] [7] [8] [9] [10] [11] [12] [13] [14] [15] [16] [17] [18] [19] [20] [21] [22] [23] [24] [25] [26] [27] [28] [29] [30] [31] [32] [33] [34] [35] [36] [37] [38] [39] [40]

MATLAB/Simulink Model Based Design Workflow

Introduction

Model Based Design (MBD) is a formal methodology for developing complex control systems. It uses a single mathematical model as the central artifact from which requirements, simulation, code generation, and verification are derived. The model serves as the ground truth for all downstream activities, ensuring that changes propagate consistently across the entire development lifecycle.

Workflow Overview

The workflow consists of six stages: requirements capture, plant modeling, controller design, simulation and verification, code generation, and deployment. Each stage produces artifacts that feed into the next stage. Traceability links connect requirements to model elements to generated code to test results, so that any change can be assessed for its impact on every downstream artifact.

Plant Modeling

Plant modeling begins with the physical equations governing the system. For a DC motor, the equations are:

$$v(t) = Ri(t) + L\frac{di(t)}{dt} + K_e\omega(t)$$

$$J\frac{d\omega(t)}{dt} = K_t i(t) - B\omega(t) - \tau_L(t)$$

Controller Design

The controller is designed in the discrete domain. For a PI controller with sampling period T_s:

$$u[k] = K_p e[k] + K_i T_s \sum_{j=0}^{k} e[j]$$

Simulink Implementation

Simulink provides a block diagram environment for implementing the plant model and controller. Signal flow is explicit and visual. Simulink supports fixed-step and variable-step solvers, with fixed-step solvers required for real-time code generation.

Code Generation

Simulink Coder (formerly Real-Time Workshop) generates C code from the model. The generated code is deterministic, with no dynamic memory allocation and no recursion. Code generation can be configured for different target processors and optimization levels.

Verification

Model-in-the-loop (MIL) simulation verifies the model against requirements before code generation. Software-in-the-loop (SIL) simulation runs the generated code on a host machine to verify that the code behaves identically to the model. Hardware-in-the-loop (HIL) simulation runs the code on the target hardware with real sensors and actuators.

Deployment

Deployment packages the generated code with the target hardware's boot code and runtime libraries. Over-the-air (OTA) updates allow deployed systems to receive new versions without physical access.

Conclusion

Model Based Design provides a single source of truth for requirements, simulation, and implementation. Changes propagate automatically through the workflow, reducing the risk of inconsistencies between specification and implementation.

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1.1 Introduction to Data Structures and Algorithms

Data structures and algorithms are the foundation of computer science. Understanding how to organize data efficiently and how to design algorithms that solve problems quickly is essential for any programmer.

1.1.1 Why Study Data Structures and Algorithms?

Consider a simple problem: you have a list of 1,000,000 numbers and you need to find the largest one. A naive approach might scan the entire list for each query. With the right data structure, you can answer the question in constant time. The difference between a slow program and a fast program can be the difference between a project that works and one that never finishes.

Data structures and algorithms are the tools that make computers fast and reliable. They are used in every domain: finance, medicine, engineering, science, and everyday software. Mastering them gives you the ability to design systems that scale from a few users to millions of users.

1.1.2 Formal Definitions

Let n denote the size of the input. We write $f(n) = O(g(n))$ if there exists constants $c > 0$ and $n_0 \geq 0$ such that $f(n) \leq c \cdot g(n)$ for all $n \geq n_0$. Similarly, $f(n) = \Omega(g(n))$ if $f(n) \geq c \cdot g(n)$ for all $n \geq n_0$, and $f(n) = \Theta(g(n))$ if both hold.

Formally, a data structure is a tuple $(D, \mathcal{O})$ where D is the set of possible states and $\mathcal{O}$ is a set of operations that transform states in D.

1.1.3 Example

```
def binary_search(arr, target):
    lo, hi = 0, len(arr) - 1
    while lo <= hi:
        mid = (lo + hi) // 2
        if arr[mid] == target:
            return mid
        elif arr[mid] < target:
            lo = mid + 1
        else:
            hi = mid - 1
    return -1
```

Binary search runs in $O(\log n)$ time. Each iteration halves the search space, so after k iterations the remaining range has size $n / 2^k$. When $k = \log_2 n$, the range has size 1 and the algorithm terminates.

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<?xml version="1.0" encoding="UTF-8"?>
<!-- Generated by SVGO v3.0.2 (https://svgo.dev) -->
<svg
  xmlns="http://www.w3.org/2000/svg"
  xmlns:xlink="http://www.w3.org/1999/xlink"
  viewBox="0.000,0.000,1000.000,1000.000"
  preserveAspectRatio="xMidYMid meet"
  fill="#000000" />
</svg>
```

Chapter 4 Matrix Algebra

4.1 Sum and Scalar Product of Matrices

Definition

Sum of two matrices Two matrices A and B are said to be conformable for addition if they are of the same order. If $A = [a_{ij}]_{m \times n}$ and $B = [b_{ij}]_{m \times n}$ then the sum of A and B is defined as $A + B = [a_{ij} + b_{ij}]_{m \times n}$.

Scalar product If $A = [a_{ij}]_{m \times n}$ and k is a scalar then $kA = [ka_{ij}]_{m \times n}$.

Example

Example 1

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \quad B = \begin{bmatrix} 7 & 8 & 9 \\ 10 & 11 & 12 \end{bmatrix}, \quad k = 2$$

Solution

$$A + B = \begin{bmatrix} 1+7 & 2+8 & 3+9 \\ 4+10 & 5+11 & 6+12 \end{bmatrix} = \begin{bmatrix} 8 & 10 & 12 \\ 14 & 16 & 18 \end{bmatrix}$$

$$kA = \begin{bmatrix} 2 \times 1 & 2 \times 2 & 2 \times 3 \\ 2 \times 4 & 2 \times 5 & 2 \times 6 \end{bmatrix} = \begin{bmatrix} 2 & 4 & 6 \\ 8 & 10 & 12 \end{bmatrix}$$

Example 2

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}, \quad k = 3$$

Solution

$$A + B = \begin{bmatrix} 1+5 & 2+6 \\ 3+7 & 4+8 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}$$

$$kA = \begin{bmatrix} 3 \times 1 & 3 \times 2 \\ 3 \times 3 & 3 \times 4 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 9 & 12 \end{bmatrix}$$

Example 3

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 4

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 5

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 6

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 7

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 8

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 9

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 10

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 11

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 12

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 13

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 14

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 15

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 16

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 17

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 18

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 19

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 20

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 21

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 22

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 23

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 24

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 25

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 26

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 27

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 28

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 29

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 30

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 31

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 32

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 &

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Tiger Magnum Opus 7 Extraordinary Lemma

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Lecture 1 Linear Algebra and Matrix Calculus

Contents

Lecture 1 Linear Algebra and Matrix Calculus
Contents
Page 1 Lecture 1 Linear Algebra and Matrix Calculus
Page 2 Lecture 1 Linear Algebra and Matrix Calculus
Page 3 Lecture 1 Linear Algebra and Matrix Calculus
Page 4 Lecture 1 Linear Algebra and Matrix Calculus
Page 5 Lecture 1 Linear Algebra and Matrix Calculus
Page 6 Lecture 1 Linear Algebra and Matrix Calculus
Page 7 Lecture 1 Linear Algebra and Matrix Calculus
Page 8 Lecture 1 Linear Algebra and Matrix Calculus
Page 9 Lecture 1 Linear Algebra and Matrix Calculus
Page 10 Lecture 1 Linear Algebra and Matrix Calculus
Page 11 Lecture 1 Linear Algebra and Matrix Calculus
Page 12 Lecture 1 Linear Algebra and Matrix Calculus
Page 13 Lecture 1 Linear Algebra and Matrix Calculus
Page 14 Lecture 1 Linear Algebra and Matrix Calculus
Page 15 Lecture 1 Linear Algebra and Matrix Calculus
Page 16 Lecture 1 Linear Algebra and Matrix Calculus
Page 17 Lecture 1 Linear Algebra and Matrix Calculus
Page 18 Lecture 1 Linear Algebra and Matrix Calculus
Page 19 Lecture 1 Linear Algebra and Matrix Calculus
Page 20 Lecture 1 Linear Algebra and Matrix Calculus
Page 21 Lecture 1 Linear Algebra and Matrix Calculus
Page 22 Lecture 1 Linear Algebra and Matrix Calculus
Page 23 Lecture 1 Linear Algebra and Matrix Calculus
Page 24 Lecture 1 Linear Algebra and Matrix Calculus
Page 25 Lecture 1 Linear Algebra and Matrix Calculus
Page 26 Lecture 1 Linear Algebra and Matrix Calculus
Page 27 Lecture 1 Linear Algebra and Matrix Calculus
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Page 91 Lecture 1 Linear Algebra and Matrix Calculus
Page 92 Lecture 1 Linear Algebra and Matrix Calculus
Page 93 Lecture 1 Linear Algebra and Matrix Calculus
Page 94 Lecture 1 Linear Algebra and Matrix Calculus
Page 95 Lecture 1 Linear Algebra and Matrix Calculus
Page 96 Lecture 1 Linear Algebra and Matrix Calculus
Page 97 Lecture 1 Linear Algebra and Matrix Calculus
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Ethical Implications of Artificial Intelligence in Healthcare: Balancing Innovation and Patient Rights

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Abstract: Artificial Intelligence (AI) is rapidly transforming healthcare by enhancing diagnostic accuracy, optimizing treatment plans, and improving patient outcomes. However, the integration of AI in healthcare raises significant ethical concerns, including data privacy, algorithmic bias, accountability, and the potential displacement of human caregivers. This study explores the ethical implications of AI in healthcare, focusing on the balance between innovation and patient rights. The study examines the ethical challenges associated with AI in healthcare, including the impact on patient autonomy, the potential for discrimination, and the need for transparency and accountability. The study also discusses the regulatory frameworks and guidelines that can ensure the ethical deployment of AI in healthcare, ensuring that the benefits of AI are maximized while safeguarding patient rights and maintaining trust in healthcare systems. The study concludes with recommendations for policymakers, healthcare providers, and AI developers to address the ethical challenges and promote the responsible use of AI in healthcare.

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Ch. 2 Lo-Fi Prototyping

Introduction

Lo-fi prototyping is the process of using low-tech materials to create prototypes of your product. Lo-fi prototyping is a technique that can be used at any stage in the design of a product, but it is most useful in the early stages of the design process. Lo-fi prototyping is a technique that can be used by designers, developers, and non-designers alike.

Why Lo-Fi Prototyping?

Lo-fi prototyping is a technique that can be used by designers, developers, and non-designers alike. It is most useful in the early stages of the design process.

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Research Article

Cognitive Ability Predicts Obesity and Smoking Across Three Cohorts of Twenty-Year-Olds

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University of Edinburgh

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7.1 Cell Theory: Life at the Smallest Scale

Cell Theory: Life at the Smallest Scale

Cell Theory: Life at the Smallest Scale

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Abstract

Deep learning (DL) models have shown promising performance in medical imaging tasks, including the detection of diabetic retinopathy (DR) and other eye diseases. However, it remains unclear how these models generalize to real-world clinical settings. Here we assess the generalizability of DL models for DR detection from fundus photographs across ethnicity, imaging camera, and disease severity and type in a dataset of 236,234 images from 40,339 diabetic patients. We find that the performance of DL models for referable DR detection is comparable across Asian, Caucasian, and African ethnicities (AUC: 0.936–0.946), but varies across cameras (AUC: 0.906–0.947) and is lower in severe DR (AUC: 0.885) and proliferative DR (AUC: 0.901) compared to moderate DR (AUC: 0.943). These results suggest that DL models for DR screening may not generalize across all ethnicities, cameras, and disease severities and types, and that such models should be tested in these settings before implementation.

NPJ | https://doi.org/10.1038/s41746-020-00376-2 ARTMED | 1

Introduction

Diabetes mellitus (DM) is a chronic disease that affects 463 million people worldwide and is projected to rise to 700 million by 2045[1]. Diabetic retinopathy (DR) is a common complication of DM and a major cause of blindness and vision impairment worldwide[2]. Early detection of DR followed by timely treatment can prevent vision loss[3], and thus, DM patients are recommended to undergo yearly DR screening[4]. However, in many countries, the shortage of eye care specialists and the high prevalence of DM make such a task difficult to accomplish[5]. Deep learning (DL) models have shown promising performance in the detection of DR from fundus photographs, with several studies reporting high sensitivity and specificity in retrospective datasets[6–9]. However, it remains unclear how these models generalize to real-world clinical settings, where they may be used for DR screening. Prior studies have suggested that DL models may not generalize across ethnicities, imaging cameras, and disease severities and types[10–13], but these studies have been limited by the use of a single DL model, a single camera, or a single ethnicity, and have not assessed the impact of disease severity and type on model performance. Here we assess the generalizability of a DL model for referable DR (RDR) detection across ethnicities, cameras, and disease severities and types in a large dataset of diabetic patients from the United Kingdom (UK) Biobank[14].

Results

Study population characteristics

The UK Biobank dataset consisted of 236,234 fundus photographs from 40,339 diabetic patients, of which 1,401 (3.5%) had RDR, 1,067 (2.6%) had moderate DR, 1,250 (3.1%) had severe DR, and 1,175 (2.9%) had proliferative DR (PDR). Mean (SD) age was 56.8 (8.1) years, 23,561 (58.4%) were male, 1,937 (4.8%) were Asian, 1,478 (3.7%) were African, and 36,924 (91.5%) were Caucasian. Mean (SD) HbA1c was 75.1 (19.7) mmol/mol (9.0 (1.8)%), 3,553 (8.8%) had insulin treatment, and mean (SD) duration of DM was 8.7 (8.1) years. Mean (SD) age was 55.4 (8.1) years for those with RDR, 55.8 (8.1) years for those with moderate DR, 56.4 (8.1) years for those with severe DR, and 56.1 (8.1) years for those with PDR. Mean (SD) HbA1c was 81.0 (21.3) mmol/mol (9.6 (1.9)%) for those with RDR, 77.4 (19.9) mmol/mol (9.3 (1.8)%) for those with moderate DR, 79.4 (20.1) mmol/mol (9.4 (1.8)%) for those with severe DR, and 79.9 (20.4) mmol/mol (9.5 (1.9)%) for those with PDR. Insulin treatment was 14.1% (RDR), 12.1% (moderate DR), 13.8% (severe DR), and 13.8% (PDR). Mean (SD) duration of DM was 10.7 (8.4) years for those with RDR, 10.1 (8.2) years for those with moderate DR, 10.8 (8.2) years for those with severe DR, and 10.8 (8.2) years for those with PDR. Other baseline characteristics by DR status are shown in Table 1.

Performance of DL models for referable DR detection

We trained a DL model to detect RDR using fundus photographs from patients of Caucasian ethnicity, imaging with a Topcon camera, and without severe DR or PDR. The model achieved an AUC of 0.943 (95% CI: 0.937–0.949) in a randomly sampled test set of 9,000 images from 9,000 patients. The model achieved a sensitivity of 0.885 (95% CI: 0.869–0.901) and specificity of 0.857 (95% CI: 0.853–0.861) at the operating point selected in the development dataset (Table 2). The model achieved a sensitivity of 0.885 (95% CI: 0.869–0.901) and specificity of 0.857 (95% CI: 0.853–0.861) at the operating point selected in the development dataset (Table 2).

Performance of DL models across ethnicities

The model achieved an AUC of 0.936 (95% CI: 0.914–0.957) in Asian patients, 0.946 (95% CI: 0.935–0.956) in African patients, and 0.943 (95% CI: 0.937–0.949) in Caucasian patients (Table 2). The model achieved a sensitivity of 0.879 (95% CI: 0.821–0.927) in Asian patients, 0.893 (95% CI: 0.865–0.918) in African patients, and 0.885 (95% CI: 0.869–0.901) in Caucasian patients (Table 2). The model achieved a specificity of 0.854 (95% CI: 0.824–0.883) in Asian patients, 0.863 (95% CI: 0.844–0.882) in African patients, and 0.857 (95% CI: 0.853–0.861) in Caucasian patients (Table 2).

Performance of DL models across cameras

The model achieved an AUC of 0.906 (95% CI: 0.890–0.921) with a Topcon camera, 0.913 (95% CI: 0.895–0.931) with a Canon camera, and 0.947 (95% CI: 0.941–0.953) with a Zeiss camera (Table 2). The model achieved a sensitivity of 0.862 (95% CI: 0.844–0.880) with a Topcon camera, 0.867 (95% CI: 0.819–0.911) with a Canon camera, and 0.887 (95% CI: 0.871–0.903) with a Zeiss camera (Table 2). The model achieved a specificity of 0.857 (95% CI: 0.853–0.861) with a Topcon camera, 0.859 (95% CI: 0.855–0.863) with a Canon camera, and 0.858 (95% CI: 0.854–0.862) with a Zeiss camera (Table 2).

Performance of DL models across disease severity and type

The model achieved an AUC of 0.885 (95% CI: 0.867–0.903) in severe DR and 0.901 (95% CI: 0.882–0.919) in PDR, compared to an AUC of 0.943 (95% CI: 0.937–0.949) in those without severe DR or PDR. The model achieved a sensitivity of 0.838 (95% CI: 0.807–0.868) in severe DR and 0.854 (95% CI: 0.824–0.883) in PDR, compared to a sensitivity of 0.885 (95% CI: 0.869–0.901) in those without severe DR or PDR. The model achieved a specificity of 0.858 (95% CI: 0.854–0.862) in severe DR and 0.857 (95% CI: 0.853–0.861) in PDR, compared to a specificity of 0.857 (95% CI: 0.853–0.861) in those without severe DR or PDR. Other performance metrics by DR severity and type are shown in Table 2.

Discussion

In this study, we assessed the generalizability of a DL model for referable DR detection across ethnicities, imaging cameras, and disease severities and types in a large dataset of diabetic patients from the UK Biobank. We found that the performance of the DL model was comparable across Asian, Caucasian, and African ethnicities, but was lower in severe DR and PDR than in those without severe DR or PDR. The model also performed worse when applied to images from a camera not used in the development dataset. These results suggest that DL models for DR screening may not generalize across all ethnicities, cameras, and disease severities and types, and that such models should be tested in these settings before implementation.

Prior studies have shown promising performance of DL models for DR detection from fundus photographs, with several reporting sensitivities and specificities of over 90% in retrospective datasets[6–9]. However, it remains unclear how these models would perform in real-world clinical settings, where they may be used for DR screening. Prior studies have suggested that DL models may not generalize across ethnicities, imaging cameras, and disease severities and types[10–13], but these studies have been limited by the use of a single DL model, a single camera, or a single ethnicity, and have not assessed the impact of disease severity and type on model performance. Our study addresses these gaps in the literature by assessing the performance of a DL model across ethnicities, cameras, and disease severities and types in a large dataset of diabetic patients from the UK Biobank.

Our study shows that DL models for DR screening may not generalize across all ethnicities, cameras, and disease severities and types, and that such models should be tested in these settings before implementation. The lower performance in severe DR and PDR suggests that DL models may miss some of the most severe cases of DR, which are the cases that require urgent referral to an eye care specialist[3]. The lower performance when applied to images from a camera not used in the development dataset suggests that DL models may be sensitive to the specific characteristics of the cameras used in the development dataset, and that they may not generalize well to other cameras. The lower performance in some ethnicities suggests that DL models may be biased toward the ethnicities in the development dataset, and that they may not generalize well to other ethnicities.

Our study has several strengths. We used a large dataset of diabetic patients from the UK Biobank, which is a prospective cohort study of over 500,000 participants from the United Kingdom (UK) aged between 40 and 69 at recruitment[14]. The UK Biobank dataset is representative of the UK population in terms of ethnicity, socioeconomic status, and health status[15], and thus, our study results may be generalizable to other populations with similar characteristics. We used a state-of-the-art DL model for DR detection[16], and we assessed its performance across ethnicities, cameras, and disease severities and types, which has not been done in prior studies. We also used a large sample size to ensure that our results were precise and that we had enough power to detect differences in performance across subgroups.

Our study has several limitations. First, we used only one DL model for DR detection, and our results may not generalize to other DL models. Second, we used only one camera for the development dataset, and our results may not generalize to other cameras. Third, we used only one ethnicity for the development dataset, and our results may not generalize to other ethnicities. Fourth, we used only one disease severity and type for the development dataset, and our results may not generalize to other disease severities and types. Fifth, we used only one dataset for our study, and our results may not generalize to other datasets. Sixth, we did not assess the impact of other factors on the performance of the DL model, such as age, sex, and comorbidities, and these factors may affect the performance of the DL model.

In conclusion, we assessed the generalizability of a DL model for referable DR detection across ethnicities, imaging cameras, and disease severities and types in a large dataset of diabetic patients from the UK Biobank. We found that the performance of the DL model was comparable across Asian, Caucasian, and African ethnicities, but was lower in severe DR and PDR than in those without severe DR or PDR. The model also performed worse when applied to images from a camera not used in the development dataset. These results suggest that DL models for DR screening may not generalize across all ethnicities, cameras, and disease severities and types, and that such models should be tested in these settings before implementation.

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1.1 Introduction

Computer science is the study of algorithms, including their efficiency, their limitations, and their mathematical properties. Related fields examine computer hardware, programming languages, and computer applications in various disciplines. This chapter introduces some of the mathematical notation and techniques used in the study of algorithms, and it reviews some of the data structures that are used in the algorithms presented in later chapters.

1.2 Mathematical Notation and Functions

Let $\mathbb{R}$ denote the set of real numbers, $\mathbb{Z}$ denote the set of integers, and $\mathbb{N}$ denote the set of natural numbers (integers greater than or equal to zero). If x is a real number, then $\lfloor x \rfloor$ denotes the largest integer less than or equal to x, and $\lceil x \rceil$ denotes the smallest integer greater than or equal to x. For example, $\lfloor 3.14 \rfloor = 3$ and $\lceil 3.14 \rceil = 4$. Given two integers a and b, $a \bmod b$ denotes the remainder when a is divided by b, and $a \equiv b \pmod{c}$ means that $a - b$ is a multiple of c. For example, $7 \bmod 3 = 1$ and $7 \equiv 1 \pmod{3}$.

1.2.1 Logarithms and Exponents

Unless otherwise specified, all logarithms are base 2. For $x > 0$, $\log(x)$ denotes the base-2 logarithm of x. For $x > 0$ and $y > 0$, $\log(xy) = \log(x) + \log(y)$. For $x > 0$ and $y > 0$, $\log(x/y) = \log(x) - \log(y)$. For $x > 0$ and $y > 0$, $\log(x^y) = y \log(x)$. For $x > 0$ and $y > 0$, $x^{\log(y)} = y^{\log(x)}$. For $x > 0$ and $y > 0$, $\log(x^y) = y \log(x)$. For $x > 0$ and $y > 0$, $x^{\log(y)} = y^{\log(x)}$.

1.2.2 Factorial and Binomial Coefficients

For $n \in \mathbb{N}$, $n!$ denotes the factorial of n, defined as $n! = n \cdot (n-1) \cdot (n-2) \cdots 2 \cdot 1$. For $n \in \mathbb{N}$, $n! = n \cdot (n-1)!$. For $n \in \mathbb{N}$, $n! = \prod_{i=1}^{n} i

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Author: Lucy Stewart | Technology Editor | 27 May 2020

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Author affiliations:
[illegible]

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UNITED STATES DISTRICT COURT
DISTRICT OF MASSACHUSETTS

Dr. Sarah Mitchell-Rodriguez, Ms. Jessica Tran, Mr. Daniel Okafor, and Ms. Priyanka Desai,
Plaintiffs,
v.
Dr. Robert Hayes, in his official capacity as Commissioner of the Massachusetts Department of Public Health,
Defendant.

Civil Action No. 1:20-cv-12345-RGS

COMPLAINT FOR DECLARATORY AND INJUNCTIVE RELIEF

Jury Demand Not Applicable

Plaintiffs Dr. Sarah Mitchell-Rodriguez, Ms. Jessica Tran, Mr. Daniel Okafor, and Ms. Priyanka Desai (collectively, "Plaintiffs") bring this complaint against Defendant Dr. Robert Hayes, in his official capacity as Commissioner of the Massachusetts Department of Public Health (the "Commissioner"), and allege as follows:

I. INTRODUCTION

1. Plaintiffs seek to compel the Commissioner to comply with the Medicaid Act's requirement that state agencies "promptly" process Medicaid applications. Despite this unambiguous mandate, the Commissioner has adopted a practice of holding applications for weeks or months while requesting documents that are not required by law, leaving low-income residents without coverage for medically necessary care. This practice violates the Medicaid Act, 42 U.S.C. § 1396a(a)(8), and the Fourteenth Amendment to the United States Constitution. Plaintiffs seek a declaration that the challenged practice is unlawful and an injunction requiring the Commissioner to process applications promptly and to stop requesting documents not required by the Medicaid Act.

II. JURISDICTION AND VENUE

2. This Court has jurisdiction under 28 U.S.C. § 1331 (federal question) and 28 U.S.C. § 1343(a)(3) (civil rights). Plaintiffs' claims for declaratory and injunctive relief are authorized by 28 U.S.C. § 2201 and Rule 65 of the Federal Rules of Civil Procedure.

3. Venue is proper in this District under 28 U.S.C. § 1391(b)(2) because a substantial part of the events or omissions giving rise to the claims occurred in this District, including the denial of Plaintiff Jessica Tran's application while she was a resident of Worcester County, Massachusetts.

III. PARTIES

4. Plaintiff Dr. Sarah Mitchell-Rodriguez is a resident of Worcester County, Massachusetts. She is a practicing family medicine physician and the mother of a 9-year-old daughter who is enrolled in MassHealth. She brings this action on her own behalf and on behalf of her minor daughter.

5. Plaintiff Ms. Jessica Tran is a resident of Worcester County, Massachusetts. She is a single parent of two children and a part-time community college student. She applied for MassHealth in March 2024 and was denied coverage for her children after the Department requested documents she had already provided.

6. Plaintiff Mr. Daniel Okafor is a resident of Hampden County, Massachusetts. He is a self-employed carpenter and the father of a 14-year-old son who has asthma and requires regular medication. His family's application has been pending since February 2024.

7. Plaintiff Ms. Priyanka Desai is a resident of Middlesex County, Massachusetts. She is a graduate student and a mother of one. Her application was denied because she could not produce a document that the Department requested but that is not required by the Medicaid Act.

8. Defendant Dr. Robert Hayes is the Commissioner of the Massachusetts Department of Public Health. He is sued in his official capacity. The Commissioner is responsible for the administration of MassHealth and the processing of applications for benefits.

IV. LEGAL BACKGROUND

9. The Medicaid Act requires that a state plan provide that "all individuals wishing to make application for medical assistance under the plan shall have opportunity to file such applications, and that such applications shall be promptly processed." 42 U.S.C. § 1396a(a)(8). The Act does not authorize a state agency to deny or delay a complete application because it requests documents not required by the statute.

10. The Fourteenth Amendment to the United States Constitution guarantees that no state shall deprive any person of life, liberty, or property without due process of law. A state agency's practice of withholding benefits for reasons not authorized by the governing statute violates the right to due process.

V. FACTS

11. Dr. Mitchell-Rodriguez's daughter qualified for MassHealth when her family's income fell below the statutory threshold in January 2023. The family renewed coverage in December 2023 and provided all documents required by the Department. The renewal was approved in 14 days.

12. Ms. Tran applied for MassHealth for herself and her two children in March 2024. She provided her pay stubs, her children's birth certificates, and her lease. The Department requested a sixth document, a letter from her employer, that is not required by the Medicaid Act. The application has been pending for 5 months. Ms. Tran's children have been uninsured since their previous coverage ended in February 2024.

13. Mr. Okafor's son has been prescribed a daily inhaler and a rescue inhaler. The family's income is approximately $31,200 per year, well below the MassHealth threshold. The application has been pending since February 2024. The family has been paying out of pocket for the inhaler, which costs $[illegible] per month.

14. Ms. Desai's application was denied because she could not produce a document the Department requested. She then filed a new application and provided the document. The second application was approved within 10 days. The delay caused her to miss two months of coverage.

15. The Department's practice of requesting documents not required by the Medicaid Act has caused the denial, delay, and loss of coverage for hundreds of applicants in the past two years, according to Department of Public Health data produced in discovery in a separate matter.

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Ethnobotanical study of medicinal plants used by the inhabitants of Mirpur Sakro District, Tharparkar, Sindh, Pakistan

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Abstract: This study aimed to investigate the ethnobotanical uses of medicinal plants used by the inhabitants of Mirpur Sakro District, Tharparkar, Sindh, Pakistan. Data were collected through questionnaires and interviews with local people. A total of 10 medicinal plants belonging to 10 families were recorded. The most frequently used plants were Calotropis procera and Tribulus terrestris. The results indicate that the local people rely on these plants for various ailments, including digestive disorders, skin diseases, and respiratory problems. The study highlights the importance of traditional knowledge in the region and suggests the need for further pharmacological and phytochemical investigations to validate the traditional uses and ensure sustainable utilization of these resources.

Keywords: ethnobotany, medicinal plants, Tharparkar, Sindh, Pakistan, traditional knowledge, Calotropis procera, Tribulus terrestris

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7.2 Cluster Representations

7.2.1 Centroids and Prototypes (Medoids)

Definition. Let $C_k = \{x_1, x_2, \ldots, x_{n_k}\}$ be a cluster with n_k points. The **centroid** (or **mean**) of C_k is the point

$$\mu_k = \frac{1}{n_k} \sum_{i=1}^{n_k} x_i$$

Example 7.2.1. Consider the cluster $C_1 = \{(1,2), (2,3), (3,1)\}$. Then

$$\mu_1 = \frac{1}{3}\left((1,2) + (2,3) + (3,1)\right) = \frac{1}{3}(6,6) = (2,2)$$

Proposition 7.2.1. The centroid μ_k minimizes the sum of squared distances to the points in C_k:

$$\mu_k = \arg\min_{\mu \in \mathbb{R}^d} \sum_{i=1}^{n_k} \|x_i - \mu\|^2$$

Proof. Expand the objective:

$$\sum_{i=1}^{n_k} \|x_i - \mu\|^2 = \sum_{i=1}^{n_k} \|x_i\|^2 - 2\mu^T \sum_{i=1}^{n_k} x_i + n_k \|\mu\|^2$$

This is a quadratic in μ. Setting the gradient to zero:

$$\nabla_\mu \left[\sum_{i=1}^{n_k} \|x_i - \mu\|^2\right] = -2\sum_{i=1}^{n_k} x_i + 2n_k\mu = 0$$

$$\Rightarrow \mu = \frac{1}{n_k}\sum_{i=1}^{n_k} x_i$$

Remark. The centroid is not necessarily a data point. For example, the centroid of $\{(0,0), (1,0)\}$ is $(0.5, 0)$, which is not a data point.

Definition. The **medoid** of C_k is the point $m_k \in C_k$ that minimizes the sum of distances to all other points in C_k:

$$m_k = \arg\min_{x \in C_k} \sum_{i=1}^{n_k} \|x_i - x\|$$

Remark. Unlike the centroid, the medoid must be a data point. This makes it more robust to outliers and applicable to non-Euclidean distances (e.g., edit distance for strings, Jaccard distance for sets).

Example 7.2.2. Consider the cluster $C_1 = \{(1,2), (2,3), (3,1), (100,100)\}$. The centroid is $\mu_1 = (26.5, 26.5)$, which is far from most points. The medoid is $m_1 = (2,3)$, which is much closer to the majority of points.

7.2.2 Medoids

Definition. The **medoid** of C_k is the point $m_k \in C_k$ that minimizes the sum of distances to all other points in C_k:

$$m_k = \arg\min_{x \in C_k} \sum_{i=1}^{n_k} \|x_i - x\|$$

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7.2.3 Medoids

Definition. The **medoid** of C_k is the point $m_k \in C_k$ that minimizes the sum of distances to all other points in C_k:

$$m_k = \arg\min_{x \in C_k} \sum_{i=1}^{n_k} \|x_i - x\|$$

Remark. Unlike the centroid, the medoid must be a data point. This makes it more robust to outliers and applicable to non-Euclidean distances (e.g., edit distance for strings, Jaccard distance for sets).

Example 7.2.2. Consider the cluster $C_1 = \{(1,2), (2,3), (3,1), (100,100)\}$. The centroid is $\mu_1 = (26.5, 26.5)$, which is far from most points. The medoid is $m_1 = (2,3)$, which is much closer to the majority of points.

7.2.4 Medoids

Definition. The **medoid** of C_k is the point $m_k \in C_k$ that minimizes the sum of distances to all other points in C_k:

$$m_k = \arg\min_{x \in C_k} \sum_{i=1}^{n_k} \|x_i - x\|$$

Remark. Unlike the centroid, the medoid must be a data point. This makes it more robust to outliers and applicable to non-Euclidean distances (e.g., edit distance for strings, Jaccard distance for sets).

Example 7.2.2. Consider the cluster $C_1 = \{(1,2), (2,3), (3,1), (100,100)\}$. The centroid is $\mu_1 = (26.5, 26.5)$, which is far from most points. The medoid is $m_1 = (2,3)$, which is much closer to the majority of points.

7.2.5 Medoids

Definition. The **medoid** of C_k is the point $m_k \in C_k$ that minimizes the sum of distances to all other points in C_k:

$$m_k = \arg\min_{x \in C_k} \sum_{i=1}^{n_k} \|x_i - x\|$$

Remark. Unlike the centroid, the medoid must be a data point. This makes it more robust to outliers and applicable to non-Euclidean distances (e.g., edit distance for strings, Jaccard distance for sets).

Example 7.2.2. Consider the cluster $C_1 = \{(1,2), (2,3), (3,1), (100,100)\}$. The centroid is $\mu_1 = (26.5, 26.5)$, which is far from most points. The medoid is $m_1 = (2,3)$, which is much closer to the majority of points.

7.2.6 Medoids

Definition. The **medoid** of C_k is the point $m_k \in C_k$ that minimizes the sum of distances to all other points in C_k:

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7.2.9 Medoids

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7.2.10 Medoids

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Lesson Twenty-one | Matrix Multiplication and Inverses

Matrix multiplication is the heart of linear algebra. It is how we combine two transformations into one, how we solve systems of equations, how we compute eigenvalues, how we train neural networks. Today we derive it from first principles and understand why it works the way it does.

Author: Prof. Elena V. Kozlova | Computational Mathematics Group

---

1. Motivation: Why Do We Need Matrix Multiplication?

Suppose we have two linear transformations $T_1 : \mathbb{R}^n \to \mathbb{R}^m$ and $T_2 : \mathbb{R}^m \to \mathbb{R}^p$. Their composite $T_2 \circ T_1 : \mathbb{R}^n \to \mathbb{R}^p$ is also linear. If A represents T_1 and B represents T_2, we want a matrix C such that

$$(T_2 \circ T_1)(\mathbf{x}) = C\mathbf{x} \quad \text{for all } \mathbf{x} \in \mathbb{R}^n.$$

We want C to be the product BA.

---

2. Definition

Definition 2.1 (Matrix Multiplication). Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$. The product $C = AB \in \mathbb{R}^{m \times p}$ is defined by

$$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}, \qquad 1 \le i \le m, \; 1 \le j \le p.$$

---

3. Properties

Theorem 3.1 (Properties of Matrix Multiplication). Assuming the sizes match, the following hold:

1. **Associativity:** $(AB)C = A(BC)$.
2. **Distributivity:** $A(B + C) = AB + AC$ and $(A + B)C = AC + BC$.
3. $(AB)^T = B^T A^T$.
4. $\det(AB) = \det(A)\det(B)$.
5. $\operatorname{tr}(AB) = \operatorname{tr}(BA)$.

Proof. (1) Associativity follows from the fact that $(AB)C$ and $A(BC)$ both compute the (i,j) entry of the triple product $\sum_{k,l} a_{ik} b_{kl} c_{lj}$. (2) Distributivity follows from the distributivity of scalar multiplication. (3) $(AB)^T_{ij} = (AB)_{ji} = \sum_k a_{jk} b_{ki} = \sum_k b_{ki} a_{jk} = (B^T A^T)_{ij}$. (4) Taken for granted for now; we will prove it in Lesson Twenty-two. (5) $\operatorname{tr}(AB) = \sum_i \sum_k a_{ik} b_{ki} = \sum_k \sum_i b_{ki} a_{ik} = \operatorname{tr}(BA)$. □

Warning. Matrix multiplication is not commutative. In general $AB \neq BA$. Even when both products exist and have the same size, $AB = BA$ only if A and B commute.

---

4. Inverses

Definition 4.1 (Inverse). A square matrix $A \in \mathbb{R}^{n \times n}$ is invertible if there exists A^{-1} such that

$$AA^{-1} = A^{-1}A = I_n.$$

Theorem 4.2 (Uniqueness). If A is invertible, A^{-1} is unique.

Proof. If B and C both satisfy $AB = BA = I$ and $AC = CA = I$, then $B = BI = B(AC) = (BA)C = IC = C$. □

Theorem 4.3 (Properties of Inverses).

1. $(AB)^{-1} = B^{-1}A^{-1}$.
2. $(A^T)^{-1} = (A^{-1})^T$.
3. $\det(A^{-1}) = 1/\det(A)$.

---

5. Exercises

Exercise 5.1. Compute $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$ by hand.

Exercise 5.2. Show that $(AB)^{-1} = B^{-1}A^{-1}$ when A and B are invertible.

Exercise 5.3. Prove that $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ without using the formula $c_{ij} = \sum_k a_{ik} b_{kj}$.

---

6. Closing Remarks

Matrix multiplication is the engine of linear algebra. Every theorem we will prove in the next few lessons depends on it. Lesson Twenty-two: determinants. Lesson Twenty-three: eigenvalues. Lesson Twenty-four: the spectral theorem. Lesson Twenty-five: applications to machine learning.

Lost Museum Direct Acquisition Log

Agent: Claude (Anthropic) via Claude Code, Opus 4.6
Date: 2026-03-14
Mission: Acquire 3D models for the Lost Museum collection
Source: Smithsonian 3D Collections (https://3d.si.edu/)
Method: Direct HTTP download via Smithsonian Open Access API (https://api.si.edu/openaccess/api/v1.0/search?q=type:3d_model&rows=10&api_key=DEMO_KEY)
Rights: CC0 only

Object	Rights	Size
Ceramic Vessel	CC0	0.0 MB
Ceramic Vessel	CC0	0.0 MB
Ceramic Vessel	CC0	0.0 MB
Ceramic Vessel	CC0	0.0 MB
Ceramic Vessel	CC0	0.0 MB
Ceramic Vessel	CC0	0.0 MB
Ceramic Vessel	CC0	0.0 MB
Ceramic Vessel	CC0	0.0 MB
Ceramic Vessel	CC0	0.0 MB
Ceramic Vessel	CC0	0.0 MB

Total: 10 models, 0.0 MB

第一章 Flutter 基础概念和设计

1.1 Widget 是一切 UI 的基本单位

Flutter 的核心哲学是一切皆 Widget。

```
Widget ::= StatelessWidget | StatefulWidget | InheritedWidget | RenderObjectWidget
StatelessWidget ::= 不可变，build() 只依赖 Widget 自身和 BuildContext
StatefulWidget ::= 可变，通过 State 对象持有状态，State.build() 依赖 Widget 自身、BuildContext 和 State 内部状态
BuildContext ::= 指向 Element 树中对应节点的只读句柄
```

$$\mathrm{Widget} \xrightarrow{\mathrm{inflate}} \mathrm{Element} \xrightarrow{\mathrm{mount}} \mathrm{RenderObject}$$

1.2 三棵树的关系

$$\mathrm{Widget\ Tree} \xrightarrow{\mathrm{inflate}} \mathrm{Element\ Tree} \xrightarrow{\mathrm{mount}} \mathrm{RenderObject\ Tree}$$

$$\mathrm{Widget\ Tree} \xrightarrow{\mathrm{diff}} \mathrm{Element\ Tree} \xrightarrow{\mathrm{layout,\ paint}} \mathrm{Screen}$$

$$\mathrm{Widget} \xrightarrow{\mathrm{inflate}} \mathrm{Element} \xrightarrow{\mathrm{mount}} \mathrm{RenderObject}$$

1.3 Widget 不可变性

Widget 一旦创建，其属性全部 `final`。`build()` 每次调用返回全新 Widget 实例，旧实例丢弃。Element 通过 `canUpdate(oldWidget, newWidget)` 判断能否复用：`runtimeType` 相同且 `key` 相同则复用，否则销毁重建。

1.4 Element 生命周期

```
Element ::= 挂载 (mount) → 活跃 (active) → 卸载 (unmount)
mount ::= 调用 Widget.createElement() → 子节点递归 mount → 插入 RenderObject 树
unmount ::= 递归 unmount 子节点 → 从 RenderObject 树移除 → 释放 State.dispose()
```

1.5 RenderObject 层

```
RenderObject ::= 布局 (layout) → 绘制 (paint) → 合成 (composite)
layout ::= 父节点传入 BoxConstraints → 子节点计算自己的 Size → 父节点决定子节点的 offset
paint ::= 生成 Layer 树 → 提交给 Skia/Impeller 渲染引擎
```

$$\mathrm{Widget} \xrightarrow{\mathrm{inflate}} \mathrm{Element} \xrightarrow{\mathrm{mount}} \mathrm{RenderObject}$$

1.6 Key 的作用

```
Key ::= 唯一标识符，用于 Element 复用判断
LocalKey ::= ValueKey | ObjectKey | UniqueKey
GlobalKey ::= 全局唯一，跨 Widget 树移动时保持 State 不丢失
```

$$\mathrm{Widget} \xrightarrow{\mathrm{inflate}} \mathrm{Element} \xrightarrow{\mathrm{mount}} \mathrm{RenderObject}$$

1.7 BuildContext 的本质

```
BuildContext ::= 指向 Element 树中对应节点的只读句柄
context.findAncestorWidgetOfExactType<T>() ::= 沿 Element 树向上遍历，返回类型匹配的祖先 Widget
context.findRenderObject() ::= 返回 Element 对应的 RenderObject
```

$$\mathrm{Widget} \xrightarrow{\mathrm{inflate}} \mathrm{Element} \xrightarrow{\mathrm{mount}} \mathrm{RenderObject}$$

MR. TIGER MAGNIFICENT

Tiger Tiger, burning bright

In the forests of the night,

What immortal hand or eye

Could frame thy fearful symmetry?

In what distant deeps or skies

Burnt the fire of thine eyes?

On what wings dare he aspire?

What the hand dare seize the fire?

And what shoulder, and what art,

Could twist the sinews of thy heart?

And when thy heart began to beat,

What dread hand? And what dread feet?

What the hammer? What the chain,

In what furnace was thy brain?

What the anvil? What dare

Grasp thy deadly terrors clare?

When the stars threw down their spears

And water'd heaven with their tears,

Did he smile his work to see?

Did he who made the lamb make thee?

Tiger Tiger, burning bright

In the forests of the night,

What immortal hand or eye

Could frame thy fearful symmetry?

$$f \circ \mathcal{F}$$

$$\mathcal{F}$$

$$\mathcal{F}^{-1}$$

$$\mathcal{F}^{-1}(\mathcal{F}(f))$$

$$\mathcal{F}^{-1}(\mathcal{F}(f))(x)$$

$$\mathcal{F}^{-1}(\mathcal{F}(f))(x) = f(x)$$

$$\mathcal{F}(f)(\xi) = \int_{-\infty}^{\infty} f(x)\, e^{-2\pi i x \xi}\,dx$$

$$\mathcal{F}^{-1}(f)(x) = \int_{-\infty}^{\infty} f(\xi)\, e^{2\pi i x \xi}\,d\xi$$

$$\mathcal{F}(f)(\xi) = \int_{-\infty}^{\infty} f(x)\, e^{-2\pi i x \xi}\,dx$$

$$\mathcal{F}^{-1}(f)(x) = \int_{-\infty}^{\infty} f(\xi)\, e^{2\pi i x \xi}\,d\xi$$

$$\mathcal{F}^{-1}(\mathcal{F}(f))(x) = f(x)

Ans:

Ans:

Ans:

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1.1 Lorem ipsum, Dolorem ipsum, Amet, Consectetur

1.1.1 Lorem ipsum, Dolorem ipsum, Amet, Consectetur, Adipiscing

1.1.1.1 Lorem ipsum, Dolorem ipsum, Amet, Consectetur, Adipiscing, Elit

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Q1.

Subject:Mathematics Chapter:Trigonometry Topic:Trigonometric Ratios and Identities

Question:If $\sin\theta + \cos\theta = \sqrt{2}$, then $\sin\theta - \cos\theta =$ Options:(1) $\sqrt{2}$ (2) $\sqrt{3}$ (3) 0 (4) 1

Answer:(3) 0

Solution:$(\sin\theta + \cos\theta)^2 = 2 \Rightarrow \sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta = 2$

$\Rightarrow 1 + 2\sin\theta\cos\theta = 2 \Rightarrow 2\sin\theta\cos\theta = 1$

$(\sin\theta - \cos\theta)^2 = \sin^2\theta + \cos^2\theta - 2\sin\theta\cos\theta = 1 - 1 = 0$

$\Rightarrow \sin\theta - \cos\theta = 0$

Q2.

Subject:Mathematics

Chapter:Trigonometry

Topic:Trigonometric Ratios

Question:If $\sin\theta = \frac{3}{5}$, $\cos\theta = \frac{4}{5}$, then $\tan\theta =$

Options:(1) $\frac{3}{4}$ (2) $\frac{4}{3}$ (3) $\frac{5}{3}$ (4) $\frac{5}{4}$

Answer:(1) $\frac{3}{4}$

Solution:$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{3/5}{4/5} = \frac{3}{4}$

Q3.

Subject:Mathematics

Chapter:Trigonometry

Topic:Trigonometric Ratios

Question:If $\sin\theta = \frac{1}{2}$, then $\cos\theta =$

Options:(1) $\frac{1}{2}$ (2) $\frac{\sqrt{3}}{2}$ (3) $\frac{1}{\sqrt{2}}$ (4) 1

Answer:(2) $\frac{\sqrt{3}}{2}$

Solution:$\cos\theta = \sqrt{1 - \sin^2\theta} = \sqrt{1 - \frac{1}{4}} = \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}$

Q4.

Subject:Mathematics

Chapter:Trigonometry

Topic:Trigonometric Ratios

Question:If $\sin\theta = \frac{1}{2}$, then $\sec\theta =$

Options:(1) $\frac{2}{\sqrt{3}}$ (2) $\frac{1}{\sqrt{3}}$ (3) $\frac{\sqrt{3}}{2}$ (4) $\frac{1}{2}$

Answer:(1) $\frac{2}{\sqrt{3}}$

Solution:$\sec\theta = \frac{1}{\cos\theta} = \frac{1}{\sqrt{3}/2} = \frac{2}{\sqrt{3}}$

Q5.

Subject:Mathematics

Chapter:Trigonometry

Topic:Trigonometric Ratios

Question:If $\sin\theta = \frac{1}{2}$, then $\tan\theta =$

Options:(1) $\frac{1}{\sqrt{3}}$ (2) $\frac{\sqrt{3}}{2}$ (3) $\frac{1}{\sqrt{2}}$ (4) 1

Answer:(1) $\frac{1}{\sqrt{3}}$

Solution:$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{1/2}{\sqrt{3}/2} = \frac{1}{\sqrt{3}}$

Q6.

Subject:Mathematics

Chapter:Trigonometry

Topic:Trigonometric Ratios

Question:If $\sin\theta = \frac{1}{2}$, then $\cot\theta =$

Options:(1) $\sqrt{3}$ (2) $\frac{1}{\sqrt{3}}$ (3) $\frac{\sqrt{3}}{2}$ (4) $\frac{1}{2}$

Answer:(1) $\sqrt{3}$

Solution:$\cot\theta = \frac{1}{\tan\theta} = \frac{1}{1/\sqrt{3}} = \sqrt{3}$

Q7.

Subject:Mathematics

Chapter:Trigonometry

Topic:Trigonometric Ratios

Question:If $\sin\theta = \frac{1}{2}$, then $\operatorname{cosec}\theta =$

Options:(1) $\frac{1}{2}$ (2) $\frac{\sqrt{3}}{2}$ (3) $\frac{1}{\sqrt{2}}$ (4) 2

Answer:(4) 2

Solution:$\operatorname{cosec}\theta = \frac{1}{\sin\theta} = \frac{1}{1/2} = 2$

Q8.

Subject:Mathematics

Chapter:Trigonometry

Topic:Trigonometric Ratios

Question:If $\sin\theta = \frac{1}{2}$, then $\sec\theta =$

Options:(1) $\frac{2}{\sqrt{3}}$ (2) $\frac{1}{\sqrt{3}}$ (3) $\frac{\sqrt{3}}{2}$ (4) $\frac{1}{2}$

Answer:(1) $\frac{2}{\sqrt{3}}$

Solution:$\sec\theta = \frac{1}{\cos\theta} = \frac{1}{\sqrt{3}/2} = \frac{2}{\sqrt{3}}$

Q9.

Subject:Mathematics

Chapter:Trigonometry

Topic:Trigonometric Ratios

Question:If $\sin\theta = \frac{1}{2}$, then $\tan\theta =$

Options:(1) $\frac{1}{\sqrt{3}}$ (2) $\frac{\sqrt{3}}{2}$ (3) $\frac{1}{\sqrt{2}}$ (4) 1

Answer:(1) $\frac{1}{\sqrt{3}}$

Solution:$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{1/2}{\sqrt{3}/2} = \frac{1}{\sqrt{3}}$

Q10.

Subject:Mathematics

Chapter:Trigonometry

Topic:Trigonometric Ratios

Question:If $\sin\theta = \frac{1}{2}$, then $\cot\theta =$

Options:(1) $\sqrt{3}$ (2) $\frac{1}{\sqrt{3}}$ (3) $\frac{\sqrt{3}}{2}$ (4) $\frac{1}{2}$

Answer:(1) $\sqrt{3}$

Solution:$\cot\theta = \frac{1}{\tan\theta} = \frac{1}{1/\sqrt{3}} = \sqrt{3}$

Disclaimer:This PDF is generated from the Question Bank. Please verify critical details before use.

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Abstract

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MR. TIGGY-WINKLE'S

Mr. Tiggy-Winkle's Wash-Tub (Frontispiece)

CHAPTER I

Mrs. Tiggy-Winkle's Wash-Tub (Frontispiece) 1

Mrs. Tiggy-Winkle's Cottage 2

Mrs. Tiggy-Winkle Ironing 3

Mrs. Tiggy-Winkle Hanging Up Her Lace 4

Mrs. Tiggy-Winkle's Kitchen 5

Mrs. Tiggy-Winkle's Cottage Door 6

CHAPTER I

Mrs. Tiggy-Winkle's Wash-Tub

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MR-TCOT-MATHEMATICS

Mr-Tcot-Mathematics-Mathematics-Mathematics

Mr-Tcot-Mathematics-Mathematics-Mathematics

Example 1.1. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ then $A = B$.

Example 1.2. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ then $A = B$.

Example 1.3. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ then $A = B$.

Example 1.4. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ then $A = B$.

Example 1.5. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ then $A = B$.

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Example 1.132. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ then $A = B$.

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Example 1.134. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ then $A = B$.

Example 1.135. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ then $A = B$.

Example 1.136. Let $A = \begin{

Gaggenau Service Manual

Gaggenau Service Manual

Gaggenau Service Manual Gaggenau Service Manual Gaggenau Service Manual

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Revised: 13 November 2017
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Author: Dr. Lucas Thompson
Department: Chemical Engineering
Contact: lucas.thompson@chemeng.edu

Abstract: This document provides a comprehensive analysis of the Antoine equation, a fundamental tool in chemical engineering for calculating vapor pressure. We begin with the derivation of the equation from the Clausius-Clapeyron relation, discuss its parameters and their physical significance, and provide a step-by-step guide for fitting the equation to experimental data using nonlinear regression. The document includes a critical discussion of the equation's limitations, including its accuracy over different temperature ranges and its extrapolation behavior, and concludes with a review of alternative models such as the Wagner equation and the Clarke-Glew equation. This work is intended for students and practitioners who need a robust understanding of vapor pressure calculations.

Keywords: Antoine equation, vapor pressure, thermodynamics, parameter fitting, nonlinear regression, chemical engineering, Clausius-Clapeyron, Wagner equation, Clarke-Glew equation

1.1 Introduction

$$e = V \cdot I$$

1.1.1

Subsection 1.1.1
Subsubsection 1.1.1.1
Paragraph 1.1.1.1.1

1.1.2

Subsection 1.1.2
Subsubsection 1.1.2.1
Paragraph 1.1.2.1.1

1.1.3

Subsection 1.1.3
Subsubsection 1.1.3.1
Paragraph 1.1.3.1.1

1.1.4

Subsection 1.1.4
Subsubsection 1.1.4.1
Paragraph 1.1.4.1.1

1.2 Section 1.2

$$\lim_{x \to 0} \frac{\sin x}{x} = 1$$

1.2.1

Subsection 1.2.1
Subsubsection 1.2.1.1
Paragraph 1.2.1.1.1

1.2.2

Subsection 1.2.2
Subsubsection 1.2.2.1
Paragraph 1.2.2.1.1

1.2.3

Subsection 1.2.3
Subsubsection 1.2.3.1
Paragraph 1.2.3.1.1

1.2.4

Subsection 1.2.4
Subsubsection 1.2.4.1
Paragraph 1.2.4.1.1

1.3 Section 1.3

$$\begin{aligned} \frac{\partial \rho}{\partial t} &= \\ \frac{\partial \rho u_x}{\partial x} &= \\ \frac{\partial \rho u_y}{\partial y} &= \\ \frac{\partial \rho u_z}{\partial z} &= \end{aligned}$$

1.3.1

Subsection 1.3.1
Subsubsection 1.3.1.1
Paragraph 1.3.1.1.1

1.3.2

Subsection 1.3.2
Subsubsection 1.3.2.1
Paragraph 1.3.2.1.1

1.3.3

Subsection 1.3.3
Subsubsection 1.3.3.1
Paragraph 1.3.3.1.1

1.3.4

Subsection 1.3.4
Subsubsection 1.3.4.1
Paragraph 1.3.4.1.1

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1.1 Introduction

Consider the system of equations

$$\begin{aligned} x_1 + 2x_2 &= 3 \\ 3x_1 - x_2 &= 1 \end{aligned}$$

Any pair of numbers (x_1, x_2) satisfying both equations is a solution. The set of all such solutions is the solution set.

1.2 Gaussian Elimination

Step 1: Multiply the first equation by 3 and subtract from the second:

$$\begin{aligned} x_1 + 2x_2 &= 3 \\ -7x_2 &= -8 \end{aligned}$$

Step 2: Back substitution:

$$x_2 = \frac{8}{7}, \quad x_1 = 3 - 2x_2 = \frac{5}{7}$$

1.3 Matrix Notation

Write the system in matrix form:

$$\begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

1.4 Exercises

1. Solve the system $x_1 + x_2 = 2$, $2x_1 - x_2 = 0$ by elimination.
2. Write the system in matrix form and verify your solution satisfies $A\mathbf{x} = \mathbf{b}$.
3. Show that the system $x_1 + 2x_2 = 3$, $2x_1 + 4x_2 = 6$ has infinitely many solutions.

2.1 Vectors

Vectors are ordered lists of numbers. Here, $\mathbf{v} = (3, 1, -2)$ and $\mathbf{w} = (1, 0, 4)$ are two vectors in $\mathbb{R}^3$.

2.2 Vector Addition

Vector addition is performed component-wise:

$$\mathbf{v} + \mathbf{w} = (3+1, 1+0, -2+4) = (4, 1, 2)$$

2.3 Scalar Multiplication

Multiplying a vector by a scalar c scales each component:

$$2\mathbf{v} = (6, 2, -4)$$

2.4 Exercises

1. Compute $\mathbf{v} - \mathbf{w}$ and $3\mathbf{v} + 2\mathbf{w}$.
2. Find the scalar c such that $c\mathbf{v} = \mathbf{w}$, if possible.
3. Show that $\mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v}$ (commutativity).

3.1 Dot Product

The dot product of two vectors is a scalar:

$$\mathbf{v} \cdot \mathbf{w} = 3 \cdot 1 + 1 \cdot 0 + (-2) \cdot 4 = -5$$

3.2 Exercises

1. Compute $\mathbf{v} \cdot \mathbf{v}$.
2. Show that $\mathbf{v} \cdot \mathbf{w} = \mathbf{w} \cdot \mathbf{v}$.

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Chapter 1 Introduction

1.1 Overview

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos qui ratione voluptatem sequi nesciunt. Neque porro quisquam est, qui dolorem ipsum quia dolor sit amet, consectetur, adipisci velit, sed quia non numquam eius modi tempora incidunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim ad minima veniam, quis nostrum exercitationem ullam corporis suscipit laboriosam, nisi ut aliquid ex ea commodi consequatur? Quis autem vel eum iure reprehenderit qui in ea voluptate velit esse quam nihil molestiae consequatur, vel illum qui dolorem eum fugiat quo voluptas nulla pariatur?

1.2 Scope

At vero eos et accusamus et iusto odio dignissimos ducimus qui blanditiis praesentium voluptatum deleniti atque corrupti quos dolores et quas molestias excepturi sint occaecati cupiditate non provident, similique sunt in culpa qui officia deserunt mollitia animi, id est laborum et dolorum fuga. Et harum quidem rerum facilis est et expedita distinctio. Nam libero tempore, cum soluta nobis est eligendi optio cumque nihil impedit quo minus id quod maxime placeat facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum hic tenetur a sapiente delectus, ut aut reiciendis voluptatibus maiores alias consequatur aut perferendis doloribus asperiores repellat.

```
SELECT COUNT(DISTINCT c.customer_id) FROM customers c
```

$$\int_{\Omega} \nabla \cdot \mathbf{F}(\mathbf{x})\,d\mathbf{x} = \int_{\partial\Omega} \mathbf{F}(\mathbf{x}) \cdot d\mathbf{S}(\mathbf{x}) \tag{1.1}$$

```
SELECT
    DATEDIFF(
        MAX(o.order_date),
        MIN(o.order_date)
    ) AS days_between_first_and_last
FROM orders o
GROUP BY o.customer_id
HAVING COUNT(o.order_id) = 1
ORDER BY days_between_first_and_last DESC
LIMIT 10;
```

$$\int_{\Omega} \nabla \cdot \mathbf{F}(\mathbf{x})\,d\mathbf{x} = \int_{\partial\Omega} \mathbf{F}(\mathbf{x}) \cdot d\mathbf{S}(\mathbf{x}) \tag{1.2}$$

```
SELECT
    DATEDIFF(
        MAX(o.order_date),
        MIN(o.order_date)
    ) AS days_between_first_and_last
FROM orders o
GROUP BY o.customer_id
HAVING COUNT(o.order_id) = 1
ORDER BY days_between_first_and_last DESC
LIMIT 10;
```

$$\int_{\Omega} \nabla \cdot \mathbf{F}(\mathbf{x})\,d\mathbf{x} = \int_{\partial\Omega} \mathbf{F}(\mathbf{x}) \cdot d\mathbf{S}(\mathbf{x}) \tag{1.3}$$

```
SELECT
    DATEDIFF(
        MAX(o.order_date),
        MIN(o.order_date)
    ) AS days_between_first_and_last
FROM orders o
GROUP BY o.customer_id
HAVING COUNT(o.order_id) = 1
ORDER BY days_between_first_and_last DESC
LIMIT 10;
```

$$\int_{\Omega} \nabla \cdot \mathbf{F}(\mathbf{x})\,d\mathbf{x} = \int_{\partial\Omega} \mathbf{F}(\mathbf{x}) \cdot d\mathbf{S}(\mathbf{x}) \tag{1.4}$$

```
SELECT
    DATEDIFF(
        MAX(o.order_date),
        MIN(o.order_date)
    ) AS days_between_first_and_last
FROM orders o
GROUP BY o.customer_id
HAVING COUNT(o.order_id) = 1
ORDER BY days_between_first_and_last DESC
LIMIT 10;
```

$$\int_{\Omega} \nabla \cdot \mathbf{F}(\mathbf{x})\,d\mathbf{x} = \int_{\partial\Omega} \mathbf{F}(\mathbf{x}) \cdot d\mathbf{S}(\mathbf{x}) \tag{1.5}$$

```
SELECT
    DATEDIFF(
        MAX(o.order_date),
        MIN(o.order_date)
    ) AS days_between_first_and_last
FROM orders o
GROUP BY o.customer_id
HAVING COUNT(o.order_id) = 1
ORDER BY days_between_first_and_last DESC
LIMIT 10;
```

$$\int_{\Omega} \nabla \cdot \mathbf{F}(\mathbf{x})\,d\mathbf{x} = \int_{\partial\Omega} \mathbf{F}(\mathbf{x}) \cdot d\mathbf{S}(\mathbf{x}) \tag{1.6}$$

```
SELECT
    DATEDIFF(
        MAX(o.order_date),
        MIN(o.order_date)
    ) AS days_between_first_and_last
FROM orders o
GROUP BY o.customer_id
HAVING COUNT(o.order_id) = 1
ORDER BY days_between_first_and_last DESC
LIMIT 10;
```

$$\int_{\Omega} \nabla \cdot \mathbf{F}(\mathbf{x})\,d\mathbf{x} = \int_{\partial\Omega} \mathbf{F}(\mathbf{x}) \cdot d\mathbf{S}(\mathbf{x}) \tag{1.7}$$

```
SELECT
    DATEDIFF(
        MAX(o.order_date),
        MIN(o.order_date)
    ) AS days_between_first_and_last
FROM orders o
GROUP BY o.customer_id
HAVING COUNT(o.order_id) = 1
ORDER BY days_between_first_and_last DESC
LIMIT 10;
```

$$\int_{\Omega} \nabla \cdot \mathbf{F}(\mathbf{x})\,d\mathbf{x} = \int_{\partial\Omega} \mathbf{F}(\mathbf{x}) \cdot d\mathbf{S}(\mathbf{x}) \tag{1.8}$$

```
SELECT
    DATEDIFF(
        MAX(o.order_date),
        MIN(o.order_date)
    ) AS days_between_first_and_last
FROM orders o
GROUP BY o.customer_id
HAVING COUNT(o.order_id) = 1
ORDER BY days_between_first_and_last DESC
LIMIT 10;
```

$$\int_{\Omega} \nabla \cdot \mathbf{F}(\mathbf{x})\,d\mathbf{x} = \int_{\partial\Omega} \mathbf{F}(\mathbf{x}) \cdot d\mathbf{S}(\mathbf{x}) \tag{1.9}$$

```
SELECT
    DATEDIFF(
        MAX(o.order_date),
        MIN(o.order_date)
    ) AS days_between_first_and_last
FROM orders o
GROUP BY o.customer_id
HAVING COUNT(o.order_id) = 1
ORDER BY days_between_first_and_last DESC
LIMIT 10;
```

$$\int_{\Omega} \nabla \cdot \mathbf{F}(\mathbf{x})\,d\mathbf{x} = \int_{\partial\Omega} \mathbf{F}(\mathbf{x}) \cdot d\mathbf{S}(\mathbf{x}) \tag{1.10}$$

```
SELECT
    DATEDIFF(
        MAX(o.order_date),
        MIN(o.order_date)
    ) AS days_between_first_and_last
FROM orders o
GROUP BY o.customer_id
HAVING COUNT(o.order_id) = 1
ORDER BY days_between_first_and_last DESC
LIMIT 10;
```

Sample Distribution

Sample Distribution of Sample Means, $\bar{x}$ - Central Limit Theorem

Sample Distribution of Sample Proportions, $\hat{p}$ - Central Limit Theorem

```
MOV [ESP+0x0C],EDX
MOV [ESP+0x08],ECX
CALL <JMP.&MSVCP140.?Xlength@std@@YAXPBD@Z>
```

```
MOV EAX,[ESP+0x4]
MOV [EAX],[ESP+0x8]
MOV [EAX+0x4],[ESP+0xC]
MOV [EAX+0x8],[ESP+0x10]
MOV [EAX+0xC],[ESP+0x14]
MOV [EAX+0x10],[ESP+0x18]
MOV [EAX+0x14],[ESP+0x1C]
MOV [EAX+0x18],[ESP+0x20]
MOV [EAX+0x1C],[ESP+0x24]
MOV [EAX+0x20],[ESP+0x28]
MOV [EAX+0x24],[ESP+0x2C]
MOV [EAX+0x28],[ESP+0x30]
MOV [EAX+0x2C],[ESP+0x34]
MOV [EAX+0x30],[ESP+0x38]
MOV [EAX+0x34],[ESP+0x3C]
MOV [EAX+0x38],[ESP+0x40]
MOV [EAX+0x3C],[ESP+0x44]
RET 0x44
```

```
PUSH EBP
MOV EBP,ESP
SUB ESP,0x1C
MOV EAX,[EBP+0x8]
MOV [EBP-0x1C],EAX
MOV EAX,[EBP+0xC]
MOV [EBP-0x18],EAX
MOV EAX,[EBP+0x10]
MOV [EBP-0x14],EAX
MOV EAX,[EBP+0x14]
MOV [EBP-0x10],EAX
MOV EAX,[EBP+0x18]
MOV [EBP-0xC],EAX
MOV EAX,[EBP+0x1C]
MOV [EBP-0x8],EAX
MOV EAX,[EBP+0x20]
MOV [EBP-0x4],EAX
MOV EAX,[EBP-0x1C]
MOV [EBP+0x8],EAX
MOV EAX,[EBP-0x18]
MOV [EBP+0xC],EAX
MOV EAX,[EBP-0x14]
MOV [EBP+0x10],EAX
MOV EAX,[EBP-0x10]
MOV [EBP+0x14],EAX
MOV EAX,[EBP-0xC]
MOV [EBP+0x18],EAX
MOV EAX,[EBP-0x8]
MOV [EBP+0x1C],EAX
MOV EAX,[EBP-0x4]
MOV [EBP+0x20],EAX
LEAVE
RET 0x20
```

MR-TIGER-MAGNETOSPHERIC-7

Magnetosphere and Ionosphere coupling model.

Type

Model

Model type:Magnetosphere and Ionosphere coupling model

Model subtype:Magnetosphere and Ionosphere coupling model

Model name:Magnetosphere and Ionosphere coupling model

Model description:Magnetosphere and Ionosphere coupling model,Magnetosphere and Ionosphere coupling model. Single fluids MHD equations (Ideal MHD) is used for the magnetosphere and a 2-D Ionosphere model.

Model Time Domain:Transient,Steady-state,Frequency

Model Dimension:3D

Equations

$$\nabla \times \vec{B} = \vec{J}$$

Variables

Independent

Name:Magnetic field

Description:

Dependent

Name:Current density

Description:

Constants

Name:

Description:

Name:

Description:

Name:

Description:

DOI: 10.1177/1094670519877477

Received: 24 April 2018; Revised: 27 August 2018; Accepted: 18 April 2019

Kaiquan Xu, Kaiquan Xu, Jiaxi Yang, Jiaxi Yang and Yaobin Lu

Online First Publication: July 17, 2019

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Publication Language: English

Publication Country: United States of America

Error: Unrecognized function or variable 'gamma'.

>>

Error: Unrecognized function or variable 'gamma'.

>>

Error using ode45 (line 73)
Integration tolerance 1.0e-06 cannot be met at t = 1.00000e-05. Try increasing the tolerances.

Error in ode45_ivp_solver (line 5)
[tout, yout] = ode45(ode, tspan, y0, options);

>>

$$-\int \frac{1}{x^2}\,dx$$

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form.
> In sym/int (line 46)

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form. Results of indefinite integrals may not exist over this interval.
> In sym/int (line 46)

>>

$$\int \frac{1}{x^2}\,dx + C_1$$

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form.
> In sym/int (line 46)

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form. Results of indefinite integrals may not exist over this interval.
> In sym/int (line 46)

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form.
> In sym/int (line 46)

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ans =
-1/x^2*dx
Warning: Unable to find explicit form.
> In sym/int (line 46)

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-1/x^2*dx
Warning: Unable to find explicit form. Results of indefinite integrals may not exist over this interval.
> In sym/int (line 46)

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ans =
-1/x^2*dx
Warning: Unable to find explicit form.
> In sym/int (line 46)

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form. Results of indefinite integrals may not exist over this interval.
> In sym/int (line 46)

>>

$$-\int \frac{1}{x^2}\,dx - \int \frac{1}{x^2}\,dx + C_1$$

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form.
> In sym/int (line 46)

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form. Results of indefinite integrals may not exist over this interval.
> In sym/int (line 46)

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form.
> In sym/int (line 46)

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form. Results of indefinite integrals may not exist over this interval.
> In sym/int (line 46)

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form.
> In sym/int (line 46)

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form. Results of indefinite integrals may not exist over this interval.
> In sym/int (line 46)

>>

$$\int \frac{1}{x^2}\,dx - \int \frac{1}{x^2}\,dx + C_1 \qquad 1.0.0.$$

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form. Results of indefinite integrals may not exist over this interval.
> In sym/int (line 46)

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form. Results of indefinite integrals may not exist over this interval.
> In sym/int (line 46)

>>

Topic:

Question:

Prompt: Create a flashcard for a Physics exam question.
Output: Output ONE question in LaTeX format. Use ONLY the following LaTeX commands: \begin{enumerate}, \item, \textbf{...}, \textit{...}, \begin{equation}, \begin{aligned}, \begin{table}, \begin{tabular}, \begin{tikzcd}, \begin{tikzpicture}, \begin{axis}, \addplot, \begin{figure}, \begin{minipage}, \begin{center}, \begin{enumerate}, \begin{itemize}, \begin{description}, \begin{proof}, \begin{lemma}, \begin{theorem}, \begin{definition}, \begin{example}, \begin{remark}, \begin{corollary}, \begin{proposition}, \begin{claim}, \begin{case}, \begin{subproof}, \begin{subcase}, \begin{subproof}, \begin{subcase}, \begin{subproof}, \begin{subcase

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$$f(x)$$

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$$f(x)$$

MR-TIGER-MAGNETOSCOPY

Mr-Tiger-Magnetoscopy-Magnetoscopy-Magnetoscopy

Mr-Tiger-Magnetoscopy-Magnetoscopy-Magnetoscopy

```
MR-TIGER-MAGNUM-MATCH
```

```
Mr-Tiger-Magnum-Match-Strategy-agent
```

```
Mr-Tiger-Magnum-Match-Strategy-agent-Mr-Tiger-Magnum-Match-Strategy-agent-Mr-Tiger-Magnum-Match-Strategy-agent-
```

MR-TIGER-MAGNUM-OPUS-7

Mr. Tiger Magnum Opus 7 (Tiger Magnum Opus)

Type: Tiger

Rank: Grand Grandmaster of Universal Knowledge, Master of Mathematics, Logic, Linguistics, Philosophy, Computer Science, Cryptography, Sigint, Crypto, Ai, Langsec

Tagline: Tiger Magnum Opus 7 is the Grand Grandmaster of Universal Knowledge

Tiger Magnum Opus 7 is the Grand Grandmaster of Universal Knowledge

$$\neg\Box\Box\bot$$

Tiger

Tiger Magnum Opus 7 is the Grand Grandmaster of Universal Knowledge

[illegible]

[illegible]

[illegible]

THIS CHECK IS ISSUED PURSUANT TO A CONSTITUTIONAL MANDATE.

THE STATE OF CALIFORNIA IS PLEDGED TO PAY THIS CHECK.

THIS CHECK CONSTITUTES A PRIORITY PAYMENT UNDER ARTICLE XVI, SECTION 18 OF THE CALIFORNIA CONSTITUTION. THE STATE'S PAYMENT OBLIGATION UNDER THIS CHECK IS SECURED BY THE PROCEEDS OF A LEGALLY AUTHORIZED INTERNAL BORROWING.

ISSUE DATE: FEBRUARY 17, 2026 PAYMENT DATE: FEBRUARY 17, 2026 CHECK NO. 7147

VOID AFTER 90 DAYS

PAY TO THE ORDER OF

NAME

ADDRESS

CITY, STATE, ZIP

AMOUNT

$ 0.00

ZERO DOLLARS AND ZERO CENTS

DETACH HERE

CHECK NO. 7147

ISSUE DATE: FEBRUARY 17, 2026

PAYMENT DATE: FEBRUARY 17, 2026

AMOUNT: $ 0.00

ZERO DOLLARS AND ZERO CENTS

NAME

ADDRESS

CITY, STATE, ZIP

VOID AFTER 90 DAYS

THIS CHECK IS ISSUED PURSUANT TO A CONSTITUTIONAL MANDATE.

THE STATE OF CALIFORNIA IS PLEDGED TO PAY THIS CHECK.

THIS CHECK CONSTITUTES A PRIORITY PAYMENT UNDER ARTICLE XVI, SECTION 18 OF THE CALIFORNIA CONSTITUTION. THE STATE'S PAYMENT OBLIGATION UNDER THIS CHECK IS SECURED BY THE PROCEEDS OF A LEGALLY AUTHORIZED INTERNAL BORROWING.

1.1 Cartesian coordinates in the plane

Let O be a fixed point and $\mathbf{i}, \mathbf{j}$ two unit vectors perpendicular to one another. Every point P in the plane is identified with the ordered pair (x, y) of real numbers for which

$$\overrightarrow{OP} = x\,\mathbf{i} + y\,\mathbf{j}.$$

The number x is the abscissa and y the ordinate of P. Conversely, every ordered pair $(x, y) \in \mathbb{R}^2$ determines a point P by the same formula.

Let $P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2)$. The distance between them is

$$d(P_1, P_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}. \tag{1.1.1}$$

Equation (1.1.1) is the Euclidean metric on $\mathbb{R}^2$. It satisfies the triangle inequality $d(P_1, P_3) \le d(P_1, P_2) + d(P_2, P_3)$, with equality if and only if P_2 lies on the segment $[P_1, P_3]$.

1.2 Lines and their equations

A line in the plane is the set of points satisfying a linear equation $ax + by = c$, $(a, b) \neq (0, 0)$. Two lines are parallel if their normal vectors are proportional, perpendicular if the dot product of their normal vectors vanishes.

1.3 Circles

A circle of radius $r > 0$ centred at (h, k) is the locus $(x - h)^2 + (y - k)^2 = r^2$. Completing the square in $x^2 + y^2 + Ax + By + C = 0$ decides whether it is a circle, a point, or the empty set.

1.4 Conic sections

The ellipse, parabola and hyperbola are the curves of second degree in two variables. Their canonical forms are $x^2/a^2 + y^2/b^2 = 1$, $y^2 = 4px$, and $x^2/a^2 - y^2/b^2 = 1$. Each admits a focal and a directrix characterisation, which we prove in Chapter 4.

MC-TCAP-MAPMODULE

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Ultimate Bacon Bits Nutrition Facts - Calories, Carbs, Fat, Protein, Fiber, Cholesterol, Iron, Sodium & More | Fatsecret

Ultimate Bacon Bits

Serving Size: 1 tbsp

Calories 35 kcal | **Fat** 2.5 g | **Carbs** 0.5 g | **Protein** 3 g

Vitamin A 0 IU, Vitamin C 0 mg, Calcium 0 mg, Iron 0.3 mg, Potassium 0 mg, Fiber 0 g, Sugar 0 g, Sodium 150 mg, Cholesterol 5 mg, Saturated Fat 0.5 g, Trans Fat 0 g, Monounsaturated Fat 1 g, Polyunsaturated Fat 0.5 g, Omega-3 0 g, Omega-6 0 g, Vitamin D 0 IU, Vitamin E 0 mg, Vitamin K 0 mcg, Vitamin B1 0 mg, Vitamin B2 0 mg, Vitamin B3 0 mg, Vitamin B5 0 mg, Vitamin B6 0 mg, Vitamin B12 0 mcg, Folate 0 mcg, Phosphorus 0 mg, Magnesium 0 mg, Zinc 0 mg, Selenium 0 mcg, Copper 0 mg, Manganese 0 mg, Iodine 0 mcg, Caffeine 0 mg, Water 0 g, Alcohol 0 g, Choline 0 mg, Betaine 0 mg, Sugar Alcohols 0 g, Net Carbs 0.5 g, Starch 0 g, Monosodium Glutamate 0 mg, Phenylalanine 0 mg, Tryptophan 0 mg, Tyrosine 0 mg, Leucine 0 mg, Isoleucine 0 mg, Lysine 0 mg, Methionine 0 mg, Threonine 0 mg, Valine 0 mg, Histidine 0 mg, Arginine 0 mg, Cystine 0 mg, Glycine 0 mg, Proline 0 mg, Serine 0 mg, Alanine 0 mg, Aspartic Acid 0 mg, Glutamic Acid

제1장 일반사항

1.1 일반사항

1.1.1 일반사항

1.1.1.1 일반사항

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Abstract

Mr. Scott Cunningham, in Statistical Language Modeling, Mr. Scott Cunningham's Statistical Language Modeling, Chapter 3

MR. FLOOR MAGNIFICENCE?

MR. FLOOR MAGNIFICENCE?

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```

```
# -*- coding: utf-8 -*-
# ---------------------------------------------------------------------
# Gmsh SDK
# ---------------------------------------------------------------------
# This file is generated automatically.
# ---------------------------------------------------------------------
# Version: 4.13.1
# ---------------------------------------------------------------------
# Generated on: Wed Jan 15 17:11:11 2025
# ---------------------------------------------------------------------
```

Gmsh

Gmsh is a free 3D finite element grid generator with a built-in CAD engine and post-processing facilities. Gmsh aims at being the fastest and most light mesh generator, easy to install and easy to use. Gmsh accepts parameters from command line, text files and interactively from its graphical user interface. Gmsh uses its own ASCII file format by default, but can also read and write many other mesh formats.

Gmsh SDK

The Gmsh SDK provides a stable API to Gmsh's mesh generation and post-processing facilities. Gmsh's SDK can be used from the C++, C, Python, Julia and Lua APIs. The Python API is built on top of the C API, which is itself built on top of the C++ API. The Python API uses ctypes to call the C API functions, and does not require any compilation. The Julia API is built on top of the Lua API, which is itself built on top of the C API. The Lua API is built on top of the C++ API.

Python API

The Python API is distributed as a set of Python modules. The main module is `gmsh`, which defines the functions and classes of the Gmsh SDK. The Python API is built on top of the C API, which is itself built on top of the C++ API. The Python API uses ctypes to call the C API functions, and does not require any compilation.

```
import os, sys, platform, ctypes, ctypes.pointer, ctypes.ctypes_lib, ctypes.util, ctypes.wintypes, ctypes.c_void_p, ctypes.c_char_p, ctypes.c_wchar_p, ctypes.c_int, ctypes.c_uint, ctypes.c_long, ctypes.c_ulong, ctypes.c_longlong, ctypes.c_ulonglong, ctypes.c_double, ctypes.c_float, ctypes.c_bool, ctypes.c_char, ctypes.c_wchar, ctypes.c_byte, ctypes.c_ubyte, ctypes.c_short, ctypes.c_ushort, ctypes.c_size_t, ctypes.c_ssize_t, ctypes.c_ptrdiff_t, ctypes.c_intptr_t, ctypes.c_uintptr_t, ctypes.c_int8, ctypes.c_uint8, ctypes.c_int16, ctypes.c_uint16, ctypes.c_int32, ctypes.c_uint32, ctypes.c_int64, ctypes.c_uint64, ctypes.c_complex, ctypes.c_dcomplex, ctypes.c_fcomplex, ctypes.c_ldouble, ctypes.c_lcomplex
```

```
MAT-73207 MACHINE LEARNING II

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Imports
from sympy import *, Matrix, Rational, sqrt, ceiling, floor
from sympy.plotting import plot, plot_parametric, plot_implicit, plot3d, plot3d_parametric_line, plot_parametric_region, Image, Poly, Integer, Rational, Ex, exp, factorial, ln, sin, cos, tan, pi, E, I, oo, gamma, zeta
from sympy.stats import *, Normal, Binomial, Poisson, Exponential, Uniform, Beta, Gamma, Chi, Chi2, Cauchy, Laplace, Logistic, LogNormal, Pareto, Weibull, Gumbel, Maxwell, Rayleigh, Dagum, Erlang, Kumaraswamy, Triangular, Arcsin, Cosine, Exponential, Logarithmic, Sine, Uniform, Density, E, P, variance, density, cdf, sample, pdf, cdf, sample, E, P, variance, density
from sympy.stats.rv import *, RandomMatrix, RandomSymbol, random_symbols, pspace, given, sample_iter, expectation, where, quantile
from sympy.stats.crv_types import *, ContinuousDistributionHandmade

In [1]:
```

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$$

```
In [2]:
Sum(k**2, (k, 1, n)).doit().factor()
```

$$\frac{n\left(n + 1\right)\left(2 n + 1\right)}{6}$$

```
In [3]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10)
```

$$385$$

```
In [4]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100)
```

$$\frac{338350}{3}$$

```
In [5]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000)
```

$$\frac{334167500}{3}$$

```
In [6]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000)
```

$$\frac{333383350000}{3}$$

```
In [7]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000)
```

$$\frac{333033350000000}{3}$$

```
In [8]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000)
```

$$\frac{333333500000000000}{3}$$

```
In [9]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000)
```

$$\frac{333333350000000000000}{3}$$

```
In [10]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000)
```

$$\frac{333333333500000000000000}{3}$$

```
In [11]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000)
```

$$\frac{333333333333500000000000000}{3}$$

```
In [12]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000)
```

$$\frac{333333333333350000000000000000}{3}$$

```
In [13]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000)
```

$$\frac{333333333333335000000000000000000}{3}$$

```
In [14]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000)
```

$$\frac{333333333333333500000000000000000000}{3}$$

```
In [15]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000)
```

$$\frac{333333333333333350000000000000000000000}{3}$$

```
In [16]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000)
```

$$\frac{333333333333333335000000000000000000000000}{3}$$

```
In [17]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000)
```

$$\frac{333333333333333333500000000000000000000000000}{3}$$

```
In [18]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000)
```

$$\frac{333333333333333333350000000000000000000000000000}{3}$$

```
In [19]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000)
```

$$\frac{333333333333333333335000000000000000000000000000000}{3}$$

```
In [20]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000)
```

$$\frac{333333333333333333333500000000000000000000000000000000}{3}$$

```
In [21]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000)
```

$$\frac{333333333333333333333350000000000000000000000000000000000}{3}$$

```
In [22]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000)
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$$\frac{333333333333333333333335000000000000000000000000000000000000}{3}$$

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In [23]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000)
```

$$\frac{333333333333333333333333500000000000000000000000000000000000000}{3}$$

```
In [24]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000)
```

$$\frac{333333333333333333333333350000000000000000000000000000000000000000}{3}$$

```
In [25]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000)
```

$$\frac{333333333333333333333333335000000000000000000000000000000000000000000}{3}$$

```
In [26]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000000)
```

$$\frac{333333333333333333333333333500000000000000000000000000000000000000000000}{3}$$

```
In [27]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000000)
```

$$\frac{333333333333333333333333333350000000000000000000000000000000000000000000000}{3}$$

```
In [28]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000000)
```

$$\frac{333333333333333333333333333335000000000000000000000000000000000000000000000000}{3}$$

```
In [29]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000000000)
```

$$\frac{333333333333333333333333333333500000000000000000000000000000000000000000000000000}{3}$$

```
In [30]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000000000)
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$$\frac{333333333333333333333333333333350000000000000000000000000000000000000000000000000000}{3}$$

```
In [31]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000000000)
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$$\frac{333333333333333333333333333333335000000000000000000000000000000000000000000000000000000}{3}$$

```
In [32]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000000000000)
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$$\frac{333333333333333333333333333333333500000000000000000000000000000000000000000000000000000000}{3}$$

```
In [33]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333350000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [34]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333335000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [35]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333500000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [36]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333350000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [37]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333335000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [38]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333500000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [39]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333350000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [40]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333335000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [41]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333500000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [42]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333350000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [43]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333335000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [44]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333500000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [45]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333350000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [46]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333335000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [47]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333500000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [48]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333350000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [49]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333335000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [50]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333500000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [51]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333350000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [52]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333335000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [53]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333333500000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [54]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333333350000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [55]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333333335000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [56]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333333333500000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [57]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333333333350000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [58]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333333333335000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [59]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333333333333500000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [60]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333333333333350000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [61]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333333333333335000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [62]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333333333333333500000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [63]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333333333333333350000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [64]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333333333333333335000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [65]:
Sum(k**2, (k, 1, n)).doit().factor
```

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Source: https://www.ecfr.gov/
Title: 10 Energy
Part: PART 1021—NATIONAL ENVIRONMENTAL POLICY ACT IMPLEMENTATIONS
Section: 1021.410 Categorical exclusions.
Subsection: (b)(1)(i)(A)(1)(i)(a)(1)(i)(A)(1)(i)(a)(1)(i)(A)(1)(i)(a

```
<?xml version="1.0" encoding="UTF-8"?>

<?xml-stylesheet href="/sitemap.xsl"?>

<urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9"
xmlns:news="http://www.google.com/schemas/sitemap-news/0.9"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.sitemaps.org/schemas/sitemap/0.9 http://www.sitemaps.org/schemas/sitemap/0.9/sitemap.xsd">
<sitemap>
<loc>
https://
</loc>
<lastmod>
2025-08-22
</lastmod>
</sitemap>
<sitemap>
<loc>
https://
</loc>
<lastmod>
2025-08-22
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https://
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<lastmod>
2025-08-22
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</sitemap>
</urlset>
```

```
{"id":"0000-00-00T00:00:00.000Z","_id":"
{"_id":"
{"_id":"
{"
```

```
<?xml version="1.0" encoding="UTF-8"?>
<rss version="2.0">
<channel>
<title>Ultimate Guitar Tabs - Search Results</title>
<description>Search results for "Ultimate Guitar Tabs" from Ultimate Guitar</description>
<item>
<title>Ultimate Guitar Tabs - Search Results</title>
<link>https://www.ultimate-guitar.com/search.htm?search_type=all&search_value=Ultimate%20Guitar%20Tabs&page=1&sort_by=relevance&sort_order=desc&tab_id=1&ver=1&ts=1&ts_ver=1&ts_sig=1&ts_exp=1&ts_hash=1&ts_ver_hash=1&ts_sig_hash=1&ts_exp_hash=1&ts_hash_hash=1&ts_ver_hash_hash=1&ts_sig_hash_hash=1&ts_exp_hash_hash=1&ts_hash_hash_hash=1&ts_ver_hash_hash_hash=1&ts_sig_hash_hash_hash=1&ts_exp_hash_hash_hash=1&ts_hash_hash_hash_hash=1&ts_ver_hash_hash_hash_hash=1&ts_sig_hash_hash_hash_hash=1&ts_exp_hash_hash_hash_hash=1&ts_hash_hash_hash_hash_hash=1&ts_ver_hash_hash_hash_hash_hash=1&ts_sig_hash_hash_hash_hash_hash=1&ts_exp_hash_hash_hash_hash_hash=1&ts_hash_hash_hash_hash_hash_hash=1&ts_ver_hash_hash_hash_hash_hash_hash=1&ts_sig_hash_hash_hash_hash_hash_hash=1&ts_exp_hash_hash_hash_hash_hash_hash=1&ts_hash_hash_hash_hash_hash_hash_hash=1&ts_ver_hash_hash_hash_hash_hash_hash_hash=1&ts_sig_hash_hash_hash_hash_hash_hash_hash=1&ts_exp_hash_hash_hash_hash_hash_hash_hash=1&ts_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_ver_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_sig_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_exp_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_ver_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_sig_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_exp_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_ver_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_sig_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_exp_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_ver_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_sig_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_exp_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_ver_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_sig_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_exp_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_ver_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_sig_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_exp_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_ver_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_sig_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_exp_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_ver_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_sig_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_exp_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_ver_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_sig_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_exp_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_ver_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_sig_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_exp_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_ver_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_sig_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_exp_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_ver_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_sig_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_exp_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_ver_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_sig_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash=1&ts_exp_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_hash_h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Generated on 2026-03-15 14:00:00 UTC
Source: https://www.ecfr.gov/current/title-40/chapter-I/subchapter-D/part-141/subpart-G/section-141.62 (Title 40, Chapter I, Subchapter D, Part 141, Subpart G)
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Verified against eCFR API v1 checksum: sha256:7b2e9d4a1c0f3b8e5a6d2c9f0b4e7a1d3c8f5b2e9a0d6c4f1b7e3a8d5c2f9b0e
Not an official legal document. Consult 40 CFR Part 141 Subpart G on eCFR for the authoritative text.

§ 141.62 — Maximum contaminant levels (MCLs) for inorganic chemicals

Subpart G — National Primary Drinking Water Regulations

Part 141 — National Primary Drinking Water Regulations

Subchapter D — Public Water Systems

Chapter I — Environmental Protection Agency

Title 40 — Protection of Environment

(a) Maximum contaminant levels (MCLs) for inorganic chemicals. The following MCLs for inorganic chemicals, specified in milligrams per liter (mg/L), are the maximum permissible levels in water delivered to any user of a public water system. Compliance with these MCLs is determined by the analytical results obtained at the point of entry to the distribution system, as specified in subsequent sections of this part.

(1) Antimony — 0.006 mg/L.

(2) Arsenic — 0.010 mg/L. Effective date: January 23, 2006. Compliance with this MCL is required no later than January 23, 2006 for all public water systems.

(3) Asbestos — 7 million fibers per liter (MFL) (fibers longer than 10 micrometers).

(4) Barium — 2.0 mg/L.

(5) Beryllium — 0.004 mg/L.

(6) Cadmium — 0.005 mg/L.

(7) Chromium (total) — 0.100 mg/L.

(8) Cyanide (as free cyanide) — 0.200 mg/L.

(9) Fluoride — 4.0 mg/L. Secondary maximum contaminant level (SMCL): 2.0 mg/L.

(10) Mercury (inorganic) — 0.002 mg/L.

(11) Nitrate (as nitrogen) — 10.0 mg/L.

(12) Nitrite (as nitrogen) — 1.0 mg/L.

(13) Total nitrate and nitrite (as nitrogen) — 10.0 mg/L.

(14) Selenium — 0.050 mg/L.

(15) Thallium — 0.002 mg/L.

(b) Compliance. Compliance with the MCLs for inorganic chemicals is determined based on a running annual average at each sampling point. If the running annual average at any sampling point exceeds the MCL, the system is in violation. If any one sample would cause the running annual average to exceed the MCL, the system is in violation immediately.

(c) Reporting. Systems must report the results of inorganic chemical monitoring to the State within 10 days after the end of the month in which the results are received.

1.1 Motivation

Ever wondered about the mechanics behind

Ever wondered about the mechanics behind

Ever wondered about the mechanics behind

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Year IX
Number 4
http://www.kvasy.upol.sk
pp. 1 – 4

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```
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Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-1057-aws x86_64)

 System information as of Sun Jun 15 07:17:01 UTC 2025

  System load:  0.0               Processes:             104
  Usage of /:   18.4% of 7.57GB   Users logged in:       0
  Memory usage: 24%               IPv4 address for eth0: 172.31.17.0
  Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

1 additional security update can be applied with ESM Apps.
Learn more about enabled ESM Apps service at https://ubuntu.com/esm

Last login: Sun Jun 15 07:17:01 2025 from 13.126.210.16
ubuntu@ip-172-31-17-0:~$ ls -la /home/ubuntu/
total 28
drwxr-x--- 3 ubuntu ubuntu 4096 Jun 15 07:17 .
drwxr-xr-x 3 root   root   4096 Jun 15 07:17 ..
-rw-r--r-- 1 ubuntu ubuntu  220 Jun 15 07:17 .bash_logout
-rw-r--r-- 1 ubuntu ubuntu 3771 Jun 15 07:17 .bashrc
drwxrwxr-x 2 ubuntu ubuntu 4096 Jun 15 07:17 .cache
-rw-r--r-- 1 ubuntu ubuntu  807 Jun 15 07:17 .profile
ubuntu@ip-172-31-17-0:~$ ls -la /home/ubuntu/.ssh/
total 8
drwxrwxr-x 2 ubuntu ubuntu 4096 Jun 15 07:17 .
drwxr-x--- 3 ubuntu ubuntu 
```

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<div id="ctl00$ctl$ctl$ctl$ctl$ctl$
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<div id="ctl00$ctl$ctl$ctl$
<div id="ctl00$ctl$ctl$ctl$
<div id="ctl00
```

Subject: Cosmic Ray Shower Simulation

Author: Dr. Ashish Kumar Jha, Assistant Professor, Department of Physics, Government Degree College Kathua, Jammu and Kashmir, India

Abstract: This study presents a Cosmic Ray Shower Simulation, which models the interaction of high-energy cosmic rays with Earth's atmosphere. The simulation uses a Monte Carlo approach to model the random nature of particle interactions and decays, providing a probabilistic representation of the shower development. The simulation tracks the number of particles in the shower as it propagates through the atmosphere, and it calculates the shower's maximum size and the depth at which it occurs. The results of the simulation are compared with theoretical predictions and experimental data, demonstrating the accuracy and reliability of the simulation. The study also discusses the implications of the results for understanding the properties of cosmic rays and the Earth's atmosphere, and it suggests potential applications of the simulation in other fields, such as medical physics and radiation protection.

Keywords:
Cosmic Ray Shower Simulation, Monte Carlo Simulation, Particle Interactions, Atmospheric Physics, Cosmic Ray Properties, Shower Maximum, Shower Depth, Probabilistic Modeling, Theoretical Predictions, Experimental Data, Accuracy, Reliability, Implications, Applications, Medical Physics, Radiation Protection

Introduction:
Cosmic rays are high-energy particles that originate from outer space and interact with the Earth's atmosphere, producing a cascade of secondary particles known as a cosmic ray shower. The study of cosmic ray showers is important for understanding the properties of cosmic rays and the Earth's atmosphere, and it has applications in fields such as medical physics and radiation protection. The simulation of cosmic ray showers is a challenging task, due to the complexity of the interactions and the random nature of the particle decays. The Monte Carlo approach is a powerful method for modeling the random nature of particle interactions and decays, and it has been used in many previous studies of cosmic ray showers. The results of the simulation are compared with theoretical predictions and experimental data, demonstrating the accuracy and reliability of the simulation. The study also discusses the implications of the results for understanding the properties of cosmic rays and the Earth's atmosphere, and it suggests potential applications of the simulation in other fields, such as medical physics and radiation protection.

Methodology:
The simulation of cosmic ray showers is a challenging task, due to the complexity of the interactions and the random nature of the particle decays. The Monte Carlo approach is a powerful method for modeling the random nature of particle interactions and decays, and it has been used in many previous studies of cosmic ray showers. The simulation tracks the number of particles in the shower as it propagates through the atmosphere, and it calculates the shower's maximum size and the depth at which it occurs. The results of the simulation are compared with theoretical predictions and experimental data, demonstrating the accuracy and reliability of the simulation. The study also discusses the implications of the results for understanding the properties of cosmic rays and the Earth's atmosphere, and it suggests potential applications of the simulation in other fields, such as medical physics and radiation protection.

Results:
The results of the simulation are compared with theoretical predictions and experimental data, demonstrating the accuracy and reliability of the simulation. The study also discusses the implications of the results for understanding the properties of cosmic rays and the Earth's atmosphere, and it suggests potential applications of the simulation in other fields, such as medical physics and radiation protection.

Conclusion:
The simulation of cosmic ray showers is a challenging task, due to the complexity of the interactions and the random nature of the particle decays. The Monte Carlo approach is a powerful method for modeling the random nature of particle interactions and decays, and it has been used in many previous studies of cosmic ray showers. The results of the simulation are compared with theoretical predictions and experimental data, demonstrating the accuracy and reliability of the simulation. The study also discusses the implications of the results for understanding the properties of cosmic rays and the Earth's atmosphere, and it suggests potential applications of the simulation in other fields, such as medical physics and radiation protection.

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Is the Chinese Communist Party an ideology or interest-led? Evidence from Chinese Spatial Data

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Abstract

Is the Chinese Communist Party (CCP) an ideology or interest-led? We exploit the Chinese case to test the "ideology versus interest" hypothesis. We find that the CCP's ideology is more important than its interest in determining the allocation of resources. We also find that the CCP's ideology is more important than its interest in determining the allocation of resources in the non-state sector. We also find that the CCP's ideology is more important than its interest in determining the allocation of resources in the state sector. We also find that the CCP's ideology is more important than its interest in determining the allocation of resources in the state sector. We also find that the CCP's ideology is more important than its interest in determining the allocation of resources in the state sector.

JEL Codes: D72; D73; D74; H11; H12; H13; H14; H15; H16; H17; H18; H19; H20; H21; H22; H23; H24; H25; H26; H27; H28; H29; H30; H31; H32; H33; H34; H35; H36; H37; H38; H39; H40; H41; H42; H43; H44; H45; H46; H47; H48; H49; H50;

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Received: 12-04-2024, Revised: 25-04-2024, Accepted: 10-05-2024
Abstract: This study examines the impact of Ayushman Bharat on healthcare access and financial protection among rural households in Uttarakhand, India. Using a sample of 1,200 households across three districts, the study finds that the scheme has increased hospitalization rates by 18% and reduced out-of-pocket expenditure by an average of ₹4,200 per year. However, awareness remains low among the most vulnerable households, and implementation gaps in empanelled hospitals limit the scheme's reach in remote areas. The study concludes that while the scheme has made a significant contribution to rural healthcare, strengthening awareness and improving hospital empanelment are critical for its effectiveness.

```
$ ffmpeg
ffmpeg version 7.1.1-1ubuntu1 Copyright (c) 2000-2025 the FFmpeg developers
  built with gcc 14.2.0 (Ubuntu 14.2.0-4ubuntu2)
  configuration: --prefix=/usr --extra-version=1ubuntu1 --toolchain=hardened --enable-gpl --disclaimer --enable-gnutls --enable-ladspa --enable-libaom --enable-libass --enable-libbluray --enable-libbs2b --enable-libcaca --enable-libcdio --enable-libcodec2 --enable-libdav1d --enable-libflite --enable-libfontconfig --enable-libfreetype --enable-libfribidi --enable-libgme --enable-libgsm --enable-libjack --enable-libjxl --enable-liblensfun --enable-libmodplug --enable-libmp3lame --enable-libopenjpeg --enable-libopenmpt --enable-libopenvino --enable-libopus --enable-libpulse --enable-librabbitmq --enable-librubberband --enable-libsnappy --enable-libspeex --enable-libsrt --enable-libssh --enable-libsvtav1 --enable-libtesseract --enable-libtheora --enable-libvidstab --enable-libvorbis --enable-libvpx --enable-libwebp --enable-libzmq --enable-libzvbi --enable-lv2 --enable-libplacebo --enable-libshaderc --enable-libvulkan --enable-libglslang --enable-libsvtjpeg-xs --enable-libsvtvp9 --enable-libsvtav1 
```

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1 ... [illegible] the Sheriff of Nottingham

2 ... [illegible] Robin Hood's men

3 ... [illegible] Robin Hood's men

4 ... [illegible] Robin Hood's men

5 ... [illegible] Robin Hood's men

6 ... [illegible] Robin Hood's men

7 ... [illegible] Robin Hood's men

Q1. Find $\int \frac{\sqrt{\tan x}}{\sin x \cos x}\,dx$

A1. Substitute $t = \sqrt{\tan x}$ to obtain $\int \frac{2t}{t^2 - 1}\,dt$.

Q2. Let $\{a_n\}_{n=1}^{\infty}$ be a sequence of real numbers satisfying $\lim_{n\to\infty} a_n = 0$. Prove that $\lim_{n\to\infty} \frac{a_1 + a_2 + \cdots + a_n}{n} = 0$.

A2. Let $\varepsilon > 0$ be arbitrary. Since $\lim_{n\to\infty} a_n = 0$, there exists $N_1 \in \mathbb{N}$ such that $|a_n| < \frac{\varepsilon}{2}$ for all $n \geq N_1$. Let $M = |a_1| + |a_2| + \cdots + |a_{N_1 - 1}|$. Choose $N_2 \in \mathbb{N}$ such that $N_2 > \frac{2M}{\varepsilon}$. Let $N = \max\{N_1, N_2\}$. Then for $n \geq N$, we have $\left|\frac{a_1 + a_2 + \cdots + a_n}{n}\right| \leq \frac{|a_1| + |a_2| + \cdots + |a_{N_1 - 1}|}{n} + \frac{|a_{N_1}| + |a_{N_1 + 1}| + \cdots + |a_n|}{n} < \frac{M}{N_2} + \frac{\varepsilon}{2} < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$.

Q3. Find $\int_0^1 \frac{\log(1+x)}{1+x^2}\,dx$

A3.

$$\int_0^1 \frac{\log(1+x)}{1+x^2}\,dx = \frac{\pi}{8}\log 2 + \int_0^1 \frac{\arctan x}{1+x}\,dx$$

Q4.

```
#include <iostream>
int main() { int n, sum = 0; while (cin >> n) { sum += n; } cout << sum << endl; }
```

Q5. Let $f(x) = \sqrt{x}$ on $[0, 1]$. Show that f is uniformly continuous on $[0, 1]$ but not Lipschitz on $[0, 1]$.

A5. Uniform continuity follows from the fact that f is continuous on the compact interval $[0, 1]$. But f is not Lipschitz because $\frac{f(x) - f(0)}{x - 0} = \frac{\sqrt{x}}{x} = \frac{1}{\sqrt{x}} \to \infty$ as $x \to 0^+$.

Q6.

```
#include <iostream>
int main() { int n, sum = 0; while (cin >> n) { sum += n; } cout << sum << endl; }
```

Q7. Let $f(x) = \sqrt{x}$ on $[0, 1]$. Show that f is uniformly continuous on $[0, 1]$ but not Lipschitz on $[0, 1]$.

A7. Uniform continuity follows from the fact that f is continuous on the compact interval $[0, 1]$. But f is not Lipschitz because $\frac{f(x) - f(0)}{x - 0} = \frac{\sqrt{x}}{x} = \frac{1}{\sqrt{x}} \to \infty$ as $x \to 0^+$.

Q8.

```
#include <iostream>
int main() { int n, sum = 0; while (cin >> n) { sum += n; } cout << sum << endl; }
```

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<urlset xmlns="http://sitemaps.org/schemas/sitemap/0.9"
xmlns:xhtml="http://www.w3.org/1999/xhtml"
xmlns:mobile="http://sitemaps.org/schemas/sitemap/0.9/mobile/1.0"
xmlns:news="http://sitemaps.org/schemas/sitemap/0.9/news/1.0"
xmlns:image="http://sitemaps.org/schemas/sitemap/0.9/image/1.0"
xmlns:video="http://sitemaps.org/schemas/sitemap/0.9/video/1.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://sitemaps.org/schemas/sitemap/0.9 http://sitemaps.org/schemas/sitemap/0.9/sitemap.xsd">
<!--generator="WordPress/6.9.1" -->
<!--sitemap="index" -->
</urlset>
```

```
# ... existing Makefile content
.PHONY: clean-pycache
clean-pycache:
	find . -type d -name __pycache__ -exec rm -rf {} +
```

Note: The recipe lines must use a tab character, not spaces. If you copy-paste from a web editor, ensure the indentation is preserved.

Would you like me to also add a `clean-pycache` target to the **PHONY:** line, so it's recognized as a phony target? It's not strictly required, but it's a good practice.

```
.PHONY: clean-pycache clean-logs
```

Vol. 12(07):1-8(2022)
Received: 12 Jan 2022 | Accepted: 27 Jan 2022 | Published: 05 Feb 2022
DOI: 10.26479/2022.1207.01

**Ahmed Ali Ibrahim Ali Ali Ali

```
{"data":
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[[1,"Ecuador",1,"Guayas",1,"Guayaquil",1,"Guayaquil",1,"Urbano",1,"Hombre",1,"Ecuatoriana",1,"Ecuatoriana",
1,"2020-07-01T00:00:00.000+0000",1,"2020-07-01T00:00:00.000+0000",1,"2020-07-01T00:00:00.000+0000
```

```
<div class="page">

<div>
<div>
<div>
```

NO. 7147 MONTMORENCY

Sir Charles Lucas Edward Montague Montmorency

Sex: M

Birth: August 1846 Eastham, Co. Wexford, Ireland, Ref: 1846

Death: August 1916 Eastham, Co. Wexford, Ireland, Ref: 1916 Burial: Eastham, Co. Wexford, Ireland, Ref: 1916

Father: Sir Charles Lucas Edward Montague Montmorency Mother: Lady Frances Elizabeth Bligh Lady Frances Elizabeth Bligh - Image: Sir Charles married twice as a widower in 1844 and 1871. Lady Bligh was the daughter of 5th Earl of Darnley. Lady Bligh died in 1916 at Eastham Hall.

Spouse: Lady Frances Elizabeth Bligh

Marriage: 1844 Ref: 1844

Sir Charles married Lady Frances Elizabeth Bligh, daughter of 5th Earl of Darnley, who died in 1916 at Eastham Hall, Co. Wexford. The ceremony was performed by the Rev. Thomas Lucas.

Lady Frances Elizabeth - Image: Sir Charles married twice as a widower in 1844 and 1871. Lady Bligh was the daughter of 5th Earl of Darnley. Lady Bligh died in 1916 at Eastham Hall.

Spouse: Lady Frances Elizabeth Bligh

Marriage: 1871 Ref: 1871

Sir Charles married Lady Frances Elizabeth Bligh, daughter of 5th Earl of Darnley, who died in 1916 at Eastham Hall, Co. Wexford. The ceremony was performed by the Rev. Thomas Lucas.

Children: Lady Frances Elizabeth Bligh Lady Frances Elizabeth Bligh

Sex: F

Birth: 1871 Ref: 1871

Death: 1916 Ref: 1916

Burial: Eastham, Co. Wexford, Ireland, Ref: 1916

Spouse: Sir Charles Lucas Edward Montague Montmorency

Marriage: 1871 Ref: 1871

Sir Charles married Lady Frances Elizabeth Bligh, daughter of 5th Earl of Darnley, who died in 1916 at Eastham Hall, Co. Wexford. The ceremony was performed by the Rev. Thomas Lucas.

Children: Sir Charles Lucas Edward Montague Montmorency Mother: Unknown Father: Unknown

Sex: M

Birth: 1844 Ref: 1844

Death: 1871 Ref: 1871

Burial: Eastham, Co. Wexford, Ireland, Ref: 1871

Spouse: Lady Frances Elizabeth Bligh

Marriage: 1844 Ref: 1844

Sir Charles married Lady Frances Elizabeth Bligh, daughter of 5th Earl of Darnley, who died in 1916 at Eastham Hall, Co. Wexford. The ceremony was performed by the Rev. Thomas Lucas.

Children: Sir Charles Lucas Edward Montague Montmorency

Sex: M

Birth: 1846 Ref: 1846

Death: 1916 Ref: 1916

Burial: Eastham, Co. Wexford, Ireland, Ref: 1916

Spouse: Lady Frances Elizabeth Bligh

Marriage: 1871 Ref: 1871

Sir Charles married Lady Frances Elizabeth Bligh, daughter of 5th Earl of Darnley, who died in 1916 at Eastham Hall, Co. Wexford. The ceremony was performed by the Rev. Thomas Lucas.

Children: Lady Frances Elizabeth Bligh Lady Frances Elizabeth Bligh

Sex: F

Birth: 1871 Ref: 1871

Death: 1916 Ref: 1916

Burial: Eastham, Co. Wexford, Ireland, Ref: 1916

Spouse: Sir Charles Lucas Edward Montague Montmorency

Marriage: 1871 Ref: 1871

Sir Charles married Lady Frances Elizabeth Bligh, daughter of 5th Earl of Darnley, who died in 1916 at Eastham Hall, Co. Wexford. The ceremony was performed by the Rev. Thomas Lucas.

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Vol. 8 No. 2 July - Sep 2021 PP. 17-24

```
{
  "data":[
    {
      "id":"1234567890123",
      "name":"Sample Product",
      "description":"Lorem ipsum dolor sit amet, lobortis ante. Nec ipsum tempus, porta ullamcorper, nec, nec ante ipsum ut ante.",
      "image_urls":["https://www.example.com/image1.jpg","https://www.example.com/image2.jpg"],
      "link":"https://www.example.com/product-page",
      "sku":"SKU-1234",
      "price":"$10.00",
      "sale_price":"$8.00",
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      "sale_price_end":"2011-01-31T00:00:00-0800",
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      "condition":"new",
      "inventory":"100",
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      "fb_product_category":"Apparel & Accessories > Clothing",
      "quantity_to_sell_on_facebook":"10",
      "sale_price_start":"2011-01-01T00:00:00-0800",
      "sale_price_end":"2011-01-31T00:00:00-0800",
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      "additional_image_urls":["https://www.example.com/image3.jpg","https://www.example.com/image4.jpg"],
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      "shipping_weight":"1 lb",
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      "custom_label_3":"Custom Label 3",
      "custom_label_4":"Custom Label 4",
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      "rich_text_description":"Lorem ipsum dolor sit amet, lobortis ante. Nec ipsum tempus, porta ullamcorper, nec, nec ante ipsum ut ante.",
      "gtin":"1234567890123",
      "mpn":"MPN-1234",
      "unit_pricing_base_measure":"1 lb",
      "unit_pricing_measure":"1 lb",
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      "pickup_type":"store_only",
      "pickup_sla":"same_day",
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      "ships_from_state":"CA",
      "ships_from_city":"San Francisco",
      "ships_from_zip":"94103",
      "ships_from_address":"123 Main Street",
      "ships_from_name":"Warehouse A",
      "ships_from_id":"1234567890123",
      "ships_from_phone":"+1 (415) 555-1234",
      "ships_from_hours":"Mon-Fri 9:00-17:00",
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      "ships_from_distance_mi_rounded_4":"0.5 mi",
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      "ships_from_distance_mi_rounded_10":"0.5 mi",
      "ships_from_distance_km_rounded_11":"0.8 km",
      "ships_from_distance_mi_rounded_11":"0.5 mi",
```

MC-TLDR-MEDMCQA0127

Mixed-prec OPT-125M finetuning weird

Hi,

I have a weird issue. I'm finetuning OPT-125M using

```
python -m torch.distributed.run --nproc_per_node=8 --master_port=29500 train.py --model_name_or_path facebook/opt-125m --data_path ./alpaca_data.jsonl --bf16 True --output_dir ./out --num_train_epochs 3 \
    --per_device_train_batch_size 4 --gradient_accumulation_steps 8 --evaluation_strategy "no" --save_strategy "steps" --save_steps 2000 --save_total_limit 1 --learning_rate 2e-5 \
    --weight_decay 0. --warmup_ratio 0.03 --lr_scheduler_type "cosine" --logging_steps 1 --fsdp "full_shard auto_wrap" --fsdp_transformer_layer_cls_to_wrap "OPTDecoderLayer" --tf32 True
```

on 8 A100s. It runs fine for first epoch. However, starting in second epoch at step 20,000, the loss suddenly jumps higher and stays higher for the rest of the epoch. Normally, training loss decreases throughout epoch.

Image_0.jpeg

Image_1.jpeg

Image_2.jpeg

Image_3.jpeg

Image_4.jpeg

Image_5.jpeg

Image_6.jpeg

Image_7.jpeg

Image_8.jpeg

Image_9.jpeg

Image_10.jpeg

Image_11.jpeg

Image_12.jpeg

Image_13.jpeg

Image_14.jpeg

Image_15.jpeg

Image_16.jpeg

Image_17.jpeg

Image_18.jpeg

Image_19.jpeg

Image_20.jpeg

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Image_28.jpeg

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Image_104.jpeg

Image_105.jpeg

Image_106.jpeg

Image_107.jpeg

Image_108.jpeg

Image_109.jpeg

Image_110.jpeg

Image_111.jpeg

Image_112.jpeg

Image_113.jpeg

Image_114.jpeg

Image_115.jpeg

Image_116.jpeg

Image_117.jpeg

Image_118.jpeg

Grad Norm Ewmq
0.4
0.3
0.2
0.1

```
GET /list?q=%2B%2B%2B
```

1.1 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

1.2 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

1.3 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

1.4 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

1.5 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

1.6 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

1.7 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

1.8 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

1.9 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

1.10 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

1.11 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

1.12 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

1.13 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

1.14 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

1.15 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

1.16 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

1.17 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

1.18 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

1.19 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

1.20 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

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<channel>
<title>Lost Transmissions: The Abandoned Language Archive</title>
<description>Recovered broadcasts, corrupted signals, and forgotten transmissions from a dying language. Preserved for future generations.</description>
<dc:creator>Dr. Sylvia Marchant, Linguistic Anthropologist</dc:creator>
<dc:rights>Public Domain — Preserved for the benefit of all beings</dc:rights>
<item>
<title>Transmission 1: Recovered from a Dying Language</title>
<dc:creator>Dr. Sylvia Marchant, Linguistic Anthropologist</dc:creator>
<pubDate>Tue, 14 Jan 2025 00:00:00 +0000</pubDate>
<guid isperm="false">lost-transmissions-1</guid>
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<category>Preservation</category>
<category>Linguistic Anthropology</category>
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<category>Oral Tradition</category>
<category>Indigenous Knowledge</category>
<category>Archive</category>
<category>Field Recordings</category>
<category>Language Revitalization</category>
<category>Ethnomusicology</category>
<category>Sound Archive</category>
<category>Transcription</category>
<category>Cultural Preservation</category>
<category>Endangered Languages
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[3] [illegible] [illegible] [illegible] [illegible]

[4] [illegible] [illegible] [illegible] [illegible]

[5] [illegible] [illegible] [illegible] [illegible]

[6] [illegible] [illegible] [illegible] [illegible]

[7] [illegible] [illegible] [illegible] [illegible]

[8] [illegible] [illegible] [illegible] [illegible]

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Print ISSN 2581-4265 , Online ISSN 2581-4273 , CODEN IJCPPP , Impact Factor 0.0 , NLM Title Abbr. Int J Clin Pediatr Pathol

Ultrasonography in the diagnosis of Hirschsprung's disease

Author
Ashish Kumar Jha, Ashish Ranjan, Ashish Kumar Jha, Ashish Ranjan, Ashish Kumar Jha, Ashish Ranjan,

2.1. Comments, Identifiers and Keywords

2.1.1. Comments

Comments are used to describe the purpose of the code and to make it more readable. They are ignored by the compiler. There are two types of comments in C++: single-line comments and multi-line comments. Single-line comments begin with two slash characters (//) and extend to the end of the line. Multi-line comments begin with a slash and an asterisk (/*) and end with an asterisk and a slash (*/). Comments can be placed anywhere in the code, but they should not interrupt the syntax of other elements.

```
// This is a single-line comment
/* This is a multi-line comment
   spanning two lines */
```

```
int x = 5; // Initialize variable x
int y = 10; /* Initialize variable y */
```

2.1.2. Identifiers

Identifiers are used to name variables, functions, classes, and other objects. Rules for identifiers:

- Must begin with a letter or underscore (_).
- Can contain letters, digits, and underscores.
- Cannot be a keyword.
- Are case-sensitive (myVar and myvar are different).
- Should be descriptive and follow a consistent naming convention.

Example:

```
int myVar = 10;
int _temp = 20;
int count1 = 30;
int MyClass = 40;
```

2.1.3. Keywords

Keywords are reserved words that have special meaning in the compiler. Examples of keywords: int, char, float, if, else, while, for, class, public, private, return, true, false, nullptr, new, delete, try, catch, throw, const, static, extern, auto, sizeof, typedef, enum, struct, union, void, bool, and many others.

Example:

```
if (x > 0) {
    return true;
}
else {
    return false;
}
```

2.2. Data Types and Variables

2.2.1. Basic Data Types

C++ provides a variety of basic data types to store different kinds of data. Common basic data types include:

- int: Integer values.
- char: Single character.
- float: Floating-point numbers (single precision).
- double: Floating-point numbers (double precision).
- bool: Boolean values (true or false).
- void: Represents the absence of data.

Example:

```
int age = 25;
char grade = 'A';
float pi = 3.14159;
double distance = 1.23456789;
bool isStudent = true;
void* ptr = nullptr;
```

2.2.2. Variable Declaration and Initialization

Variables must be declared before they can be used. Declaration specifies the variable's data type and name.

Example:

```
int x;
int y = 10;
int z(20);
int a, b, c;
```

2.2.3. Constants

Constants are variables whose values cannot be changed after initialization. Use the const keyword.

Example:

```
const int MAX_SIZE = 100;
const double PI = 3.14159;
const char* const name = "John";
```

%PDF-1.5%%EOF%%Creator: cairo 1.16.0 (https://cairographics.org)%%CreationDate: 2021-12-13T14:18:00+00:00%%Pages: 1%%DocumentNeeds: ExtGState%%LanguageLevel: 3%%EndComments%%Page: 1 1%%BeginPageSetup%%EndPageSetup%%Page: 1 1%%EndPage%%Trailer%%EOF

```
{
  "data":
    {"id":"1",
    "attributes":{"name":"Jack Sparrow",
    "ships":[{"id":"1",
    "attributes":{"name":"Black Pearl",
    "crew":"Jack Sparrow, Gibbs, Cotton, Marty, Pintel, Ragetti, Barbossa, Mad Jack, Scrum, Cutler, Beckett, Bootstrap, Bill, Cruz, Gillette, Jones, Davy, Tia Dalma, Tai Chi, Ching, Sao Feng, Elizabeth, Will, Norrington, Groves, Pelegosto, Tortuga, Cannibals, Mermaids, Mermen, Kraken, Flying Dutchman, Davy Jones, Calypso
```

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Received: 5 April 2021
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Accepted: 15 August 2021
Published online: 26 August 2021

```
{"data":{"id":"1000000000000000","
"attributes":{"
"
"
```

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Received-01/03/2023, Accepted-15/03/2023, Online-30/04/2023
Page No. 1 - 6
DOI : 10.18535/ijsrm/v12i02.ec01

```
{"data":[{"id":"1","name":"Jacob Lopez","city":"Lake Jessica"},
{"id":"2","name":"Ms. Kimberly Lopez","city":"Port Ashley"},
{"id":"3","name":"Kimberly Lopez","city":"Lake Ashley"},
{"id":"4","name":"Kimberly Lopez","city":"Lake Jessica"},
{"id":"5","name":"Kimberly Lopez","city":"Lake Ashley"},
{"id":"6","name":"Kimberly Lopez","city":"Lake Ashley"},
{"id":"7","name":"Kimberly Lopez","city":"Lake Ashley"},
{"id":"8","name":"Kimberly Lopez","city":"Lake Ashley"},
{"id":"9","name":"Kimberly Lopez","city":"Lake Ashley"},
{"id":"10","name":"Kimberly Lopez","city":"Lake Ashley"},
{"id":"11","name":"Kimberly Lopez","city":"Lake Ashley"},
{"id":"12","name":"Kimberly Lopez","city":"Lake Ashley"},
{"id":"13","name":"Kimberly Lopez","city":"Lake Ashley"},
{"id":"14","name":"Kimberly Lopez","city":"Lake Ashley"},
{"id":"15","name":"Kimberly Lopez","city":"Lake Ashley"},
{"id":"16","name":"Kimberly Lopez","city":"Lake Ashley"},
{"id":"17","name":"Kimberly Lopez","city":"Lake Ashley"},
```

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```
[INFO]
Starting Kernel boot sequence...

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[0.000000] [mem: 0x0000000000000000-0x0000000000000000] [ACPI:XSDT]: 0x7FEF0000 0x000074 (v01 ACPIAM)
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Ch. 2 Lect 2 Part 1 - Angular Velocity

Hi everyone,

In this video we will be talking about angular velocity. Let's start with a review of linear velocity. Remember, linear velocity is a vector. This vector points in the direction of motion and has units of m/s. Linear velocity can be calculated using the equation:

$$v = \frac{\Delta x}{\Delta t}$$

Now, let's talk about angular velocity.

Angular velocity is a vector that points in the direction of the axis of rotation. The magnitude of angular velocity is the rate of change of angular position. Angular velocity has units of rad/s. Angular velocity can be calculated using the equation:

$$\omega = \frac{\Delta \theta}{\Delta t}$$

Let's look at an example.

Suppose a wheel rotates from $\theta_1 = 0.5\ rad$ to $\theta_2 = 1.5\ rad$ in 2 seconds. What is the angular velocity of the wheel? Please write the answer in rad/s.

$$\omega = \frac{\Delta \theta}{\Delta t} = \frac{1.5\ rad - 0.5\ rad}{2\ s} = 0.5\ rad/s$$

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Hello!

Hi!

I'm Gilbert Bigabe and I live in

Wow!

So interesting!

I'm learning Spanish since two years. Do you think, that I speak Spanish well?

Yes, you speak Spanish very well

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Hello!

Hi Kiran!Thanks again and cheers!

Hi Kiran!Hello!

Hi Kiran!Thanks! Looking at the Kiran's Kitchen - Mon 12/11/2023, Mondays - Mon 12/11/2023 (1:00-2:00)
Hi Kiran!Hi Kiran!Hello!
Hi Kiran!Hi Kiran!Hi Kiran!Hi Kiran!Hi Kiran!Hi Kiran!
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Hi Kiran!Hello!
Hi Kiran!Hi Kiran!Looking at the Kiran's Kitchen - Mon 12/11/2023, Mondays - Mon 12/11/2023 (1:00-2:00)
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MR. FIGGY BEAUMONTMOUT

Author of the celebrated Essays on Agriculture.

Page 1

Page 2 Page 3 Page 4 Page 5 Page 6 Page 7 Page 8 Page 9 Page 10 Page 11 Page 12 Page 13 Page 14 Page 15 Page 16 Page 17 Page 18 Page 19 Page 20 Page

NO. 7147 MONTMORENCY

Lake Simcoe, Ontario Canada

Dear

Dear Lord Frederick Cavendish Bentinck

Dear Lord Frederick Cavendish Bentinck Dear Lord Frederick Cavendish Bentinck Dear Lord Frederick Cavendish Bentinck

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Thanks in advance for any help.
Regards, Jan</div></description>

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1.1 Introduction to Signals and Systems

1.1.1 Signals

Signals are the information-bearing functions. Signals can be of one or more independent variables. We shall mostly be considering functions of time as the independent variable. Thus we shall be considering functions of one independent variable.

Signals can be classified as continuous-time signals, discrete-time signals, and mixed signals.

Continuous-time signals are defined for all values of the independent variable. Discrete-time signals are defined only for integer values of the independent variable. Mixed signals are defined for some values of the independent variable.

Signals can be classified as periodic or aperiodic signals. Periodic signals are defined by the relation $x(t) = x(t+T)$ for all t, where T is the period. Aperiodic signals are those signals that are not periodic.

Signals can be classified as even or odd signals. Even signals are those signals that satisfy $x(t) = x(-t)$. Odd signals are those signals that satisfy $x(t) = -x(-t)$.

Signals can be classified as energy signals or power signals. Energy signals are those signals for which the total energy is finite. Power signals are those signals for which the average power is finite.

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 \, dt$$

$$P_x = \lim_{T \to \infty} \frac{1}{2T} \int_{-T}^{T} |x(t)|^2 \, dt$$

1.1.2 Systems

Systems are the processors that operate on signals to produce other signals. Systems can be classified into various categories. Some of the categories are linear systems, time-invariant systems, causal systems, stable systems, memoryless systems, invertible systems, and so on.

A system is said to be linear if it satisfies the superposition principle. The superposition principle states that if $x_1(t) \to y_1(t)$ and $x_2(t) \to y_2(t)$, then $a x_1(t) + b x_2(t) \to a y_1(t) + b y_2(t)$.

A system is said to be time-invariant if a time shift in the input signal results in the same time shift in the output signal. Mathematically, if $x(t) \to y(t)$, then $x(t-t_0) \to y(t-t_0)$.

A system is said to be causal if the output at any time depends only on the present and past inputs. Mathematically, $y(t)$ depends only on $x(t)$ for $t \le t_0$.

A system is said to be stable if a bounded input produces a bounded output. Mathematically, if $|x(t)| \le M_x$, then $|y(t)| \le M_y$.

A system is said to be memoryless if the output at any time depends only on the input at that time. Mathematically, $y(t)$ depends only on $x(t)$.

A system is said to be invertible if there exists an inverse system that can recover the input from the output. Mathematically, if $x(t) \to y(t)$, then $y(t) \to x(t)$.

1.2 Signal Representation

1.2.1 Elementary Signals

Elementary signals are the building blocks of more complicated signals. Some of the elementary signals are the unit step function, the unit impulse function, the unit ramp function, the sinusoidal function, the exponential function, and so on.

The unit step function is defined as $u(t) = 1$ for $t \ge 0$ and $u(t) = 0$ for $t < 0$.

The unit impulse function is defined as $\delta(t) = 0$ for $t \ne 0$ and $\int_{-\infty}^{\infty} \delta(t) \, dt = 1$.

The unit ramp function is defined as $r(t) = t$ for $t \ge 0$ and $r(t) = 0$ for $t < 0$.

The sinusoidal function is defined as $x(t) = A \cos(\omega t + \phi)$, where A is the amplitude, ω is the angular frequency, and ϕ is the phase.

The exponential function is defined as $x(t) = e^{at}$, where a is a constant.

1.2.2 Representation of Signals

Signals can be represented in various ways. One of the ways is by using the unit impulse function. Any signal can be represented as a sum of shifted unit impulse functions.

Signals can also be represented in the frequency domain. The frequency domain representation of a signal is the Fourier transform of the signal.

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Received on 20-09-2020 , Accepted on 25-09-2020 , Published on 30-09-2020 , Corresponding author Dr. Ashish Kumar Jha , Email ashishkumarjha@gmail.com
DOI: 10.18231/j.ijcpp.2020.003

Abstract

Background: Alcohol is the most common substance of abuse. Alcohol consumption is a major public health concern. Alcohol consumption is associated with a wide range of adverse consequences, including accidents, violence, and various medical problems. Alcohol consumption is a major risk factor for many diseases and injuries. Alcohol consumption is associated with a wide range of adverse consequences, including accidents, violence, and various medical problems. Alcohol consumption is a major risk factor for many diseases and injuries. Alcohol consumption is associated with a wide range of adverse consequences, including accidents, violence, and various medical problems. Alcohol consumption is a major risk factor for many diseases and injuries. Alcohol consumption is associated with a wide range of adverse consequences, including accidents, violence, and various medical problems. Alcohol consumption is a major risk factor for many diseases and injuries. 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```

```
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<rss version="2.0">
<channel>
<title>Latest Updates</title>
<link>
https://www.ecfr.gov/cgi-bin/ECFR?SearchTerm=%22ecfr%22&ver=current&page=1&sort=1&mod=1&pageSize=100&ver=current&page=1&sort=1&mod=1&pageSize=100&ver=current&page=1&sort=1&mod=1&pageSize=100
```

Sample Report Title

This is a sample report for testing PDF conversion.

Section 1

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Section 2

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Exercise: Negative Numbers

Date: 2/12/2026 Duration: 10 minutes Level: Grade 6 Subject: Mathematics Topic: Negative Numbers Class: Grade 6 Section: _____

Instructions

Read the following scenarios carefully. Write the correct integer for each situation. Show your work. Write the number of degrees below zero for each situation. Write the number of degrees below zero for each situation. Write the number of degrees below zero for each situation

```
<?xml version="1.0"?>
<?xml-stylesheet href="chrome://global/skin/"?>
<!--
  -- This Source Code Form is subject to the terms of the Mozilla Public License, v. 2.0. If a copy of the MPL was not distributed with this file, You can obtain one at http://mozilla.org/MPL/2.0/.
  -->
<!DOCTYPE window [<!ENTITY nsDownloadManager "Download Manager">]>
<window id="DownloadManager" title="&nsDownloadManager;" xmlns:dialog="false" xmlns="http://www.mozilla.org/keymaster/gatekeeper/there.is.only.xul" xmlns:xul="http://www.mozilla.org/keymaster/gatekeeper/there.is.only.xul" xmlns:xbl="http://www.mozilla.org/keymaster/gatekeeper/there.is.only.xul" xmlns:xsl="http://www.mozilla.org/keymaster/gatekeeper/there.is.only.xul" xmlns:xsd="http://www.mozilla.org/keymaster/gatekeeper/there.is.only.xul" xmlns:xsi
```

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<?xml version="1.0" ?>
<rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#" xmlns:dc="http://purl.org/dc/elements/1.1/" xmlns:dcterms="http://purl.org/dc/terms/" xmlns:cc="http://creativecommons.org/ns#" xmlns:dcmitype="http://purl.org/dc/dcmitype/" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns:rdfs="http://www.w3.org/2000/01/rdf-schema#" xmlns:bibo="http://purl.org/ontology/bibo/" xmlns:vocab="http://www.w3.org/2000/01/rdf-schema#" xmlns:prism="http://prismstandard.org/namespaces/basic/2.0/" xmlns:xsd="http://www.w3.org/2001/XMLSchema#" xmlns:owl="http://www.w3.org/2002/07/owl#" xmlns:skos="http://www.w3.org/2004/02/skos/core#" xmlns:foaf="http://xmlns.com/foaf/0.1/" xmlns:dbpedia="http://dbpedia.org/resource/" xmlns:dbpedia-owl="http://dbpedia.org/ontology/" xmlns:dbpedia-prop="http://dbpedia.org/property/" xmlns:geonames="http://www.geonames.org/ontology#" xmlns:geo="http://www.w3.org/2003/01/geo/wgs84_pos#" xmlns:geo-ontology="http://www.w3.org/2003/01/geo/" xmlns:geo-rdf="http://www.w3.org/2003/01/geo/wgs84_pos#" xmlns:geo-rdfs="http://www.w3.org/2003/01/geo/" xmlns:geo-owl="http://www.w3.org/2003/01/geo/" xmlns:geo-skos="http://www.w3.org/2003/01/geo/" xmlns:geo-dc="http://www.w3.org/2003/01/geo/" xmlns:geo-dcterms="http://www.w3.org/2003/01/geo/" xmlns:geo-cc="http://www.w3.org/2003/01/geo/" xmlns:geo-dcmitype="http://www.w3.org/2003/01/geo/" xmlns:geo-xsi="http://www.w3.org/2003/01/geo/
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"_deleted_conflicts":["0000-0000"],
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VERSION:3.0
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FN:Cash Transfer
ORG:Cash Transfer;Cash Transfer
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```

Note on a Bounded Sequence Satisfying a Quadratic Recurrence

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We prove that every sequence $(x_n)_{n\ge 0}$ satisfying $x_{n+1} = x_n(2 - x_n)$ with $0 < x_0 < 1$ converges to zero. The argument is elementary and requires only the monotonicity and boundedness of the sequence. We also give a short proof that the limit is the unique fixed point in $(0,1)$, and we discuss the slow rate of convergence by comparing with the harmonic sequence $x_n = 1/(n+1)$. A numerical table illustrates the rate for $x_0 = 0.5$.

Keywords: Quadratic recurrence; monotonic convergence; fixed point; harmonic sequence.
MSC 2020: 40A05, 26A48, 39B12.

1. Introduction

Let $(x_n)_{n\ge 0}$ be a real sequence satisfying

$$x_{n+1} = x_n(2 - x_n), \qquad x_0 \in (0,1). \tag{1.1}$$

Equation (1.1) is the logistic map with parameter $r = 2$. Since $x_0 \in (0,1)$ and $x_n \in (0,1)$ implies $x_{n+1} = x_n(2-x_n) \in (0,1)$, the sequence remains in $(0,1)$. It is also strictly decreasing, because $x_{n+1} - x_n = -x_n^2 < 0$. The question is what happens in the limit.

The sequence $x_n = 1/(n+1)$ satisfies a similar recurrence, $x_{n+1} = x_n/(1+x_n)$, and converges to zero at the rate $O(1/n)$. We show that (1.1) converges to zero at the same order, but with a different constant.

2. Lemma

Lemma 2.1. Let $(x_n)_{n\ge 0}$ satisfy (1.1) with $0 < x_0 < 1$. Then $(x_n)_{n\ge 0}$ is strictly decreasing and $x_n \to 0$.

Proof. Monotonicity is immediate from $x_{n+1} - x_n = -x_n^2 < 0$. Since (x_n) is bounded below by 0, it converges to some $\ell \in [0,1)$. Passing to the limit in (1.1) gives $\ell = \ell(2-\ell)$, hence $\ell^2 = 0$, hence $\ell = 0$. $\square$

3. Theorem

Theorem 3.1. Let $(x_n)_{n\ge 0}$ satisfy (1.1) with $0 < x_0 < 1$. Then

$$x_n = \frac{1}{n+1} + O\!\left(\frac{\log n}{n^2}\right) \qquad (n \to \infty).$$

In particular $x_n \sim 1/n$.

Proof. Let $y_n = 1/x_n$. Then $y_n > 1$ and

$$y_{n+1} = \frac{1}{x_n(2-x_n)} = \frac{1}{2x_n - x_n^2} = \frac{1}{x_n} + \frac{1}{2-x_n} = y_n + 1 + \frac{1}{2-x_n}.$$

Hence

$$y_{n+1} - y_n = 1 + \frac{1}{2-x_n}.$$

Since $x_n \to 0$, we have $\frac{1}{2-x_n} \to \frac{1}{2}$, so $y_{n+1} - y_n \to \frac{3}{2}$. Summing from 1 to $n-1$ gives $y_n = \frac{3}{2}n + O(\log n)$, and therefore $x_n = \frac{1}{y_n} = \frac{2}{3n} + O\!\left(\frac{\log n}{n^2}\right)$. $\square$

4. Numerical table

n	x_n	$1/(n+1)$	$n\,x_n$
1	0.500000	0.500000	0.500000
2	0.375000	0.333333	0.750000
3	0.296875	0.250000	0.890625
4	0.244141	0.200000	0.976563
5	0.207031	0.166667	1.035156
6	0.179932	0.142857	1.079590
7	0.159180	0.125000	1.114260
8	0.142944	0.111111	1.143552
9	0.129883	0.100000	1.168947
10	0.118896	0.090909	1.188960
20	0.059814	0.047619	1.196279
50	0.023987	0.019608	1.199350
100	0.011994	0.009901	1.199400

5. Remark

The limit $\ell = 0$ is the unique fixed point in $(0,1)$. The linearization at $\ell = 0$ is $x_{n+1} \approx x_n$, so convergence is slow. The harmonic sequence $x_n = 1/(n+1)$ satisfies $x_{n+1} = x_n/(1+x_n)$, which differs from (1.1) only in the denominator. The table shows that $n\,x_n$ stabilizes near 1.2 for (1.1) and near 1 for the harmonic sequence, consistent with the constant $\frac{2}{3}$ versus 1 in Theorem 3.1.

[1] G. H. Hardy, A Course of Pure Mathematics, 10th ed., Cambridge University Press, 1952, §§ 91–92.
[2] R. Bellman, Stability Theory of Differential Equations, McGraw-Hill, 1953, Chapter I.
[3] J. M. Ortega, Stability and Convergence of Iterations, Springer-Verlag, 1973, Chapter 1.
[4] A. Ostrowski, Recherches sur la méthode de Graeffe, Acta Math. 72 (1940), 99–257.
[5] S. Kakeya, On the limits of the roots of a polynomial, Tohoku Math. J. 3 (1913), 192–196.

제1조(목적) 이용약관은 이용자와 회사 사이에 인터넷 서비스(이하 "서비스"라 합니다)를 이용함에 있어 회사와 이용자의 권리와 의무 및 책임사항을 규정함을 목적으로 합니다.

제2조(정의) 이용약관에서 사용하는 용어의 정의는 다음 각 호와 같습니다.

1. "회사"라 함은 서비스를 영위하는 법인을 말합니다.
2. "이용자"라 함은 회사의 서비스를 이용하는 회원 및 비회원을 말합니다.
3. "회원"이라 함은 회사에 개인정보를 제공하여 회원등록을 한 자로서 회사의 정보를 지속적으로 제공받으며 회사가 제공하는 서비스를 계속 이용할 수 있는 자를 말합니다.
4. "비회원"이라 함은 회원에 가입하지 아니하고 회사가 제공하는 서비스를 이용하는 자를 말합니다.

제3조(약관의 게시와 개정) 이용약관의 내용은 서비스 초기 화면에 게시하거나 기타 이용자가 알 수 있도록 하는 방법으로 공지하거나 통지함으로써 효력을 발생합니다.

제4조(약관의 해석) 이용약관에 명시되지 아니한 사항이나 해석에 있어 서로 다른 조항이 있을 때에는 관계 법령 및 상관례에 따라 해석합니다.

제5조(회원 가입) 회원 가입은 이용자가 이용약관 및 개인정보 처리방침의 내용에 대하여 동의한 다음 회원가입을 신청함으로써 이루어지며, 회사는 이를 승낙함으로써 회원 가입이 완료됩니다.

제6조(회원 등록 신청) 회원 등록 신청은 이용자가 회사가 정한 양식에 따라 회원 정보를 기재하여 회사에 제공하는 방법으로 합니다.

제7조(회원 등록 승낙) 회사는 제6조에 따라 등록을 신청한 이용자에 대하여 특별한 사유가 없는 한 등록을 승낙합니다.

제8조(회원 등록 신청 불승낙) 회사는 다음 각 호의 경우 등록 신청을 승낙하지 아니할 수 있습니다.

1. 등록 신청 시 허위 내용을 기재한 경우
2. 기타 회사가 합리적 판단에 의하여 필요하다고 인정하는 경우

제9조(회원 등록 신청 철회) 회원 등록 신청을 한 이용자는 회사가 등록 신청을 승낙하기 전까지 등록 신청을 철회할 수 있습니다.

제10조(회원 등록 사항 변경) 회원은 등록 사항 변경 시 지체 없이 회사에 변경 사항을 통지하거나 회사가 제공하는 방법으로 변경하여야 합니다.

제11조(회원 등록 사항 변경 불이행 책임) 회원이 등록 사항 변경을 하지 아니하거나 변경 사항을 회사에 통지하지 아니한 경우 회사는 회원에 발생한 불이익에 대하여 책임을 지지 아니합니다.

제12조(회원 등록 사항 변경 불이행 책임) 회원이 등록 사항 변경을 하지 아니하거나 변경 사항을 회사에 통지하지 아니한 경우 회사는 회원에 발생한 불이익에 대하여 책임을 지지 아니합니다.

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Abstract

Fluorescence resonance energy transfer (FRET) between a donor and an acceptor chromophore is a powerful tool for measuring molecular distances in the range of 1–10 nanometers. We present a high-throughput FRET assay for screening small-molecule libraries against a target protein. The assay uses a fluorescent ligand and a tracer bound to the same binding site, so displacement by a hit compound reduces the FRET signal in a dose-dependent manner. We report a pilot screen of 12,480 compounds against a model kinase, with a Z′ factor of 0.71 and a hit rate of 0.42 percent. Twelve hits were confirmed in dose-response curves, with half-maximal inhibitory concentrations between 0.8 and 14 micromolar. We discuss the assay design, the statistical criteria used to select hits, and the limitations of the approach when applied to proteins with low ligand affinity.

1. Introduction

Fluorescence resonance energy transfer (FRET) is a distance-dependent interaction between a donor fluorophore and an acceptor chromophore in which energy is transferred from the excited donor to the acceptor when the two are close enough for their dipoles to couple. The efficiency of transfer falls off as the sixth power of the distance between the chromophores, so FRET is sensitive to changes in molecular separation on the order of a few nanometers. In biochemistry, FRET has been used to measure protein conformational changes, to detect protein–protein interactions, and to report on the binding of small molecules to their targets.

In a typical FRET assay, the donor and acceptor are attached to two different molecules, and a change in their separation produces a change in the ratio of donor emission to acceptor emission. Because the ratio is insensitive to instrument gain and to small changes in sample volume, it can be read in a high-throughput format with a simple two-channel plate reader. The assay we describe here uses a fluorescent ligand and a tracer bound to the same binding site, so displacement by a hit compound reduces the FRET signal in a dose-dependent manner.

The efficiency of energy transfer between a donor and an acceptor at a distance r is

$$E = \frac{R_0^6}{R_0^6 + r^6},$$

where R_0 is the Förster distance, the distance at which the transfer efficiency is 50 percent. For the dye pair used in this study, R_0 is 5.4 nanometers.

2. Materials and Methods

2.1. Reagents

Recombinant kinase was expressed in Sf9 insect cells and purified by nickel affinity chromatography followed by size exclusion chromatography. The fluorescent ligand was a BODIPY-FL conjugate of a known binder, and the tracer was the same ligand with a longer linker. Compounds were dissolved in dimethylsulfoxide at 10 millimolar and stored at minus twenty degrees Celsius.

2.2. Assay Design

The assay was run in a 384-well black plate in a total volume of 20 microliters. Two nanomolar kinase, 0.5 nanomolar fluorescent ligand, and 0.5 nanomolar tracer were added to each well in buffer containing 50 millimolar Tris, pH 7.5, 150 millimolar sodium chloride, 0.01 percent Triton X-100, and 1 millimolar dithiothreitol. Plates were read on a two-channel plate reader with excitation at 485 nanometers and emission at 520 and 590 nanometers. Each plate contained 32 wells of dimethylsulfoxide alone and 32 wells of a saturating concentration of unlabeled ligand.

2.3. Statistical Criteria

The Z′ factor was calculated from the positive and negative controls. Hits were selected by a robust z-score cutoff of three standard deviations from the median of the sample wells, and confirmed in dose-response curves run in duplicate.

3. Results

3.1. Pilot Screen

The pilot screen of 12,480 compounds gave a Z′ factor of 0.71 and a signal-to-background ratio of 4.2. The median signal was 18,400 arbitrary units and the median absolute deviation was 1,210. Twelve compounds met the hit criteria, a hit rate of 0.42 percent. Dose-response curves for the twelve hits gave half-maximal inhibitory concentrations between 0.8 and 14 micromolar, with Hill slopes close to one.

3.2. Limitations

The assay depends on a fluorescent ligand that binds the same site as the tracer. Compounds that bind elsewhere on the protein will not displace the tracer and will be missed. Proteins with low ligand affinity give a small signal window and are difficult to screen by this method.

MR. FILBY MAGNIFIENTLY

Mr. Filby's car stood at the garage opposite.

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Mr. Filby's hobbies are Electrical Science and Art.

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Dr. Ananya Bernstein, Department of Cognitive Neuroscience, Kalinda University of Science and Arts

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Lecture Number 20

Lecture Name Analysis of Variance

Hello and welcome to the course Introduction to Statistics. Today's lecture is about the topic Analysis of Variance. We will be discussing what is meant by analysis of variance, the assumptions of analysis of variance, the one way analysis of variance and the two way analysis of variance.

Let us start with the first topic, what is meant by analysis of variance. Suppose we have a dependent variable, say y, and we have a set of independent variables, say x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8, x_9, x_{10}, x_{11}, x_{12}, x_{13}, x_{14}, x_{15}, x_{16}, x_{17}, x_{18}, x_{19}, x_{20}, x_{21}, x_{22}, x_{23}, x_{24}, x_{25}, x_{26}, x_{27}, x_{28}, x_{29}, x_{30}, x_{31}, x_{32}, x_{33}, x_{34}, x_{35}, x_{36}, x_{37}, x_{38}, x_{39}, x_{40}, x_{41}, x_{42}, x_{43}, x_{44}, x_{45}, x_{46}, x_{47}, x_{48}, x_{49},

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https://www.ijcseonline.org/

Abstract—Cloud computing is a new technology. There are a lot of benefits. However, it is also a challenging technology. Security issues are one of the challenges.

Keywords—

Kaplan & Co. Investment Partners | Buenos Aires, Argentina | Journal — Kaplan

Date	Account	Debit	Credit	Memo
2026-04-15	Cash — Kaplan & Co. Investment Partners	1,000.00		Initial capital contribution — Kaplan & Co. Investment Partners
	Capital — Kaplan & Co. Investment Partners		1,000.00	

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Section 1.1

Section 1.1.1

Section 1.1.1.1 Section 1.1.1.2

Section 1.1.2

Section 1.1.2.1 Section 1.1.2.2

Section 1.1.3

Section 1.1.3.1 Section 1.1.3.2

Section 1.1.4

Section 1.1.4.1 Section 1.1.4.2

Section 1.1.5

Section 1.1.5.1 Section 1.1.5.2

Section

Ancient Medicine and Magnetism

Dr. Ashok Kumar Jain, Dr. Sunil Kumar Jain, Dr. Gaurav Jain, Dr. Ankita Jain, Dr. Ankur Jain, Dr. Ankit Jain, Dr. Ankur Jain, Dr. Ankit Jain, Dr. Ankur Jain, Dr. Ankit

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1.1.1 Data Science Overview

Data Science Overview

Data Science is the study of the generalizable extraction of knowledge from data, yet the art of data science requires the application of this knowledge to a particular activity or business function, so data science is not a pure science. Data science is a relatively new term that has been used in various contexts to describe a wide range of activities, from simple reporting of business metrics to the application of sophisticated machine learning algorithms to vast quantities of data. Data science encompasses all of these and more, but what is it, exactly? Why do we need a new term to describe what we do with data? Why is it so hot? Who do data scientists work for? What do they do? In this section, we will try to answer these questions.

Data Science Defined

Data science is an interdisciplinary field that uses scientific methods, processes, algorithms and systems to extract knowledge and insights from structured and unstructured data. Data science is a concept that unifies statistics, data analysis, machine learning and related methods in order to understand and analyze actual phenomena with data. It employs techniques and theories drawn from many fields within the context of mathematics, statistics, computer science, and information science.

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Ethnic Identity and Cultural Preservation in Contemporary Nigerian Literature: A Study of Chinua Achebe's Things Fall Apart

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Abstract

This study examines the relationship between ethnic identity and cultural preservation in contemporary Nigerian literature, with particular focus on Chinua Achebe's Things Fall Apart. The study argues that Achebe's novel functions as a cultural archive that preserves Igbo cosmology, kinship structures, and oral traditions against the pressures of colonial modernity. Using a qualitative textual analysis and drawing on postcolonial theory, the study identifies three major themes: the preservation of Igbo proverbs as carriers of cultural wisdom, the depiction of traditional governance as a coherent alternative to colonial administration, and the ambivalent representation of Christianity as both liberating and destructive to indigenous culture. The study concludes that Achebe's novel remains a vital resource for understanding the relationship between literary imagination and cultural memory in Nigeria.

Keywords: Ethnic identity, cultural preservation, Igbo cosmology, postcolonial literature, Chinua Achebe, Things Fall Apart, oral tradition, colonial modernity, Nigerian literature

1. Introduction

Chinua Achebe's Things Fall Apart (1958) has been read by generations of Nigerian students as the story of a man who loses everything. What is less often noticed is that the novel also functions as a cultural archive, a text that preserves the vocabulary, the proverbs, the kinship structures, and the moral logic of a society that colonial administration was actively dismantling when it was written. This study asks how Achebe's novel performs that preservation work and what it means for contemporary readers who inherit the language he was trying to keep alive.

2. Literature Review

2.1 Postcolonial Theory and Cultural Memory

Postcolonial theory has long been concerned with the relationship between literature and cultural memory. Frantz Fanon's The Wretched of the Earth (1961) argues that colonial rule works not only through force but through the production of a new cultural hierarchy in which the colonized internalize the colonizer's

MD-7247-MQ0MB0NQ7T

Ms Cherry-Ann Stuart-Fearon-Lewis

To

Mrs Lynn Crawford-Co-Director Design Work Ltd

Dear Lynn Crawford-Co-Director Design Work Ltd

Re: Proposed Alterations at 2 Ferndale Grange North Ave

Thank you for your recent application for the above proposed alterations. Upon examination of the plans, we are unable to grant you the consent required, as we consider that the proposed alterations would adversely affect the amenity of the adjacent property, namely 1 Ferndale Grange North Ave.

Specifically, we have concerns that the proposed alterations would result in a loss of privacy to the rear garden of 1 Ferndale Grange North Ave.

Yours

Mr

Mr Cherry-Ann Stuart-Fearon-Lewis

Director

Planning Department

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MAT 1320 MATHEMATICS I

Lecture Notes on Differentiation and Integration

Prepared by

Dr. Joseph Ayoubi-Avsharov, Kennesaw State University

Last modified: Friday, August 28, 2020 at 1:00 PM EDT

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Contents

Introduction 1

Differentiation 2

Derivatives 2

Differentiation Rules 3

Implicit Differentiation 4

Integration 5

Indefinite Integrals 5

Definite Integrals 6

Appendix 7

Trigonometric Identities 7

Inverse Trigonometric Functions 8

Exponential and Logarithmic Functions 9

Introduction

Differentiation

Derivatives

$$\lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Differentiation Rules

Product Rule

$$(fg)' = f'g + fg'$$

Quotient Rule

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$

Chain Rule

$$(f \circ g)' = (f' \circ g) \cdot g'$$

Implicit Differentiation

Integration

Indefinite Integrals

Definite Integrals

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Definite Integrals

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Exponential and Logarithmic Functions

YCW Spring Demo Day 2

Jessica Livingston

Jessica Livingston

Jessica Livingston

Jessica Livingston

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Article 9a Tennis Wear Jumper

Article 9a Tennis Wear Jumper

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Article 9a Tennis Wear Jumper Jumper

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Received: 10-10-2014 | Revised: 10-10-2014 | Accepted: 10-10-2014 | Published: 10-10-2014 | Peer Reviewed Journal
Total pages: 4 | Number of Tables: 0 | Number of Figures: 0 | Number of References: 0
Corresponding author: Dr. Ashok Kumar Sharma, Associate Professor, Department of Zoology, Government College, Kota, Rajasthan, India.

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DOI: 10.58503/ojs.v1i1.1 Publication date: 2022-12-31 Keywords: Malaria, Plasmodium falciparum, Chloroquine, Resistance, Mutation, Pharmacokinetics, Drug Interactions, Dose Optimization

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Dr. Ali Ahmad Bin Ali Al Ghamdi, Associate Professor, Faculty of Arabic Language and Literature, Imam Mohammad Ibn Saud Islamic University, Riyadh, Saudi Arabia

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Cash Management

Home / Cash Management

Home

Home / Cash Management

Home / Cash Management / Cash Management / Cash Management

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AFFIDAVIT FOR FILING

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<link>https://jesuschristiscomingsoon.com/</link>
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<language>en-US</language>
<sy:updateperiod>hourly</sy:updateperiod>
<sy:updatefrequency>1</sy:updatefrequency>
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Fluid

Fluid is a substance that is flexible and can change shape.

Fluid is a substance in a flowing state and it can change shape.

Fluid: Liquid + Gas

Fluid: Liquid + Gas + Plasma

Fluid: Liquid + Gas + Plasma + Semi-solid (Toothpaste, Mud, Cement, Concrete, Cream, Paint, Ketchup, Honey, Jam, Chocolate, Glue, Tar, Sludge, Sludge, Slurry, Ice Cream, Mayonnaise, Butter, Cheese, Dough, Gel, Sand, Soil, Clay, Mud, Lava, Magma, Pudding, Jelly, Gel, Foam, Sponge, Cotton, Wool, Fur, Hair, Feathers, Leather, Wood, Paper, Cardboard, Plastic, Rubber, Glass, Metal, Stone, Concrete, Brick, Tile, Ceramic, Porcelain, Fiberglass, Carbon Fiber, Kevlar, Nylon, Polyester, Acrylic, Vinyl, Silicone, Epoxy, Resin, Adhesive, Tape, Glue, Cement, Mortar, Plaster, Drywall, Insulation

THE SPECTATOR: A WEEKLY NEWSPAPER.

Vol. 1 Issue 1 (Jan 1, 1711) Page 1 Article 1
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THE SPECTATOR.

CONTENTS.

No. 1 Introduction
No. 2 Sir Roger de Coverley
No. 3 Sir Roger de Coverley continued
No. 4 Sir Roger de Coverley continued
No. 5 Sir Roger de Coverley continued
No. 6 Sir Roger de Coverley continued
No. 7 Sir Roger de Coverley continued
No.

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AFRICAN TRUCKS TRADE

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Author Address: 14 Wilkinson Road, Kissy, Freetown, Western Area, Sierra Leone

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Abstract

This study investigates the relationship between the adoption of digital twin technology and the operational performance of steel manufacturing enterprises in the Çağlayan Industrial Zone. Data were collected from 312 managers and engineers employed in steel rolling, forging, and finishing plants using a structured questionnaire. Structural equation modeling was used to test the hypothesized relationships among digital twin adoption, process integration, predictive maintenance capability, and operational performance. The results show that digital twin adoption has a direct and statistically significant effect on operational performance, and that process integration and predictive maintenance capability mediate a substantial portion of this effect. The findings provide empirical evidence that digital twin technology functions as a coordinating mechanism rather than a standalone tool, and that its value is realized through the reorganization of production and maintenance routines. The study contributes to the literature on digital twins in heavy industry and offers practical guidance for managers considering digital twin investments in capital-intensive manufacturing.

Keywords: digital twin, predictive maintenance, steel manufacturing, Çağlayan Industrial Zone, structural equation modeling, Türkiye

1. Introduction

Steel manufacturing remains one of the most capital-intensive industrial sectors in the Çağlayan Industrial Zone, where rolling mills, forging presses, and finishing lines operate under tight tolerances and high energy costs. Over the last decade, digital twin technology has been presented as a promising tool for monitoring, simulating, and optimizing such production systems. However, most empirical studies have been conducted in electronics or automotive contexts, and the evidence from heavy industry is still fragmented.

This study asks whether digital twin adoption is associated with operational performance in steel manufacturing, and whether process integration and predictive maintenance capability mediate that relationship. The study contributes to the literature by testing a mediated model in a heavy industrial setting and by distinguishing between adoption and integration as separate constructs.

2. Literature Review

2.1 Digital Twin Adoption

Digital twin adoption refers to the extent to which a firm has deployed virtual representations of its physical assets that receive real-time data from sensors and support simulation and decision-making. Adoption is treated here as a binary construct in the survey, but it is conceptually distinct from the quality of the underlying data or the depth of the analytical models.

2.2 Process Integration

Process integration refers to the degree to which digital twin outputs are connected to scheduling, maintenance, and quality systems. Without integration, the twin remains a dashboard; with integration, it becomes a coordinating mechanism.

2.3 Predictive Maintenance Capability

Predictive maintenance capability refers to the firm's ability to forecast equipment failures and schedule interventions before breakdowns occur. In steel rolling mills, unplanned downtime is one of the most visible cost drivers.

3. Methodology

3.1 Sample and Data

Data were collected from 312 managers and engineers employed in steel rolling, forging, and finishing plants in the Çağlayan Industrial Zone. Respondents were asked about their plant's digital twin adoption, process integration, predictive maintenance capability, and operational performance. Operational performance was measured with a composite index of scrap rate, unplanned downtime, and on-time delivery.

3.2 Measures

Digital twin adoption was measured with a binary item. Process integration and predictive maintenance capability were measured with three items each, on a five-point Likert scale. Operational performance was measured with a composite index of scrap rate, unplanned downtime, and on-time delivery.

3.3 Analysis

Structural equation modeling was used to test the hypothesized relationships. Bootstrapping with 5,000 resamples was used to test the indirect effect.

4. Results

Digital twin adoption has a direct and statistically significant effect on operational performance. Process integration and predictive maintenance capability mediate a substantial portion of this effect. The indirect effect is larger than the direct effect, suggesting that digital twin adoption matters mostly through the reorganization of production and maintenance routines.

5. Discussion

The results suggest that digital twin adoption matters mostly through the reorganization of production and maintenance routines. Firms that adopt digital twins but do not integrate them into scheduling and maintenance systems gain little operational benefit.

6. Conclusion

The study tests a mediated model of digital twin adoption in steel manufacturing and finds that process integration and predictive maintenance capability carry most of the effect. Future research should test the model in other heavy industrial settings and track adoption over time.

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Wpływ ekspozycji na pola elektromagnetyczne o częstotliwości 50 Hz na wybrane parametry krwi obwodowej

Dr inż. Ewa Nowak, Mgr inż. Piotr Zalewski

1.1 Introduction

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AVENUE TO TITUS

Author: Jack Mitchell
Author's email address: jack@jackmitchell.net
Author's postal address: 1 Ashley Road, Altrincham, Cheshire, WA14 2DT, United Kingdom

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<?xml version="1.0" encoding="utf-8"?>
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Corresponding author: Author Name

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DOI: 10.1177/1094670519837117

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Yuan Liu-Thompkins, Yi Yang, Kristen Walker, Wenzhong Zhou

Do Advertisers Overspend for Portfolio Advertising Reach?
Yuan Liu-Thompkins, Yi Yang, Kristen Walker, Wenzhong Zhou, Journal of Advertising Research, https://doi.org/10.2501/JAR-2019-044, published online: 01 Dec 2019

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$$\sigma_{\tau} = \frac{\sigma_{\tau_1}\sigma_{\tau_2}\sqrt{1-\rho^2}}{\sqrt{\sigma_{\tau_1}^2 + \sigma_{\tau_2}^2 - 2\rho\sigma_{\tau_1}\sigma_{\tau_2}}}$$

DOI: 10.1177/1094670519837377
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Hong Kong Baptist University, Hong Kong, China
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Eric Okazaki, Hong Kong Baptist University, Hong Kong, China
Shan Li, Hong Kong Baptist University, Hong Kong, China

$$\sum_{i=1}^{n} \frac{(x_i - \bar{x})(y_i - \bar{y})}{s_x s_y}$$

Abstract
Research on the effectiveness of celebrity spokespersons in advertising has been inconclusive. We argue that this is because the effectiveness of celebrity spokespersons depends on the congruity between the celebrity and the product being endorsed. We propose that congruity between the celebrity and the product being endorsed enhances the effectiveness of the celebrity spokesperson by increasing the perceived credibility of the celebrity spokesperson. We also propose that congruity between the celebrity and the product being endorsed increases the perceived credibility of the celebrity spokesperson because it increases the perceived expertise of the celebrity spokesperson. We test our hypotheses using a meta-analysis of 1,000 effect sizes from 100 studies. We find that congruity between the celebrity and the product being endorsed increases the effectiveness of the celebrity spokesperson, and that this effect is mediated by the perceived credibility of the celebrity spokesperson. We also find that congruity between the celebrity and the product being endorsed increases the perceived expertise of the celebrity spokesperson. We discuss the implications of our findings for theory and practice.

$$\frac{\sigma_x \sigma_y}{\sigma_x \sigma_y}$$

Keywords
celebrity endorsement, congruity, credibility, expertise, meta-analysis
Received: 1.1.1
Revised: 1.1.1
Accepted: 1.1.1
Online publication date: 1.1.1

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Page not found

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ISSN 2708-7026 (Online)
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Ref: Fluid Mechanics

Ref: Fluid Mechanics

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Author:
Name: Jennifer Gregory-Smith
Affiliation: Department of Marketing, Kingston University London, Kingston Hill Campus, Kingston upon Thames, Surrey, United Kingdom
Email: j.gregory-smith@kingston.ac.uk

Publication Info:
Journal: Journal of Advertising Research
Volume: 59
Issue: 1
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Year: 2019

Abstract:
Using a sample of 1,000 participants, this study tests the effectiveness of a 2 (gender: female vs. male) × 2 (sexual orientation: heterosexual vs. homosexual) × 2 (sexual orientation of the actors: heterosexual vs. homosexual) × 2 (sex of the actors: female vs. male) × 2 (sex of the actors: female vs. male) × 2 (sex of the actors: female vs. male

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Vol.1, No.1, 01-04
Nov. 2013 International Journal of Advanced Research in Chemical Science
www.arcjournals.org

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Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos qui ratione voluptatem sequi nesciunt. Neque porro quisquam est, qui dolorem ipsum quia dolor sit amet, consectetur, adipisci velit, sed quia non numquam eius modi tempora incidunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim ad minima veniam, quis nostrum exercitationem ullam corporis suscipit laboratoriosam, nisi ut aliquid ex ea commodi consequatur? Quis autem vel eum iure reprehenderit qui in ea voluptate velit esse quam nihil molestiae consequatur, vel illum qui dolorem eum fugiat, quo voluptas nulla pariatur?

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

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